PREVEX® FLAMMABILITY ANALYZER

MODEL FTA

700 Series

INSTRUCTION MANUAL

Control Instruments® Corporation

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WARNING!

THIS ANALYZER IS USED FOR THE SAFE OPERATION OF MACHINERY AND PROCESSES, AND TO PREVENT FIRES AND EXPLOSIONS.

YOU MUST READ AND UNDERSTAND THE ENTIRE INSTRUCTION MANUAL BEFORE INSTALLATION, OPERATION, CALIBRATION, OR SERVICING OF THIS ANALYZER.

OBSERVE, UNDERSTAND, AND OBEY ALL WARNINGS IN THIS MANUAL. FAILURE TO PROPERLY INSTALL, OPERATE, CALIBRATE OR SERVICE THIS

ANALYZER MAY RESULT IN A FIRE OR EXPLOSION THAT CAN CAUSE DESTRUCTION OF PROPERTY OR SERIOUS BODILY INJURY.

PREVEX® Flammability Analyzer

Model FTA Instruction Manual
For Model Numbers:
700 Series: SNR701, SNR702, SNR704, SNR705
SNR701D, SNR702D, SNR704D, SNR705D
Including -T2, T3, T4 and T6 variants
For Information on 670 series units refer to 670 series FTA Manual Part Number: H7FTA670
Software Versions 7.01 and above
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TABLE OF CONTENTS

1. Introduction 7
1.1. Warnings 7
1.2. Qualified Personnel 7
1.3. Software Versions 8
1.4. Special Notices 8
1.5. Models 11

2. Theory of Operation 14
2.1. Hazards of Flammable Gases and Vapors 14
2.2. Measuring Principle 15
2.3. Flame Cell and Sensing Flame 16
2.4. Flow Diagram 17
2.5. Electrical Diagram 18
2.6. The Controller 19
2.7. Air Dilution Option 28
2.8. Fail-safe Considerations 31

3. Specifications 35

4. Installation 41
4.1. Unpacking 41
4.2. Equipment, materials, and utilities 41
4.3. Location and Mounting 42
4.4. Sample and Exhaust Tubing 42
4.5. Utilities 49
4.6. Compressed air supply 49
4.7. Fuel supply cylinders 51
4.8. Zero and Span Test Gases 52
4.9. AC Power 52
4.10. RS-485 Serial Communications 53
4.11. Relays 54
4.12. Wiring for EMC (Electromagnetic Compatibility) 55
4.13. Remote Control Inputs 56
4.14. 4-20mA Signal Output(s) 57
4.15. Visual and Audible Alarms and Indicators 57
4.16. Security 58

5. Display, Indicators, Controls 59
5.1. Indicators 59
5.2. Menu and Select Pushbuttons 59
5.3. MENUS AND REGISTERS 61
5.4. COMMANDS 63
5.5. GAUGES MENU 64
5.6. FLOW SYSTEM 64
5.7. DILUTION MONITOR 64
5.8. FUEL MONITOR 65
5.9. PURGE MONITOR 65
5.10. CALIBRATION MENU 65
5.11. ALARMS MENU 66
5.12. ALARM TESTS MENU 67
5.13. 4-20mA OUTPUTS 67
5.14. CONTROLS 68
5.15. COM PORT 68
5.16. SETTINGS MENU 69
5.17. SECURITY MENU 69
5.18. Flashlight-activated Commands 69

6. Startup Procedure 71
6.1. Initial conditions 71

6.2. Air *71*
6.3. Fuel *71*
6.4. Power *71*
6.5. Ignition *72*
6.6. Preliminary Calibration *72*
6.7. Inputs and outputs *74*
6.8. Records *74*
6.9. Stability test *75*
6.10. Response time verification *75*
6.11. Verification of Readings *76*

7. Calibration **77**
7.1. Initial calibration *77*
7.2. Recalibration Procedure *85*

8. Operation **86**
8.1. Reading and status. *86*
8.2. Acknowledge Command. *86*
8.3. Alarm Reset Command *87*

9. Maintenance **88**
9.1. General information and precautions *88*
9.3. "Service Needed" Messages *90*
9.4. Scheduled Maintenance *92*
9.5. Readings *93*
9.6. Utilities *93*
9.7. Gauges *93*
9.8. Recalibration *94*
9.9. Leak test *94*
9.10. Alarm test *94*
9.11. System review *94*
9.12. O-Rings *94*
9.13. Purge Restrictor *95*
9.14. Flow System Preventative Maintenance *96*
9.15. Cleaning *97*
9.16. Fuel Regulator Adjustment *100*

10. Troubleshooting **101**
10.1. AC Power *101*
10.2. Flame Cell Temperature *101*
10.3. Air Inlet Pressure *101*
10.4. Fuel Inlet Pressure *101*
10.5. Spark *102*
10.6. Flame *102*
10.7. Sample Flow *102*
10.8. Calibration Flow *103*
10.9. Troubleshooting by status message *103*
10.10. Data Lost and Cold Start *106*
10.11. Drift *106*
10.12. "Service Needed – Check" Relay *107*
10.13. Cold Start *107*

11. Spare Parts **108**
11.1. Parts for Preventative Maintenance *108*
11.2. Parts for Routine Maintenance *108*
11.3. Parts for Maintenance and Repair *108*
11.4. Burners and restrictors *108*
11.5. Parts for Repair *108*
11.6. Controllers *109*

TABLE OF CONTENTS

12. Serial Communications 115
12.1. Serial Communications Specifications 115
12.2. Serial Communications Notes 115
12.3. Register Conversion Factors 116
12.4. Registers 117
13. Revision History 128

1. Introduction

Read and understand this instruction manual before installation, operation or maintenance of the analyzer.

IMPORTANT: In this manual, the marking WARNING! indicates important instructions. Failure to understand and follow these instructions can result in a fire or explosion, destruction of property or serious bodily injury.

1.1. Warnings

Observe the following warning notices and all additional warnings found within this manual:

WARNING! Read and understand this instruction manual before installation, operation or servicing.

WARNING! It is recommended to users to read the procedures described in EN60079-29-2 for reference.

WARNING! This analyzer measures the flammability of gases. It does not provide protection from toxic gases, or from the toxic effects of flammable gases. Many flammable gases have toxic effects. Some can pose an immediate danger to life at concentrations too low for this analyzer to measure.

WARNING! The analyzer contains electric circuits. Proper handling procedures for high voltage circuits must be observed. Keep all covers tight when circuits are energized. Do not remove covers, disconnect terminal blocks, or disconnect wires unless circuits are de-energized, or the atmosphere is known to be well below the lower flammable or explosive limit.

WARNING! The analyzer is purged. In hazardous areas, maintain the proper purge rate and pressurization as required by code. If purge or pressurization is lost, restore it immediately or disconnect electrical power. Purge for a minimum of 9 minutes at 15 liters per minute flow before turning on electrical power.

WARNING! Do not operate unless the flame arrestors are installed. Operation without flame arrestors may allow flame propagation that could ignite the process or atmosphere being measured.

WARNING! The analyzer uses hydrogen or propane fuel. Observe proper handling precautions. Turn the fuel supply off whenever the air supply to the analyzer is off.

WARNING! During calibration the analyzer cannot make readings or activate alarms. Perform calibration only when it will not interfere with safety of the process being monitored. During calibration, the analyzer signals cannot be used for control or safety function.

WARNING! Do not bypass, disable, or tamper with this analyzer. Secure it from unauthorized access.

WARNING! Fuel pressure failure, low oxygen conditions or flammable concentrations above the measurement range, can cause reading errors. Off-scale readings in either direction may indicate a hazardous gas concentration.

WARNING! IP ratings do not imply that the equipment will detect gas during or after exposure to those conditions. Follow the calibration frequency intervals of section 7.2 if exposed to those IP rating limits.

1.2. Qualified Personnel

Installation, operation and maintenance must be performed by qualified personnel only. This requires understanding of instrumentation, handling compressed gases, flammable fuels, electric circuits, the behavior of the gases and vapors in the process being monitored, and the relevant codes, standards and recommended practices for flammable gas detectors and the process being monitored.

WARNING! All personnel who install, operate, or maintain this device must read, and understand this instruction manual. Short-form instructions and guides are not an acceptable substitute.

All personnel who monitor, use, or depend upon this device, must understand the hazards associated with flammable gases and vapors, the meaning of the readings made by this analyzer, and the meaning of all alarms and indicators.

Use this manual in conjunction with the codes and standards that apply to the hazards of the flammable gases and vapors, and the intended use of the analyzer as a means of protection. Specifications and requirements vary in different locations. The device must be reviewed for the particular requirements of the local authority having jurisdiction.

WARNING! The user must review and ensure compliance with all applicable safety codes.

1.3. Software Versions

This manual covers software versions 7.01 and above. It is not valid for lower versions. The software version installed in the analyzer can be found in the VERSION register.

1.4. Special Notices

A permanent serial plate (nameplate) is mounted on the lower right side of the analyzer. It contains the model number, serial number, ratings, and the approval mark, if any. Models have differences based on requirements for performance and hazardous locations from their respective approvals. In particular, purging and pressurizing the enclosure for installation in hazardous areas varies between models. Installation in hazardous locations may require an additional purge device.

1.4.1. Special Notice for Factory Mutual Research (FMR) approval:

Approval of the 4-20mA output does not include or imply approval of the apparatus connected to the instrument. To maintain FMR Approval of the system, all 4-20mA or current loop instruments connected must also be FMR Approved.

Approval allows the presence and operation of serial communications software in the sensor/controller (Modbus protocol). However, the communications functions provided are not included in the FMR Approval.

In order to maintain FMR approval for Division 2 area it is important to occasionally inspect the alarm relays within the electronics assembly for package integrity.

WARNING! In order to maintain FMR approval, the danger alarm must latch, either by the LATCHES register, or through the use of an equivalent external latching relay or device.

1.4.2. Special notice for FMc approval.

ATTENTION: POUR DES RAISONS DE SECURITE, CET EQUIPEMENT DOIT ETRE UTILISE, ENTRETENU ET REPARE UNIQUEMENT PAR UN PERSONNEL QUALIFIE. ETUDIEZ LE MANUEL D'INSTRUCTIONS EN ENTIER AVANT D'UTILISER, 'ENTRETENIR OU DE REPARER L'EQUIPEMENT.

AVERTISSEMENT: RISQUE D'EXPLOSION. NE DEBRANCHEZ PAS LES BORNIERS OU LES PENDANT QUE LE CIRCUIT EST SOUS TENSION, SAUF SI LA ZONE EST EXEMPTE DE CONCENTRATIONS INFLAMMABLES.

For instructions in French or other languages, please contact the manufacturer or its representative.

1.4.3. Special notice for ATEX, UKCA or IECEX approval - for the safe functioning of equipment

WARNING! The danger alarm must latch, either by the DANGER LATCH register, or through the use of an equivalent external latching relay or device.

WARNING! The maximum operating temperature, BLOCK SETTING, must be set at or below the T-rating temperature minus 15°C.

1.4.4. Special notice for ATEX, UKCA or IECEX approval - installation in a non-hazardous zone

Models which are ATEX, UKCA or IECEX marked may be installed in a non-hazardous zone while the sample and exhaust are connected to an Ex Zone 1, explosion group IIB+H2, without the requirements of section 0, if instrument air (dry, clean compressed air free of flammables) is supplied to the analyzer with an inlet pressure of 20 psig (1.4 bar), supplying a constant stream of air through the instrument.

1.4.5. Special notices for ATEX, UKCA or IECEX approval - installation in Ex Zones

Models which are ATEX, UKCA or IECEX marked can be installed and operated in, or connected to, Ex Zone 1 or Ex Zone 2 hazardous locations, explosion group IIB+H2. The ATEX, UKCA and IRCEX labels includes the year of manufacture, the ratings, and purge information needed for installation in Ex Zone 1 or Ex Zone 2 hazardous locations in conformance with norms of the 60079-29-1.

If the analyzer itself is installed in Ex Zone 1, then an ATEX, UKCA or IECEX certified Ex p purge and pressurization device must be used to deliver protective gas to the analyzer, typically air taken from a non-hazardous zone, or an inert gas, and to disconnect the power supply and other electrical connections when the purge is not active or fails. The Ex p safety device must conform to all applicable codes. It should also meet the requirements listed here:

1. The purge apparatus and analyzer instruction manuals must be read, understood, and followed.
2. All seals of the analyzer housing must be in place and in good condition. Suitable cable glands for wiring must be installed and properly sealed. All seals must be inspected in regular intervals to ensure proper operation. Seals showing signs of wear or leakage must be replaced.
3. The analyzer must be connected to earth ground using the connector on the exterior of the enclosure.
4. The inlet for the protective gas at the analyzer is a ¼ inch compression fitting. The outlet is a 3/8 NPT thread. The tubing between the outlet at the analyzer and the purge system must provide a suitable inner diameter that ensures that the pressure drop across that tubing is at most 5 hPa.
5. Before any electrical circuits in the analyzer enclosure, including input and output connections to other devices, are energized, an initial purge must be made for a minimum of nine minutes, at a flow fifteen liters per minute or more and a minimum enclosure pressure of 0,5 hPa. Thereafter the 0,5 hPa enclosure pressure and fifteen liters per minute purge flow rate must be maintained for as long as circuits remain energized. The maximum pressure of 25 hPa must not be exceeded. These pressures are measured relative to the ambient pressure exterior to the analyzer enclosure.
6. If either the continuous purging or pressurization of the analyzer enclosure fails, the purge equipment shall give an alarm, and in the case of Ex Zone 1 installations, shall, via Ex-coupling relays, automatically remove power from all circuits in the analyzer enclosure except those which are intrinsically safe.

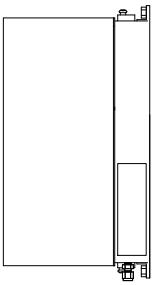
7. The analyzer has flame arrestors to prevent flashback through the sample and exhaust tubing. They must be kept in place at all times. The flame arrestors must be replaced if they show any sign of mechanical damage that might reduce their effectiveness. The flame arrestors must never be cleaned with an abrasive.
8. The fuel pressure delivered to the analyzer must not exceed the maximum specified pressure as given in the analyzer instruction manual. A relief valve should be provided at the fuel source to prevent overpressure of the fuel supply to the analyzer.
9. The fuel inlet fitting, part number SNP374 supplied with the analyzer contains a restrictor element that limits the maximum flow of fuel into the analyzer housing during failure of the fuel containment system. This fitting must be kept in place at all times. It should never be cleaned with an abrasive or mechanically damaged. If it becomes contaminated, or is otherwise damaged, it must be replaced only by the same part number.
10. The internal fuel delivery system of the analyzer is assembled using stainless steel capillaries with stainless steel compression fittings. Care must be taken to ensure that the fuel system is not damaged and is free of leaks.

1.4.6. Specific Conditions of Use for ATEX, UKEX and IECEx approval

1. Flamepath joint repair is not intended.
2. The flame cell enclosure includes four fasteners: hexagonal socket set screws, Stainless Steel Type 300 18-8, per ISO/R861, with length 80mm and diameter 6 mm.
3. Not for use in atmospheres with hazards related to Carbon Disulfide.
4. The end user shall provide safety devices with spark and particle barriers in accordance with IEC/EN60079-2 and the manufacturer's instructions. The control and pressurization system shall include "pxb" ratings to ensure suitable integrity and function. They shall themselves not be capable of causing ignition or shall be mounted outside the hazardous area. They shall facilitate and monitor the purging cycle at minimum 0,5mbar and 15 L/min for minimum 9 minutes, and thereafter maintaining minimums for internal pressure of 50Pa and flowrate of 15 L/min. They shall also prevent energizing the protected enclosure and keep flammable gas supply off, until the purging cycle has successfully completed and specified pressure and flow rates are maintained.

1.5. Models

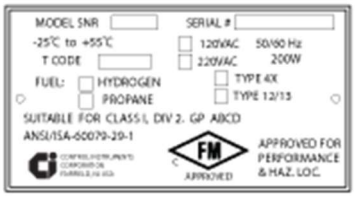
A permanent serial plate (nameplate) is mounted on the lower right side of the analyzer. It contains the model number, serial number, ratings, and the approval mark, if any. Models have differences based on requirements for performance and hazardous locations from their respective approvals. In particular, purging and pressurizing the enclosure for installation in hazardous areas varies between models. Installation in Ex Zone 1 or Division 1 areas requires an additional purge device.



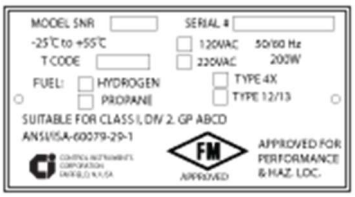
Models suitable for hazardous locations may also be installed in non-hazardous or general-purpose locations without an additional purge device.

1.5.1. FMc and FM Models

Canadian Standards Association (FMc)

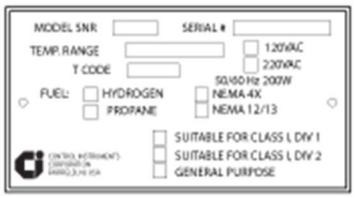


Factory Mutual Research (FM)

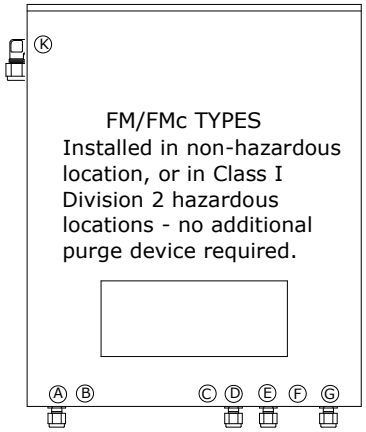


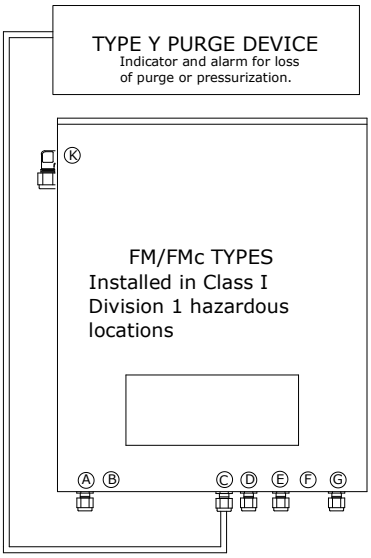
Generic Label

Models with generic labels have options not tested and approved by a third party.

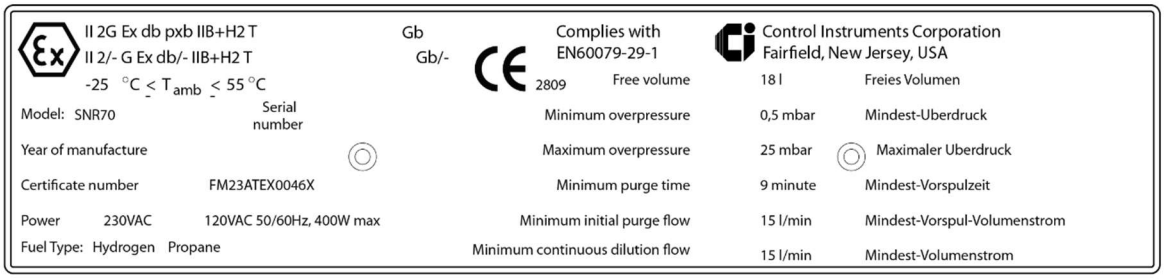


FM and FMc approved models are suitable for installation in Class I Division 2 hazardous locations and for sampling from Class I Division 1 hazardous locations. Installation in a Class I Division 1 hazardous location requires the addition of an approved purge device and is not FM approved as such. Generic types require case-by-case evaluation.

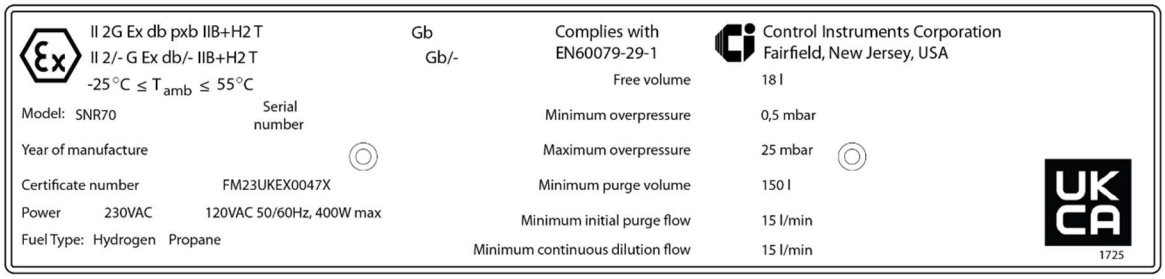




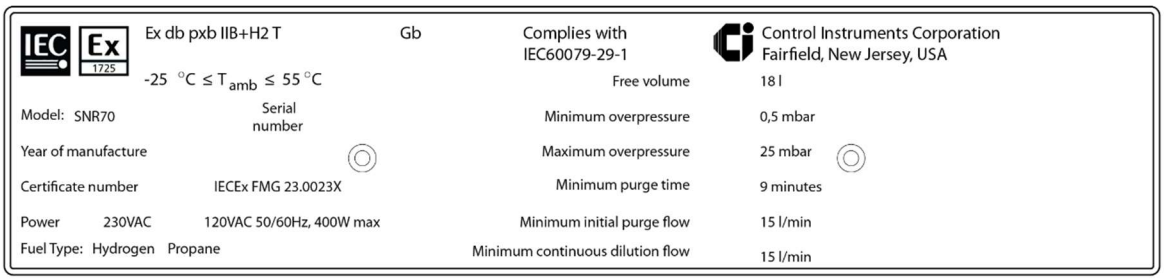
1.5.2. ATEX (CENELEC) and CE Models



ATEX Label



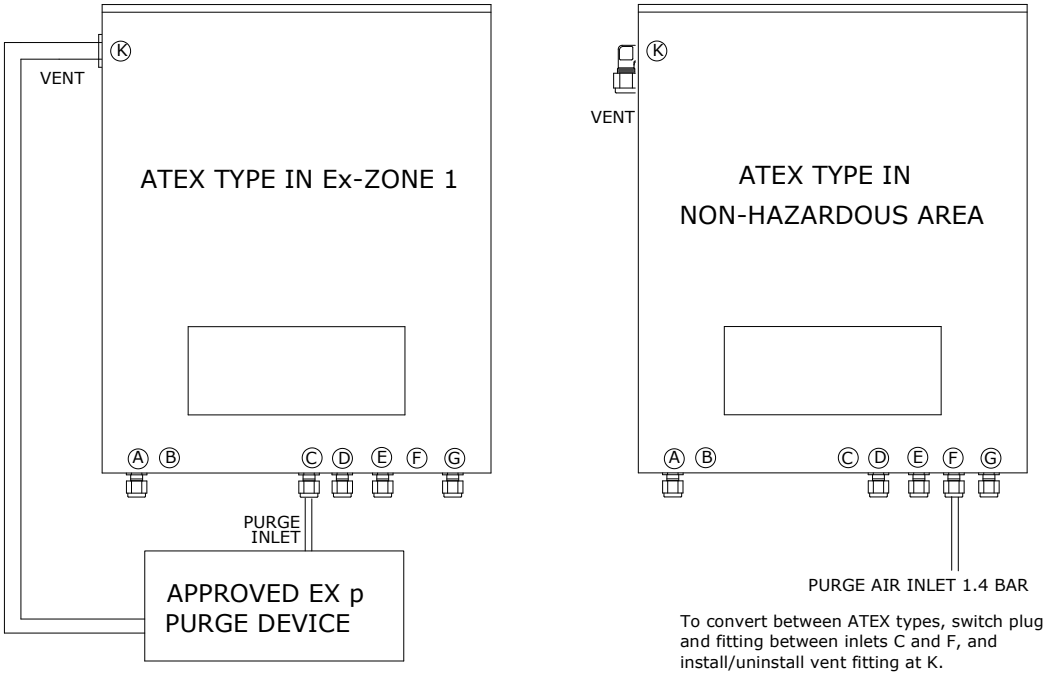
UK Label



IECEX Label

Models SNR70x-Ty where x is 1,2,4 or 5 and Ty is the temperature rating on the ATEX, UKCA, IECEX and CE marks. The label includes the year of manufacture, the ratings, and purge information needed for installation in Ex Zone 1 or Ex Zone 2 hazardous locations in conformance with norms of the ATEX directive 2014/34/EU. A latching safety thermocouple with manual reset is installed to prevent over-temperature. It removes power from the heater so the temperature does not exceed the T rating of the analyzer. The maximum operating temperature must be set at or below the T-rating temperature minus 15°C.

ATEX Purge Configurations



To convert between ATEX types, switch plug and fitting between inlets C and F, and install/uninstall vent fitting at K.

2. Theory of Operation

2.1. Hazards of Flammable Gases and Vapors

The hazards of flammable gases and vapors are fire, explosion, and toxicity. The PrevEx analyzer does not protect against short or long-term toxicity and toxic effects. It is designed to give early warning of the possibility of fire and explosion.

2.1.1. Lower Flammable Limit - LFL

The Lower Flammable Limit (LFL), is the leanest mixture of gas or vapor in air, that, when ignited, will continue to burn even after the source of ignition is removed. This means that a combustion wave, or "flame front," can travel through the mixture, releasing energy as it moves, and continuing on its own. It is possible that the combustion wave can travel long distances through the mixture, back to the source of the vapors, and cause a fire. At concentrations above the LFL, the mixture can rapidly increase in speed and the amount of energy it releases as it burns, to the point where it can cause very high pressures and an explosion. This is why the Lower Flammable Limit is sometimes called the Lower Explosive Limit (LEL), an older term used interchangeably with LFL.

Each flammable gas has its own LFL value, the percent by volume concentration in air at which that particular gas becomes flammable. At concentrations below the LFL, there is not enough flammable gas to propagate a combustion wave. By keeping an atmosphere below the LFL, it is possible to isolate sources of ignition from sources of combustion, preventing a flame front from traveling back to the source of vapor or gas and creating a fire or explosion.

There is also an Upper Flammable Limit (UFL), which is the richest mixture of gas or vapor in air such that the ignited mixture will continue to burn after the source of ignition is removed. At concentrations above the UFL, there is not enough air in the mixture to propagate a combustion wave. But once additional air is added to the mixture, it becomes flammable or explosive. The monitoring of mixtures having high concentrations above the UFL requires special considerations that are beyond the scope of the use of this device. In the use of this analyzer, all concentrations above the LFL should be treated as if they were flammable and explosive.

2.1.2. %LFL Readings

This device measures flammability in the range from "zero" air that is free of flammable gas, up to the LFL. The measuring scale is divided into percentages of the LFL, so that 0%LFL means that no flammable gas is present, and 100% LFL means that the Lower Flammable Limit has been reached. Laws for industrial processes typically forbid operation above 50% LFL, and require an immediate response to protect life and property when the flammable gas concentration exceeds 50% LFL.

WARNING! Readings above 50% LFL indicate a potential hazard to life and property.

2.1.3. Flash Point

The Flash Point is the temperature at which a flammable liquid gives off enough vapor to form an ignitable mixture with air. The liquid can form an invisible, explosive "cloud." Many flammable liquids have flash points at or below ordinary ambient temperatures. These will ignite immediately if a source of ignition is brought anywhere near the liquid. When the Flash Point is below the ambient temperature, vapor that forms above the surface of the liquid can very easily travel through the air, away from the spilled liquid, until a source of ignition is reached. The ignited vapor cloud can explode, or it can flash back to the liquid and explode, or it can ignite the surface of the liquid itself and cause a fire.

Liquids with Flash Points above ordinary ambient temperatures can still form hazardous vapor mixtures when heated, and can condense back into liquid when cooled. For accurate measurement,

every part of the sampling system and sample tubing must be heated above the Flash Point. Otherwise, condensation prevents flammable vapors from reaching the analyzer. Even a very short section of tubing, at a temperature just below the flash point, can disable the measurement[1].

2.1.4. Autoignition

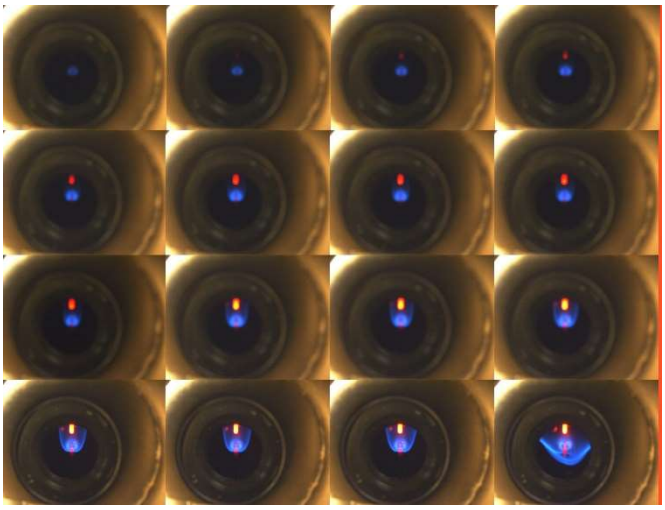
Sufficiently heated, a flammable gas mixture can spontaneously ignite. The auto ignition temperature (AIT) is the lowest temperature at which a flammable vapor spontaneously ignites. The AIT varies for each type and concentration of gas, but usually occurs at the stoichiometric concentration (perfect combustion). Heating a mixture to the AIT can cause a fire or explosion, sometimes after a time delay up to a few minutes. In effect, heated flammable gases are more flammable.

2.1.5. Temperature Dependence

Because heating a flammable gas makes it more flammable, a mixture below the LFL at one temperature can exceed the LFL when heated. For accurate measurement of heated processes, this temperature dependence must be included in the analyzer calibration. Regulations may require an increase in sensitivity of either 7.8% (USA) or as much as 14% or more (ATEX, UKEX or IECEX) for every 100ºC increase in temperature.[2]

2.2. Measuring Principle

A sample is drawn into the analyzer's flame cell. Flammable gases and vapors in the sample are burned in the sensing flame. A thermocouple located directly above the sensing flame converts the resulting temperature rise into an electrical signal, which is proportional to the concentration of flammable gas from 0 to 100% of the Lower Flammable Limit (LFL or LEL).



Sensing flames in a hydrogen-fueled analyzer. Flame size and thermocouple temperature increase from low (upper left) to high flammability (lower right - where a flame front is visible).

---

[1] Small temperature changes have a large effect on the concentration. A temperature drop as little as 10°C below the flash point can cause false low readings from condensation and prevent an alarm. For many solvents, near the flash point the maximum (saturation) vapor concentration is reduced 50% by a -10°C reduction in temperature.

[2] It is important to note that the temperature dependency of flammability is not a physical influence on the analyzer's reading or its principle of operation. Because it is thermostatically heated, the analyzer does not respond to changes in the temperature of the atmosphere being monitored. Nor is the temperature dependency of flammability due to the expansion or contraction of air. It is an effect that changes the chemical and thermodynamic properties of the flammable gas mixture.

2.3. Flame Cell and Sensing Flame

Flammability measurement occurs in the flame cell, where a small “sensing flame” sits on a burner tube. A sample is continuously drawn in through a sample tube by suction produced by an air aspirator. The aspirator runs from a regulated supply at constant pressure on the air inlet. Flammables in the sample burn in the sensing flame and produce a temperature rise, proportional to the Lower Flammable Limit, in a thermocouple that is located above the flame. The sample gases are then exhausted through the aspirator and exit through the exhaust tubing.

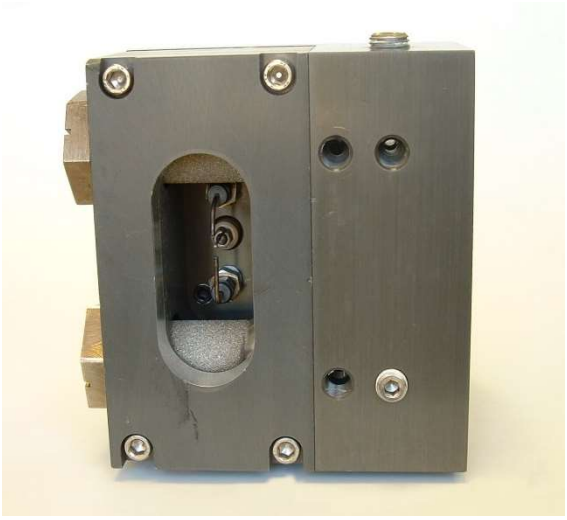
The sensing flame is ignited by a discharge from a spark electrode to the grounded burner tube. It is maintained by a constant flow of fuel to the burner. A fuel regulator feeds constant pressure through a sintered metal restrictor inside the base of the burner tube to produce a stable flame with a constant temperature in the absence of flammable gases in the sample.

The flame cell has inlet and exhaust flame arrestors to prevent propagation of the sensing flame out of the flame cell. They must be kept clean and properly installed at all times. They should never be subjected to mechanical abuse or cleaning with abrasive substances. They should be replaced if they show signs of wear or damage. Disposable filters inside the inlet and exhaust flame arrestors keep the sensor flow paths clean.

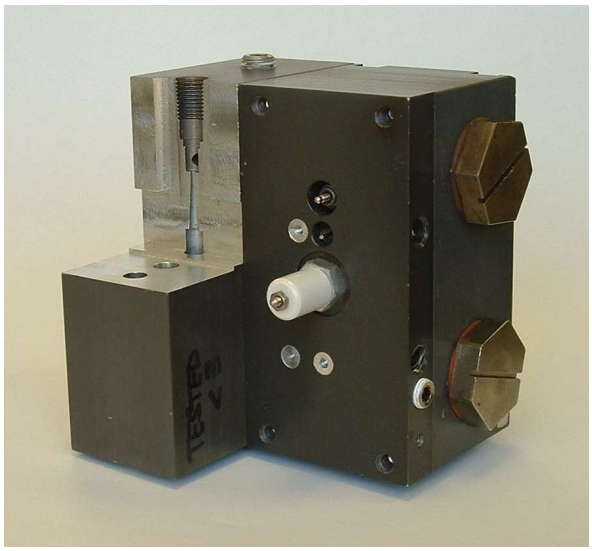
WARNING! Always operate the analyzer with undamaged flame arrestors in place.

WARNING! Flamepath joint repair is not allowed. Damaged parts must be replaced with OEM parts.

The sample flow is measured as it exits the flame cell. An orifice located behind the small sintered metal filter produces a pressure drop proportional to the flow rate. The pressure drop is measured by a transducer in the controller. If the sample flow rate decreases, a message is given to check the flow. If it continues to decrease, a “Low Flow” fault is activated.

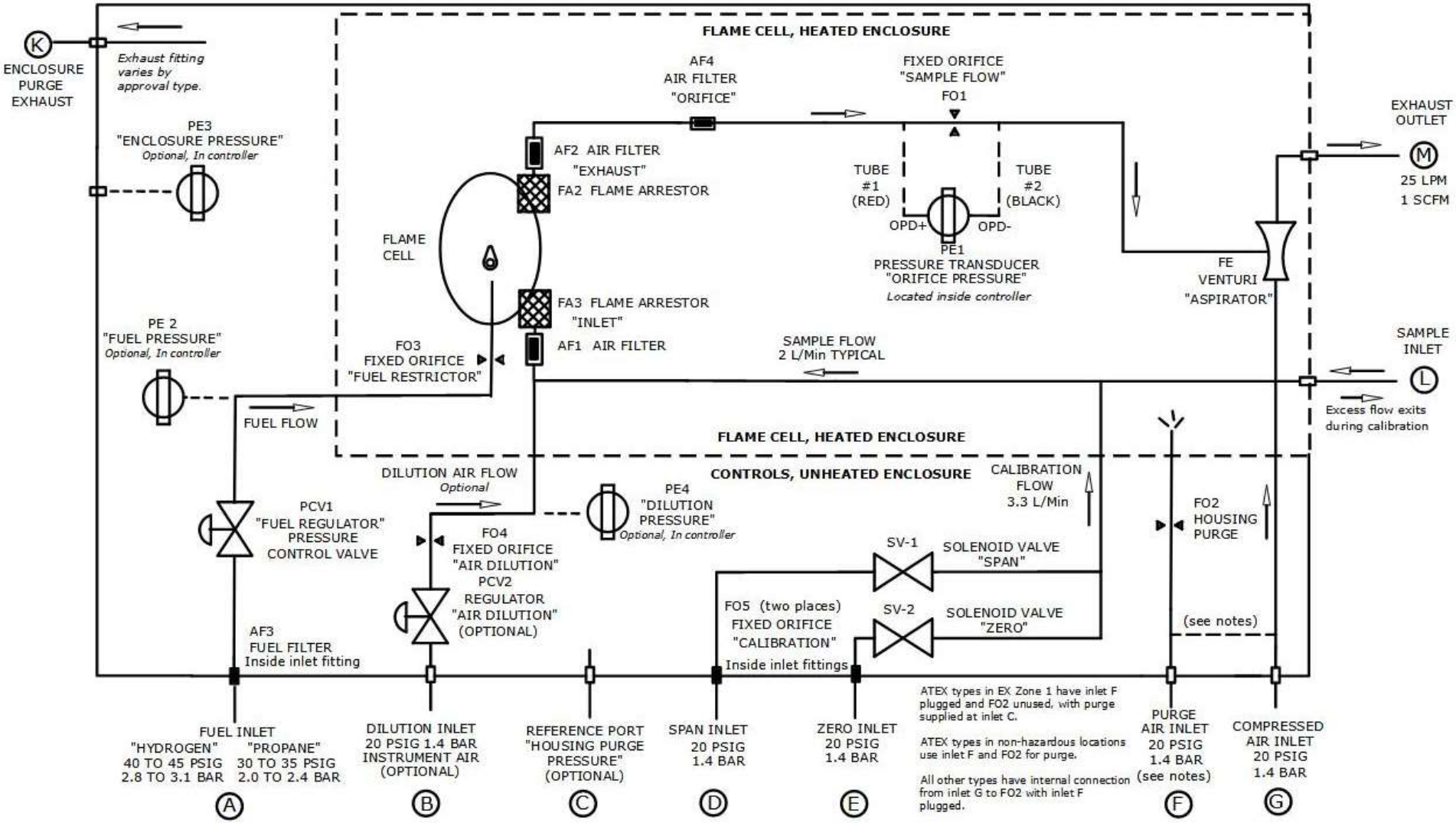


Front view of flame cell with cutaway exposing the internal parts. The flame arrestor caps can be seen along the left edge. Adjacent to the caps are the flame arrestors, positioned at the top and bottom of the flame cell. Directly in the center is the spark plug. The thermocouple is above the spark plug and the burner is below.

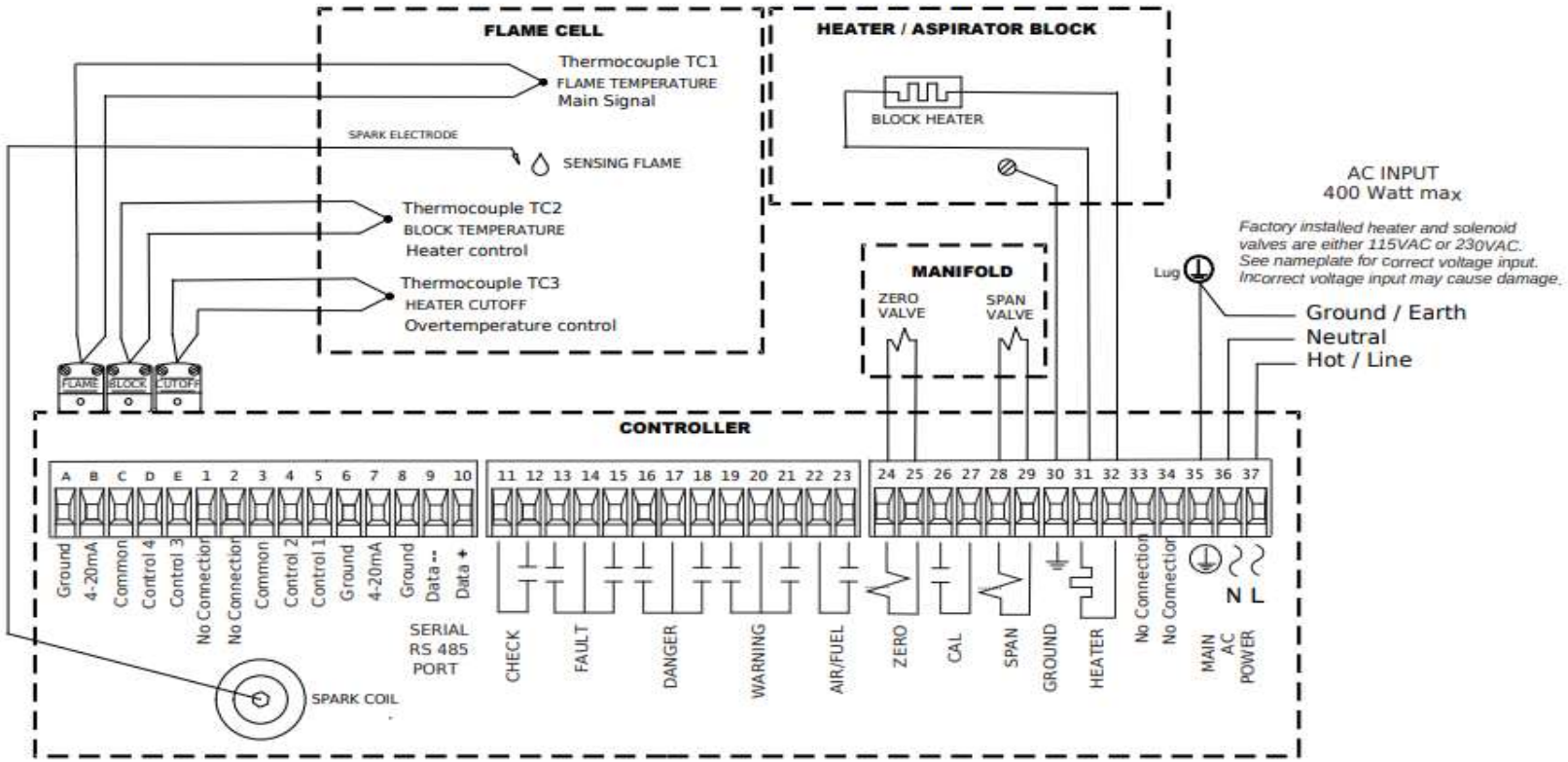


Rear view of flame cell and aspirator block with cutaway showing the location of the aspirator and flow paths. An aspirator nozzle sits inside the block. Compressed air passing through the aspirator creates suction that draws sample flow through the flame cell.

2.4. Flow Diagram

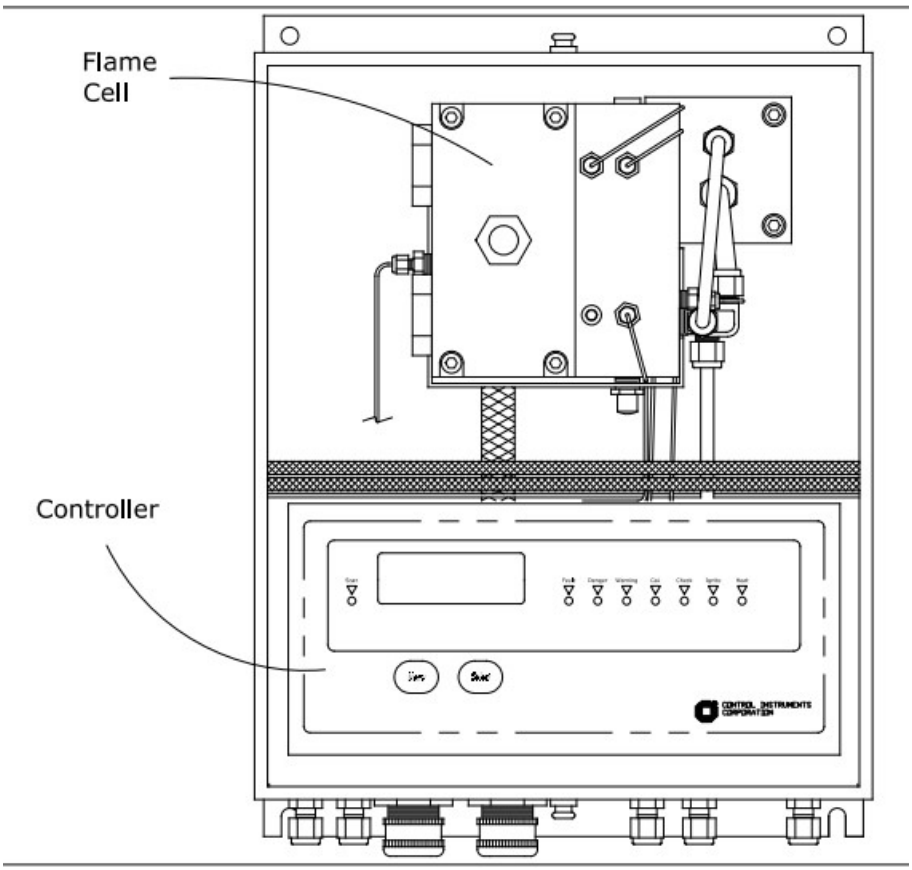


2.5. Electrical Diagram



2.6. The Controller

The controller is an electronic measuring and control device mounted inside the sensor enclosure. It conditions and converts signals from the thermocouples and transducers, and performs control, alarm, and output functions. The controller makes the flammability reading, controls alarms, regulates the flame cell temperature, monitors the sample flow, and generates faults if it detects improper operation. It amplifies the thermocouple signal and performs digital signal processing to obtain a final calibrated flammability reading in percent of the Lower Flammable Limit. It controls two level alarms: a “high” alarm (warning), and a “high high” alarm (danger). It outputs the reading and status of the sensor by means of indicators, electromechanical relays, 4 to 20 milliamp outputs, and a serial port.



Front view of typical analyzer, shown with cover(s) removed. The controller is in the lower, unheated section. Pneumatic controls are behind the controller. Pneumatic and electrical connections are at bottom.

2.6.1. Control of Flame Cell Temperature

An electric heater embedded in the sampling system keeps the flame cell temperature constant. The temperature setting is found in the BLOCK SETTING register. A type K thermocouple, called TC2, also embedded in the sampling system, produces the BLOCK TEMP reading. During warm-up the HEATER indicator is on continuously until the temperature is within a few degrees of the setting, and then it cycles on and off to control the temperature to the setting.

Another type K thermocouple called TC3 is also embedded on the outside of the flamecell. This thermocouple is known as the cutoff thermocouple (replacing the cutoff thermostat) and will shut off the heater circuit should TC2 fail to accurately read the block temperature and the temperature rises 15 degrees Celsius above the BLOCK SETTING register.

WARNING! Keep the analyzer heated to prevent condensation that could cause false low readings.

2.6.2. Monitoring Sample Flow

The ORIFICE PRESSURE across the sample flow orifice is measured and converted to SAMPLE FLOW in standard liters per minute. The orifice may require periodic cleaning to keep it clean and accurate. A reduced flow rate will slow down the response time of the sensor.

2.6.3. Making the Flammability Reading

A type K thermocouple, called TC1, passes through the wall of the flame cell and is bent in a 90° angle directly over the sensing flame, so that the tip of the thermocouple can measure the heat given off. This signal is the FLAME TEMP, and, when captured during the calibration process, it is also the ZERO FLAME TEMP and SPAN FLAME TEMP. When the FLAME TEMP equals the ZERO FLAME TEMP, the reading is 0%LFL. When FLAME TEMP equals ZERO FLAME TEMP plus SPAN FLAME TEMP, the reading is equal to the CAL READING. Other FLAME TEMP temperatures produce proportional flammability readings.

2.6.4. Calibration

Calibration uses two test gases: "zero" air with no flammables, and "span" gas containing a known concentration of flammables. Zero calibration sets the reading to 0 %LFL. Span calibration sets the reading to the CAL READING, for example 60% LFL. Zero and span solenoid valves automate calibration. When the solenoid valve energizes, pressurized test gas flows into the flame cell, completely filling it, with some excess gas flowing backwards out the sample inlet.

2.6.5. Calibration and Reading Calculations

First, Zero Gas, air completely free of flammable gases, is injected to the flame cell. After a minute or more, the FLAME TEMP temperature is captured as the ZERO FLAME TEMP, which makes the reading 0% LFL. Next, span gas is injected. It has a known flammability concentration, called the CAL READING. After a minute or more, the FLAME TEMP temperature minus the ZERO FLAME TEMP is captured as the SPAN FLAME TEMP, which makes the reading equal to CAL READING. New readings are made by comparing the FLAME TEMP to the ZERO FLAME TEMP and SPAN FLAME TEMP.

$$\text{ZERO FLAME TEMP} = \text{FLAME temperature measured while sensor is exposed to Zero Gas}$$

$$\text{SPAN FLAME TEMP} = (\text{FLAME} - \text{ZERO FLAME TEMP}), \text{measured while sensor is exposed to Span Gas}$$

$$\%\text{LFL Reading} \cong \frac{\text{FLAME} - \text{ZERO FLAME TEMP}}{\text{SPAN FLAME TEMP}} \times \text{RDNG LAST CAL}$$

RDNG LAST CAL is a copy of the CAL READING from the last successful calibration used to make the reading. A change to the CAL READING activates "Service Needed – Check Span," and has no immediate effect until a calibration is performed.

2.6.6. Warning and Danger Alarms

When the flammability exceeds the WARNING LEVEL or DANGER LEVEL, the corresponding alarm and relay is activated. Once activated, the alarms latch, and stay activated until they reset. If an external device provides the latching function, disabling the WARNING LATCH or DANGER LATCH

automatically resets the alarm a few seconds after the flammability returns to normal. If the danger alarm is not latching, an external device must be used to provide the latching function.

When both alarms are active, but their settings differ, the danger alarm controls the warning alarm. If danger is latched, warning also latches. If danger autoresets, warning also autoresets – if the flammability is below the WARNING LEVEL at the moment danger autoresets.

The danger alarm indicates a hazardous concentration of gas even if the fault alarm is active at the same time. Never ignore a danger. An explosive concentration of gas can cause a flameout fault.

WARNING! Simultaneous fault and danger alarms may indicate an explosive atmosphere.

WARNING LEVEL and DANGER LEVEL settings should be made as low as practical without giving false alarms. A lower alarm level is faster and more effective.

2.6.7. Rate-of-Rise Alarm

When a quickly rising flammability reading is expected to exceed the danger alarm, the danger alarm trips, even if the danger level has not yet been reached. This feature can be enabled or disabled using the RATE ALARM register.

2.6.8. 4 to 20 Milliamp Output Signal

The flammability reading is converted to a 4 to 20 milliamp output signal. 4 mA represents 0% LFL and 20 mA represents 100% LFL. All other readings are directly proportional.

Flammability Reading %LFL	Milliamp Output
-10	2.4
-5	3.2
0	4.0
50	12.0
100	20.0
>100	20.0
< MIN MILLIAMP setting	MIN MILLIAMP
Fault	MILLIAMP FAULT
Calibrating	MILLIAMP CAL SET

The lowest output is equal to the MIN MILLIAMP register. The highest output is 20 mA (100%LFL), except when the calibration or fault overrides are used for higher outputs, up to 24 mA.

Some monitoring devices do not accept readings below 4 milliamps. The MIN MILLIAMP register can be used to set the minimum output. It can also be used to increase the milliamp output for testing.

During calibration, or during faults, the signal can be forced to a special value. If enabled, the MILLIAMP CAL SET register sets the output during calibration, and the MILLIAMP FAULT register sets the output during faults. A value of 0mA (“OFF””) disables these overrides.

The signal does not go below 2.0 milliamps unless there is a mis-adjustment of the 4 MILLIAMP ADJUST register, or a loss of power, or a disconnected wire.

The 4MILLIAMP ADJ register should not be needed, but can be used to correct for small amounts of error in the output circuit.

The 700 Series Analyzers are fitted with dual 4-20mA chips for safety redundancy. Ensure that if only one 4-20mA chip is being used, a terminating resistor is placed across the terminal positions of the one not being used to avoid unnecessary check alarms.

2.6.9. Serial Communications

An RS 485 serial port with Modbus RTU protocol allows remote monitoring of display and control functions.

2.6.10. "Service Needed" and Faults

If operation is not within the optimal range, but the problem is not yet severe enough to be a fault, a "Service Needed" message, indicator, and relay activate.[3] The analyzer can still be operated, and can give warning and danger alarms, but maintenance is needed to help prevent a fault.

If operation is not within the acceptable range, a fault is given. During faults, warning and danger alarms are disabled. The analyzer must be serviced immediately to restore proper operation. If more than one fault exists at the same time, the most serious fault is displayed

2.6.11. Air/Fuel Cutoff Relay

The air/fuel cutoff relay can be used to control an external solenoid valve supplying compressed air and/or fuel to the analyzer. It does not have an indicator. The relay can be used:

- To stop the flow of fuel into the analyzer if the sensing flame is extinguished for too long.
- To stop the sample flow to the analyzer so that high-temperature process gases do not condense and clog the analyzer while the process is running and the analyzer is not up to temperature.
- As a general-purpose relay. By changing the setting to "de-energized" or "energized" the relay can be controlled

The RELAY SETTING register has three settings: 0 = de-energized / 1 = air/fuel cutoff / 2 energized.

This is different from the legacy method, which had a "HORN" setting for the relay. Now that the horn is not implemented, the "0 = de-energized (Off)" has been created, and "Override" is being renamed to the more general name "On – energized." The relay could become a direct remote control if the user wants to do so.

RLY_CONFIG = Off (de-energized)

The "Off" setting (RELAY SETTING = 0) de-energizes the cutoff relay regardless of any other conditions.

RLY_CONFIG = Air/Fuel cutoff

When set to "air/fuel" (RELAY SETTING =1), the relay is normally energized except when the sensing flame has been extinguished or the block temperature has been too low for more than fifteen minutes. The flameout or low temperature must persist for fifteen minutes before the cutoff occurs and the relay is de-energized. If the alarm reset command is received, then the relay is energized and the fifteen minute delay starts over again. If the block temperature and the sensing flame are momentarily correct, then the relay is energized and the fifteen minute delay begins again.

[3] "Service Needed" messages appear in the form of a "CHECK" on the display of the analyzer

RLY_CONFIG = On (energized)

The “On” (RELAY SETTING =2) setting energizes the cutoff relay regardless of any other conditions.

2.6.12. Flameout

WARNING! Flameout may indicate the presence of an explosive atmosphere.

Flame temperatures below 425°C indicate flameout. The flame may be out entirely, or too weak for valid measurements. As flameout is detected, a two-second ignition pulse causes a spark discharge to the burner tube to re-ignite the sensing flame. If successful, an ALARM RESET command clears the fault indicator and relay. If ignition fails, an ALARM RESET command will repeat ignition. If the AUTO IGNITION register is set to 1 or more, re-ignition attempts repeat at 15-second intervals. If re-ignition is successful and there are no other faults, the flameout fault will clear regardless of the FAULT LATCH setting.

2.6.13. Flameout from Flammable Mixtures - Flooding

If flameout follows a danger alarm, the flameout is assumed to be from flammable gas above 100% LFL. The special state of "flameout during a danger alarm" is called flooding.

WARNING! Flooding may indicate the presence of an explosive atmosphere.

While in this special state:
The WARNING and DANGER alarm indicators and relays are active.
The milliamp output is forced to 20 milliamps.
The FAULT alarm indicator and relay are inactive, unless a fault other than flameout occurs.
The display indicates "DANGER - FLOODED - 100% LFL"
The STATUS code in the status register is 2 "Danger-Flooded" (same as "Danger")
The FLAMMABILITY register may be invalid, for example a reading of –99% LFL.

While in this special state, if another type of fault (not flameout) occurs:
The WARNING, DANGER and FAULT alarm indicators and relays are active.
The milliamp output is either:

a) 20 milliamps, if the MA FAULT register is set to "OFF."
b) forced to the value programmed into the MA FAULT register.

The display indicates "DANGER - FLOODED - 100% LFL"
The STATUS code in the status register is 2 "Danger-Flooded" (same as "Danger")
The FLAMMABILITY register may be invalid, for example a reading of –99% LFL.

The analyzer will recover from this special state by ignition of the sensing flame.

Recovery from the special state following automatic re-ignition:
If the danger alarm is not latched, and the AUTO IGNITION setting is nonzero, the analyzer will attempt a series of re-ignition attempts. If ignition is successful, and there are no faults of any kind, and the reading is below the alarm settings, then the alarms will clear and the analyzer will return to normal status.

Recovery from the special state following a RESET command:
The fault alarm indicator and relay are active.
The warning and danger alarm indicators and relays are inactive.
The milliamp output is either:

a) 20 milliamps, if the MILLIAMP FAULT register is set to "OFF."
b) Forced to the value programmed into the MILLIAMP FAULT register.

The display is "FLAMEOUT" and the reading may indicate –99% LFL until the flame is lit.
The FLAMMABILITY register is invalid, for example a reading of –99% LFL, until the flame is lit.

WARNING! An upscale reading that resulted in a "DANGER - FLOODED - 100% LFL" message may have damaged the analyzer. A full calibration is to be completed if the analyzer reading does not return to 0%LFL after the flame has been restored and stabilized and there are no flammable gases remaining in the sample. A zero test command can also be initiated to verify the reading returns to 0%LFL.

2.6.14. Analyzer outputs and behavior for all states

The following tables show the behavior of the analyzer relays and milliamp outputs for all possible analyzer states.

Analyzer Status Table							
State of the Analyzer Message on LCD Display	Relays are normally energized, and de-energize to trip				Description	Milliamp Output	Status Code in Reg#2
	Warn	Danger	Fault	Cal			
WARMUP Power-on delay time.	W	D	F	C	Analyzer power is on. All safety relays are de-energized. All controls are inactive. After a minimum warmup delay, if no faults exist, the analyzer enters the normal state. After the 15-minute maximum warmup time, it enters the fault state.	0 mA	11
NORMAL OPERATION Operating properly, No faults, no alarms					Flammability readings are below alarm levels. No faults are present. All safety relays are energized. If a fault is detected the analyzer enters the fault state. If a calibration request occurs, calibration begins. If the reading exceeds the warning level or danger level, the analyzer enters the respective warning or danger state.	FR	0
WARNING ALARM Reading above warning level or warning alarm latched	W				Warning relay is de-energized, all others are energized. If the reading goes above the danger level, the analyzer enters the danger state. If a fault is detected, the analyzer enters the fault state. A calibration command turns zero or span gas on. If the reading is below the warning level, an alarm-reset command or auto-reset puts the analyzer into the normal state. Note: warning is always tripped whenever danger is.		1
DANGER ALARM Reading above danger level or danger alarm latched	W	D			Warning and Danger relays are both de-energized, all others are energized. If the reading goes below the danger level, and an alarm reset or an auto-reset occurs, an alarm-reset command or auto-reset puts the analyzer back to the normal or warning state. If a fault is detected, the analyzer enters the danger-fault state. A calibration command turns zero or span gas on to begin calibration or calibration-test.		2
DANGER ALARM **If fault occurs** when danger alarm is already tripped	W	D	F		If a fault is detected during a danger alarm, the Fault relay de-energizes. If the fault condition clears, the analyzer returns to the danger state. An alarm-reset command clears the Warning and Danger relays and puts the analyzer into the fault state.	FR / F	2
DANGER - FLOODED **Flameout** occurs while danger alarm is tripped	W	D	F		Warning, Danger and Fault relays are de-energized. The Cal relay is energized. Ignition attempts occur every fifteen seconds until the sensing flame ignites or the maximum number of re-ignition attempts is reached. If all attempts to ignite fail, the analyzer remains in the danger-flooded state until an alarm-reset occurs and it transitions to a fault state.	20 mA	2
ZERO GAS ON Zero calibration or zero test in progress	Warning and/or Danger Alarms present at the start of calibrations remain active until the calibration completes.			C	The zero valve injects air to the flame cell. After a delay the flame temperature is checked - if it is within limits the calibration data is updated, otherwise the analyzer enters the zero-fail state.	FR / C	4
SPAN GAS ON Span calibration or span test in progress				C	The span valve injects test gas to the flame cell. After a delay the flame temperature is checked- if it is within limits the calibration data updates, otherwise the analyzer enters appropriate span high or span low fault state.		5
CALIBRATION ENDS Calibration ending				C	The zero and span valves turn off, and after a short delay, the analyzer returns to the normal state. *Note: This replaced the "Purge" phase of calibration.*		7
ZERO FAIL Zero calibration or test failed			F		Flame temperature during zero phase of calibration or test is less than 460°C or above 525°C. The fault relay is de-energized. The calibration ends and analyzer remains in fault until an alarm reset occurs or a new calibration begins.	FR / F	8
SPAN LOW FAIL Span calibration or test failed - result too low			F		Flame temperature during span phase of calibration is at least 20°C below the span temperature setting. The calibration ends and the analyzer remains in fault until an alarm reset occurs or a new calibration begins.		9
SPAN HIGH FAIL Span calibration failed - result too high			F		Flame temperature during span phase of calibration is more than 20℃ above the span temperature setting. The calibration ends and the analyzer remains in fault until an alarm reset occurs or a new calibration begins.		10

Fault Table

Faults are detected during periodic self-test and during calibrations. Faults that can occur simultaneously are prioritized in order of importance. If the highest priority fault is "downscale" and a calibration request occurs, calibration begins. If no faults exist and an alarm reset or fault auto-reset occurs the analyzer enters the normal state.

State of the Analyzer Message on LCD Display	Relays are normally energized, and de-energize to trip				Description	Milliamp Output	Status Code in Reg#2
FAULT Analyzer fault detected			F		Fault relay de-energized, all others energized. Faults are detected by diagnostic self-tests. Faults are ranked in priority; the highest priority fault determines the status. See the "Fault Table" for a description of each type of fault.	FR / F	See below

Priority	Message on LCD Display of front panel	Description	Status Code in Reg#2
Faults related to hardware failure requiring repair or replacement			
1	ROM Fail		27
2	RAM Fail	Failed self diagnostic test	13
3	Data Lost	Nonvolatile settings have been lost (cold start)	14
4	12VDC Fail	Power supply is too high or too low	15
5	TC1 Fail	Thermocouple TC1 for sensing flame failed or open circuit / detached	24
6	TC2 Fail	Thermocouple for temperature control failed or open circuit / detached	25
Faults related to temperature that can be due to warmup, change to temperature setting, or heater failure			
7	Block Temp Low	Block temperature of the analyzer too low.	18
8	Block Temp High	Block temperature of the analyzer too high	19
Faults related to flow that can be corrected by maintenance			
9	Sample Flow Low	Sample flow rate too low	20
10	Sample Flow High	Sample flow rate too high	21
Faults related to optional monitoring features that are not present in all analyzers			
11	Dilution Low	Dilution pressure too low	16
12	Dilution High	Dilution pressure too high	17
13	Fuel Pressure Lo	Fuel pressure too low	22
14	Fuel Pressure Hi	Fuel pressure too high	23
Faults related to the sensing flame while all other faults are clear.			
15	Flameout	Sensing flame is extinguished (Below 425°C). Ignition attempts occur every fifteen seconds until the sensing flame ignites or the maximum number of attempts is reached. If all attempts fail, the analyzer remains in the fault state until an "alarm reset" occurs, which begins new ignition attempts. Once ignited, an additional alarm-reset clears the fault.	26
16	DownscaleReading	Flammability reading below the RDNG FAULT LEVEL setting, typically -10%LFL.	3
Faults that are detected only during calibrations (Zero Cal, Span Cal, Full Cal, Zero Test, Span Test, Full Test)			
N/A	Zero Fail	Zero calibration failed, result unacceptable	8
N/A	Span Fail Low	Span calibration failed, result too low	9
N/A	Span Fail High	Span calibration failed, result too high	10

Milliamp Output Table			
Flammability Reading %LFL	**Milliamp Output**	**Legend**	**Description**
< MIN MILLIAMP setting	MIN MILLIAMP	FR	The MIN MILLIAMP register sets the lowest milliamp output. It is used to keep the milliamp output above a minimum value – for example, not below 4 milliamps. It is also used for testing the milliamp output.
Fault	MILLIAMP FAULT	FR/F	If enabled, the MILLIAMP FAULT register forces the milliamp output to a special value during faults. A setting of 0mA disables the MILLIAMP FAULT override.
Calibrating	MILLIAMP CAL SET	FR/C	If enabled, the MILLIAMP CAL SET register forces the milliamp output to a special value during faults. A setting of 0mA disables the MILLIAMP CAL SET override.
-10	2.4	FR	The flammability reading is converted to a 4 to 20 milliamp output signal. 4 mA represents 0% LFL and 20 mA represents 100% LFL. All other readings are directly proportional. The milliamp output does not go below 2.0 milliamps unless there is a mis-adjustment of the 4MILLIAMP ADJUST register, or a loss of power, or a circuit failure such as a disconnected wire. The 4MILLIAMP ADJ register should not be needed but can be used to correct for small amounts of error in the output circuit.
-5	3.2		
0	4.0		
50	12.0		
100	20.0		
>100	20.0	20 mA	The highest output is 20 mA (100%LFL), except when the calibration or fault overrides (MILLIAMP CAL SET or MILLIAMP FAULT) are used for higher outputs, up to 24 mA.

2.7. Air Dilution Option

Air dilution is a factory-installed option that injects air in the sample flow to improve accuracy and prevent flameout over a wide range of sample oxygen concentrations. Half the flow into the flame cell comes from the sample, the rest is dilution air from a pressure regulator. Added air keeps the oxygen concentration in the flame cell high enough for the flame.

Identification of air dilution models

Their DEVICE TYPE contains the letter D[4]. They have a dilution air inlet (B) and dilution air regulator PCV2. The burner is part number BRN050, and the gap between the burner and thermocouple is 0.5 inches (1.3 cm).

Installation of air dilution models

Connect 20 PSIG (140 kPa) clean, dry "instrument grade" compressed air to the air dilution inlet "B." Compressed air quality will affect accuracy. It must be free of oil and condensed water to prevent flow problems and false readings. A shut-off valve at the Dilution Air Inlet is required.

Protect inert processes from dilution air in the analyzer's exhaust. Make certain that exhaust air cannot adversely affect an inert process, or take special precautions for the exhaust method.

Oxygen Effects5 **- Nitrogen-rich atmospheres**

Oxygen displacement by Nitrogen causes readings to increase approximately +0.7%LFL for each -1% by volume reduction in oxygen concentration in the process being monitored.

Oxygen Effects – Carbon-dioxide (CO2) and Steam (H2O)-rich atmospheres

Oxygen displacement by CO2 and H2O causes readings to decrease approximately -0.7%LFL for each -1% by volume reduction in oxygen concentration in process being monitored

[4] The K device type for air dilution option has been removed for the 700 series of analyzers. This is due to the inclusion of the MINIMUM SPEED option in the SETTINGS. When enabled this allows the user to set the SPEED SETTING to off. This along with the ability to set the unit for "double downscale" made the need for a separate K-type unit void

[5] See section 3 for important specifications: response time, pressure, oxygen, and flow effects.

Calibration of air dilution type analyzers

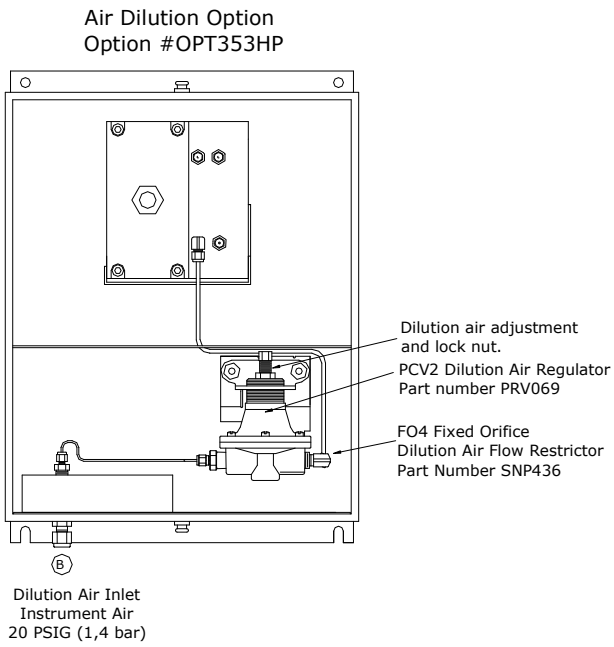
Calibration gases are also diluted. To maintain accuracy, the sample flow during the span calibration is monitored to ensure that it matches the flow during sampling.

Checking for leaks

To perform leak checks as indicated in the maintenance procedures, a valve on the Dilution Air Inlet (B) must be installed, so it can be shut for the duration of the test.

Alignment

The burner is aligned 0.5 inch (1.3cm) directly beneath the thermocouple, a gap twice as large as the standard analyzer. The spark electrode is aligned normally, directly behind and slightly above the burner. Typically the alignment does not require adjustment unless the flame cell has been opened for maintenance.



Tuning the dilution air regulator – option #OPT353HP

The dilution air regulator is factory-tuned so that the ratio of sample to dilution air is one-to-one. It does not normally require adjustment.

This procedure tunes the dilution air regulator to get 50% sample and 50% air. Use with burner BRN050, having a 0.5 inch (1.3cm) gap between the burner and thermocouple.

1. Start with cover on and analyzer stable.[6]
2. Shut off pressure to the air dilution inlet only. This requires a shutoff valve at the air dilution inlet.
3. Open the cover and perform the following as quickly as possible.
4. Set CALIBRATION TIME in the CALIBRATION MENU to 4 minutes.
5. Record the FLAME TEMP reading in air from the GAUGES menu.
6. Initiate a SPAN TEST with 1.15% Ethylene[7] in Air.[8,9]
7. After one minute, record the FLAME TEMP reading for 1.15% Ethylene without dilution.
8. With SPAN TEST still active, turn on pressure to the air dilution inlet.
9. With SPAN TEST still active, adjust the dilution air regulator[10] to get a FLAME reading of:
   For 0.50 inch gaps[11], FLAME TEMP = 62% of FLAME TEMP reading in air (step 5) plus 38% of FLAME TEMP reading for 1.15% Ethylene without dilution (step 7).
10. At the end of span test, return CALIBRATION TIME to 1 minute.
11. Replace the outer cover.
12. Wait for analyzer to stabilize.
13. Perform a FULL CALIBRATION from the COMMANDS menu using a flashlight (keep cover on).
14. For SPAN FAIL faults, SPAN TEMP SET may require adjustment.

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[6] The analyzer should be checked regularly for leaks. Analyzer leaks cause reading errors. Only leak-free analyzers should be operated or adjusted.

[7] Alternate calibration gas, 0.85% Propane, may also be used.

[8] Misalignment can cause errors similar to incorrect span calibration gas.

[9] Accuracy will depend completely on the accuracy of the span calibration gas.

[10] Following this calculation results in a SPAN FLAME TEMP that is 45% of the SPAN FLAME TEMP without air dilution for 0.25"gaps, and 38% for 0.5" gaps.

[11] For 0.25 inch gaps, FLAME TEMP = 55% of FLAME TEMP reading in air (step 5) plus **45%** of FLAME TEMP reading for 1.15% Ethylene without dilution (step 7), if for some reason a 0.25 inch (6.4mm) burner-thermocouple gap is in use.

2.8. Fail-safe Considerations

Safety equipment must give continuous protection. It should notify the operator of failures that might disable protection. The analyzer continually checks itself, and can detect many types of faults. During faults the reading is not reliable, and the warning and danger alarms are disabled. Processes that depend on the analyzer for safety must have the fault relay hard-wired to the safety shut down.

WARNING! During faults, the warning and danger alarms are disabled. Because the proper function of the analyzer is required for safety, the fault relay must be used to initiate a safety shutdown or equivalent corrective actions suitable to the process or area being monitored.

WARNING! Not all faults are detectable. Investigate unusually low readings. Perform regular leak checks. Abnormal behavior must be recorded, investigated, and resolved.

Incorrect calibration gas type or concentration causes significant errors. The calibration gases should be tested and certified within a tolerance of +/-2% or less prior to use. The ZERO FLAME TEMP and SPAN FLAME TEMP should be monitored for significant changes, especially for a new calibration gas cylinder. Keep the old cylinder as a reference until the new cylinder is verified. The span for both should be within the cylinder's stated tolerance, not more than 4% cumulative error.

WARNING! Calibration gas must be the correct type and concentration for valid readings and alarms.

Certain types of leaks in the sampling system might not be detected unless a thorough leak check is performed. Unusually low readings could be caused by a leaking sampling system.

2.8.1. Oxygen Concentration

The sensing flame requires oxygen from the sample. Oxygen deficient atmospheres shift the "zero" and change the reading. When the oxygen in the sample is displaced by nitrogen, the reading increases. Displacement by carbon dioxide or water vapor causes the reading to decrease..

The minimum recommended oxygen concentration depends on the fuel type and air dilution option:

Propane fuel	18% oxygen by volume
Hydrogen fuel	12% oxygen by volume
Air dilution	0% oxygen by volume

Many factors can affect the point at which the sensing flame is extinguished by lack of oxygen. The typical point of "flameout" is:

Propane fuel	16% oxygen by volume
Hydrogen fuel	7% oxygen by volume
Air dilution	No flameout

Oxygen concentrations above 25% by volume require special precautions beyond the scope of this device and manual. In addition, special precautions for electrical safety must be observed when the possibility exists for oxygen-enriched atmospheres.

WARNING! Low-oxygen atmospheres can prevent proper operation of the analyzer.

2.8.2. Flammable Concentration Limits

Very high concentrations, at or above the LFL, can extinguish the sensing flame. Up to the point of flame out, the readings increase, and then suddenly drop downscale.

2.8.3. Speed of Response

Effective alarms require a fast analyzer and immediate corrective action. Keep sample lines short. Use low-diameter tubing. Do not add extra sample filters. Corrective action must quickly reduce the flammable gas concentration to a minimum. Every second counts. The person responsible for overall safety and design must verify that the total time for the analyzer to respond plus the time needed for effective corrective action is less than the time it takes for any foreseeable upset condition to produce

an explosive concentration. The time of response of the entire system is determined by the time of response of all parts of equipment within the gas detection system. This includes all delays due to added equipment such as sample filters, and also delays by PLC or control system logic such as signal filtering or delays in alarm activation

WARNING! An extremely rapid rise may reach 100% LFL before the analyzer can make an alarm.

2.8.4. Fuel Pressure Limits

The sensing flame needs constant fuel pressure to operate. Low pressure causes downscale reading drift. Loss of pressure extinguishes the sensing flame. Keep the entire fuel line clean and free of leaks. Maintain fuel pressure above the minimum. In cases where the analyzer is used to control processes, a pressure switch that monitors the fuel line for a loss of pressure is required.

WARNING! Low fuel pressure can cause false low readings. Always maintain proper fuel pressure.

2.8.5. Flash Point Limitations

Flammable gases and vapors with Flash Points above the operating temperature of the sensor and sampling system will condense before they can be measured. This can result in false low readings, failure to detect a flammable or explosive concentration, or clogging in the sampling system.

WARNING! The analyzer cannot detect flammable vapors with Flash Points above the operating temperature of the analyzer or any portion of the sampling system. DO NOT ATTEMPT TO MONITOR A FLAMMABLE VAPOR WITH A FLASH POINT ABOVE THE ANALYZER'S OPERATING TEMPERATURE.

Some industrial processes can use a variety of solvents having different flash points. Each solvent to be monitored must have a flash point below the analyzer and sampling system operating temperatures. Before a new solvent is used, the flash point must be checked.

2.8.6. Condensation Limits

Hot vapors condense when cooled below the dew point. The dew point of water and other non-flammable vapors in the sample should be lower than the entire sampling system. Condensation can result in flow system fouling, erratic flow, and other sampling problems. The combination of condensing water and water-soluble solvents might cause false low readings.

WARNING! Condensation of flammable vapors in the sampling system may delay or disable alarms.

2.8.7. Ambient Temperature Effects

The analyzer reading drifts a small amount from ambient temperature changes. This error can be removed by recalibration. Recalibrate if the ambient temperature changes more than 50°C. Removal of the analyzer cover can also cause small errors. Calibrate by remote controls to prevent this error.

2.8.8. Flow Effects

Changes to the sample flow rate can affect accuracy, especially for air dilution types. During calibration, the flow rate must be observed to verify that it matches the flow rate during sampling, and the CHECK FLOW setting should be set within 0.4 LPM of the flow rate during calibration.

2.8.9. Sample Pressure Effects

When sampling from a process at increased or decreased pressures, a small amount of error is created. This error can be removed by recalibrating. The use of a pressure regulator in the sample line is not recommended. During calibration, excess calibration gases must flow freely out the

sample inlet. A pressure regulator in the sample line can interfere with the flow of calibration gases and reduce accuracy or cause faults.

2.8.10. Halogens

Halogens and halogen-containing substances such as chlorine, fluorine, bromine, hydrogen chloride, trichloroethylene, dichlororethylene, vinyl chloride, and Freon will produce halogen acids when oxidized in the sensing flame. The analyzer is constructed of stainless steel and hard coated aluminum components with corrosion resistance. Under normal conditions, these substances do not affect performance. Some halogenated hydrocarbons that are relatively non-flammable in air produce a flammability reading at very high concentrations. Some mixtures of halogens and amines may form ammonium chloride upon contact with the heated analyzer. The process chemistry must be taken into account to ensure that a representative sample reaches the analyzer's flame cell. For example, halogens in the presence of amines can form solids upon contact with a hot analyzer.

2.8.11. Combustible Dusts, Aerosols and Mists

This analyzer cannot measure combustible dusts, aerosols and mists, which can form explosive mixtures in air, but are not in the vapor state, and may be trapped in the sample filter.

WARNING! The analyzer does not provide protection from the fire and explosion hazards of combustible dusts, aerosols and mists.

2.8.12. Silicones

Performance is generally unaffected by low concentrations of silicones. Extremely high silicone concentrations require extra maintenance to remove silicone dust on the burner and exhaust filter. An optional flow system that reduces silicone effects is available. Contact Control Instruments for recommendations in handling samples with extremely high silicone content.

2.8.13. Auto ignition / Decomposition Temperature

Substances which auto ignite or decompose at or below the analyzer operating temperature will not be detected. For example, Carbon Disulfide (CS_2) has an approximate auto ignition temperature of 90°C, and so will not be detected in an analyzer operated above that temperature.

WARNING! The analyzer cannot measure substances that thermally or chemically decompose from contact with the heated sampling system.

2.8.14. Analyzers used for control

If the analyzer signal controls a process, for example to vary process speed or ventilation rate, or to modulate some process control device, a "secondary safety" must ensure that analyzer faults cannot cause an unsafe condition. Industry standards, for example "Safety Instrumented Systems" require safety devices to be independent of process control. Otherwise, a fault in the analyzer that produces false low readings could cause an unsafe condition in the process (reduced ventilation, increased speed, etc.) without warning or alarm.

WARNING! Analyzers used for process control purposes require a secondary means of safety. A redundant analyzer, or an independent physical limit on the flammable gas concentration, must be used.

Low fuel pressure can cause false low readings. Analyzers that control a process require the correct fuel pressure at all times. If the fuel pressure is below the minimum the controls must be forced to a safe state.

WARNING! Analyzers used for process control require continuous monitoring of fuel pressure, and an automatic alarm that performs a safety shutdown when the fuel is below the minimum

2.8.15. Humidity Limits

The concentration of water vapor in the sample effects the oxygen concentration. For relative humidity (%RH) in the sample at a temperature of 40ºC, humidity of 20%RH, 50%RH and 90%RH produce less than 10%LFL deviation in readings. The following additional humidity limits shall be observed:

The analyzer must be supplied with clean, dry, compressed air with a dew point below the lowest ambient temperature to which the air supply tubing is exposed.

The sample must be non-condensing. All parts of the sampling system shall be heated above the dew point of water vapor in the sample.

The oxygen concentration that results from displacement of oxygen by water vapor and steam shall be taken into consideration – see section 2.8.1.

Special considerations for sample gases shall be made, to understand the potential for condensation to effect sample handling and the composition of the sample gas itself

3. Specifications

Measurement range	0 to 100% LFL display range. 0 to 80% LFL range for stated accuracy.
Flame Cell Temperature	SNR701: 60°C SNR702: 120°C SNR704: 200°C SNR705: 250°C
Accuracy	± 3% of full scale, or 10% of applied gas concentration, whichever is greater[12]
Repeatability	± 1% of measurement range
Zero Stability	± 1% in 30 days[13]
Span Stability	± 5% per year
Response Time	Time to 50% (T50), 60% (T60) and 90% (T90) for a sudden increase in flammability

Device Type	% of step	Response time in seconds Speed setting Low	Med	High
SNR701	T50	2.6	2.4	1.9
	T60	2.8	2.5	2.1
	T90	3.6	3.3	2.9
SNR702	T50	2.3	2.1	1.9
	T60	2.4	2.3	2.0
	T90	3.3	3.0	2.8
SNR704 and SNR705	T50	2.4	2.0	1.8
	T60	2.5	2.3	1.9
	T90	3.1	2.9	2.6

Note: Air dilution adds 0.50 to 0.75 seconds to the response

Note: When speed setting is off: T50 > 10, T90 > 30 seconds

Note: The response times are the same for all gases. The type of gas does not influence response time.

Note: The K-type was removed due to the inclusion of a MINIMUM SPEED setting (K-type had the speed setting off)

Refresh Rate	All inputs, logic, displays and outputs refresh 12 times per second (Refresh period 82 milliseconds)

[12] Air dilution types (D) are accurate in the range from 0 to 60% LFL.

[13] Thermocouple ageing during the first months of operation could cause greater drift. To alleviate effects, monthly or more frequent calibration schedule should be followed.

Sample Transport Time	0.5 seconds per meter of ¼ inch OD tubing. 1.0 second per meter for air dilution option.
Stabilization Time	Calibration gas should be applied for at least 60 seconds
Sample Pressure Range	800 to 1200 millibar absolute pressure at sampling point, subject to accuracy limits when pressure changes due to change in pressure from the point of calibration. See “Sample Pressure Effect
Sample Pressure Effect	Increase in pressure decreases the reading slightly. New BRN071 significantly improves on the performance of the analyzer when subject to changes in sample pressure. Air dilution types experience a greater drift when subject to sample pressure changes compared to non-dilution analyzers.

Sample Pressure Effect

Type	Fuel	Burner	%LFL per Millibar
701	Propane	RSP018	-0.08
701	Hydrogen	RSH001	-0.08
701	Hydrogen	BRN071	-0.01
701D	Hydrogen	RSH002	-0.16
702	Hydrogen	RSH001	-0.06
702	Hydrogen	BRN071	-0.01
702D	Hydrogen	RSH002	-0.12
704	Hydrogen	RSH001	-0.04
704	Hydrogen	BRN071	-0.01
704D	Hydrogen	RSH002	-0.08
705	Hydrogen	RSH001	-0.04
705	Hydrogen	BRN071	-0.01
705D	Hydrogen	RSH002	-0.08

Sample Humidity Effect	The relative humidity (%RH) in the sample at a temperature of 40ºC produce less than 10%LFL deviation in reading at 0%RH, 20%RH, 50%RH and 90%RH. See section 2.8.15.

Ambient Temperature Range:	–25°C to 55°C
Sample Flow Rate	Typical range: 2.5 ±0.5 liters per minute[14] Low Flow: Less than 1.4 liters per minute
Sample Flow Rate Effect	The reading increases approximately 1% LFL for a 0.1 liter per minute decrease in the sample flow rate.
Sample Flow Rate Limit	Accuracy limit - maximum deviation relative to nominal (point of calibration) 0.4 liters per minute change in flow rate. The CHECK FLOW setting should therefore be set within 0.4 LPM of the flow rate during calibration. For air dilution types (D) this limit is 0.2 LPM.
Calibration Test Gas	1.15% by volume Ethylene in air or 0.85% Propane in air. Tolerance should be within +/-2% of specified concentration.
Calibration Gas Pressure	20 PSIG (1.4 bar) inlet pressure
Calibration Gas Consumption	Approximately 7 SCFH (3.5 liters per minute) during calibration
Calibration Hardware Kit	Part# PRV022 set to 20 PSIG delivery pressure at span inlet
Oxygen Limitation	Oxygen concentration within the sample should fall within limits below: 0 to 21% oxygen with air dilution option and hydrogen fuel 12 to 21% oxygen with hydrogen fuel 18 to 21% oxygen with propane fuel
Oxygen Effect	Using hydrogen fuel, the reading increases 1% LFL for each one percent (1%) decrease in oxygen concentration. The effect for propane fuel is approximately three times higher. For air dilution types the reading increases 0.7% LFL for each 1% decrease in oxygen concentration.
Dimensions	16"H x 12.1"W x 8.5"D (41cm H x 31cm W x 22cm D)
Storage Temperature Range	–40°C to 65°C
Storage Pressure Range	500 to 1500 millibar
Storage Humidity Range	0 to 90% Relative Humidity (non-condensing)
Storage Time	18 months typical
Ambient Humidity	0 to 100% Relative Humidity (non-condensing)
Ambient Pressure Range	800 to 1100 hPa (millibar). Air dilution types are within stated accuracy when pressure range is within +/- 50 hPa of the ambient pressure at the time of calibration; All other types are within stated accuracy when pressure range is within +/- 100 hPa of the ambient pressure at the time of calibration. Range for explosion proof rating: 800 to 1100 hPa (millibar).

[14] Air dilution units draw less than half their total flow rate from the sample - the "sample" flow is half that indicated by the "SAMPLE FLOW" register. The balance of flow into the flame cell is dilution air.

Dew Point / Flash Point	Must not exceed lowest temperature found in any part of sampling system and flame cell. To reach maximum Flash Point requires increase of HSG TEMP setting and recalibration.	
Fuel Required	Fuel Type:	Hydrogen, pre-purified 99.99% minimum
	Inlet pressure:	40 to 45 PSIG (2.7 to 3.1 bar)
	Consumption:	60 Liters per day typical
	or	
	Fuel Type:	Propane, 99% minimum purity
	Inlet pressure:	30 to 35 PSIG (2.0 to 2.4 bar)
	Consumption:	1 Pound per month typical
Fuel Pressure Limit	Hydrogen fuel pressure operational limit 3.1 bar.[15]	
Sample Inlet	1/4" outside diameter (OD) tubing standard, except 3/8" OD for HTR075 concentric probe and MTG058 dual tube mounting.	
Exhaust Outlet	3/8" OD tubing for wall mount (MTG053) and dual tube (MTG054, MTG058) types, 3/4” OD for concentric sample probes (HTR063, HTR075)	
Compressed Air	3/8" OD tubing or larger for main lines. Reduce to 1/4" OD tube connection at analyzer inlet.	
Fuel Inlet Connection	Requires 1/4" OD tubing	
Span Gas Inlet	Requires 1/4" OD tubing	
Electrical Connections	Two 3/4 " NPT (pipe), for independent AC and DC connections	
Compressed Air Required	Inlet pressure:	20 ±1 PSIG (140 kPa), regulated
	Consumption:	64 SCFH, 32 Liters per minute including purge
	Quality:	Per ANSI/ISA-7.0.01-1996 or equal Must be clean, dry, and oil-free.
	Dew Point:	At least 10°C below ambient
	Particulate:	Less than 40 micrometer
	Lubricant:	Less than 1 part per million
Electrical Power Required	Voltage:	Either 120 VAC or 230 VAC Factory built to accept only one range. Must use rated sensor voltage
	Tolerance:	+10%, -15% of rated voltage
	Frequency:	50 to 60 Hertz
	Power:	400 Watts maximum
	Maximum Allowable Voltage ACrms:	300V
	Maximum Allowable Voltage DC:	385V
	Clamping Voltage Max:	775V
Warm Up Time	30 to 90 minutes varying with sensor type and ambient conditions.	
Materials	Stainless Steel (300 series) and corrosion-resistant, hard-coat aluminum. Brass fittings on SNR701.	
Hazardous location Rating	See section 1.4. Refer to serial plate for rating and approval marks.	

[15] The rated burst pressure of the fuel system is 13.8 barg (200 psig)

NEMA Rating	NEMA 12 (HSG228) standard, NEMA 4X (HSG229) optional
IP Rating	65
4-20mA Output	Self-powered, non-isolated output signal. Delivers 4mA at 0%LFL, 20ma full-scale to a passive load. Maximum load resistance is 275 Ohms total, which allows 250 Ohms non-inductive load plus 25 Ohms line resistance.
Milliamp Output Range	4 to 20 milliamps for 0 to 100% LFL, full range 2.5 to 22.7 milliamps.
Milliamp Output Notice	In order to signal a special state such as faults or the flooded condition, the MA FAULT and MA CAL settings should be configured outside the normal range of the milliamp output.
Rate of rise notice	This function is outside the scope of CENELEC norms and was not examined during ATEX approval testing.
Special state notices	For special state "Flooded," refer to "Flameout from Flammable Mixtures - Flooded" under "Theory of Operation." Special states listed below in which the %LFL reading may not represent in %LFL concentration of the process being monitored.

State	Status	4-20mA output	Relays activated
Warming up	Warm up	'MA FAULT' or %LFL display	Fault
Calibrations	Zero, span or recovery	'MA CAL' or %LFL display	Calibrating
Tests	Zero, span or recovery	'MA CAL' or %LFL display	Calibrating
Fault	Fault	'MA FAULT' or %LFL display	Fault

Relay rating	60 Watts non-inductive 0.5 Amp at 125 VAC 0.2 Amp at 230 VAC 2.0 Amp at 30 VDC Refer to Safety manual (H7FTA712) for additional ratings.
Weight	35 to 40 lbs. (16 to 18 kg.) - Depending on options
Vibration	Frequency less than 25 Hz with amplitude less than 0.15mm
Orientation	Install upright only: less than 15º deviation from vertical. 0º degrees is preferred. CENELEC accuracy limit allows maximum deviation 2.5° relative to nominal (point of calibration).
Spare part storage	O-ring storage is five to ten years under clean, dry conditions. For best results: Temperature 0°C to 50°C, and Humidity 10 to 90 %RH. Atmospheric pressure. Avoid strong light, direct sunlight,

ozone and oxidizers. Use sealed polyethylene bags. All other spare parts that do not contain elastomers, no more than 10 years storage in original packaging in atmospheric indoor conditions.

4. Installation

WARNING! The accuracy and reliability of the analyzer depends completely on proper installation.

4.1. Unpacking

Check the packing list and contents of the shipment. Look for additional components: mounting gaskets, regulators, and sample probes. Retain shipping materials through installation and startup.

4.2. Equipment, materials, and utilities

AC electrical power, at 400 Watts (VA) maximum per analyzer, at a voltage matching the serial plate. The analyzer is factory-set for either 120 VAC or 230 VAC, but not both. Field wiring to be rated for the at least 65°C.

Fuel gas cylinder or generator, and pressure regulator. The analyzer is factory-set for only one fuel type, either hydrogen or propane, but not both. Pressure at the fuel inlet must match the pressure rating on the serial plate. Hydrogen is typically 45 PSIG (310 kPa), propane 35 PSIG (240 kPa).[16]

Calibration span gas cylinder and pressure regulator, typically 1.15% Ethylene in Air with 20 ±1 PSIG (140 kPa) delivery pressure at the span inlet.

Calibration zero gas cylinder or compressor, and pressure regulator for clean, dry compressed air with 20 ±1 PSIG (140 kPa) delivery pressure at the zero inlet. If a compressor is used in place of the zero gas cylinder, include a shutoff valve.

Clean, dry, regulated compressed air. Delivery pressure to the air inlet is 20 ± 1 PSIG (140 kPa). A properly sized air filter and regulator must be used.

Either **Stainless steel sample and exhaust tubing** or **Sample probe**. Optional sample probes or sample tubes may be supplied with the analyzer. Otherwise, use ¼ inch OD stainless steel tubing for the sample and 3/8 inch for the exhaust.

Copper or stainless steel tubing for utilities, for connecting the fuel, air and calibration gases.

Shutoff valves for fuel, air and calibration gases, so that the sensor can be disconnected for service or maintenance procedures without affecting other equipment using the same utilities.

Switch for AC power, so that analyzer can be serviced safely.

Mounting bolts, washers and nuts. Size 3/8 inch mounting bolts or their metric equivalent.

Either **Conduit seals** or **cable glands**, to seal electrical connections at the enclosure entrance.

Mounting supports, saddles and/or flanges as needed, depending on the particular application. Mount analyzer upright, at zero degrees around X and Z-axis.

Size AWG 14 wire (1,3mm to 1,6mm diameter) for AC power, or the metric equivalent.

Size AWG 14 to AWG 24 wire (0,5mm to 1,6mm diameters) for relays and outputs. The terminals accept these wire gauges. The wire size depends on the relay or output function and current-carrying capacities of the circuit. Stranded wire is preferred.

Size AWG 22 (0,6mm dia.) **shielded three-conductor wire for RS 485 serial communications**.

Tools required: Tube fittings are fractional inches: 9/16 inch, etc. Flame cell bolts are metric.

[16] NOTE: Should fuel be supplied through the use of a generator, that device would need to be included in SIL/PL evaluation

Purge and pressurization system. The analyzer is rated by Factory Mutual Research for Class I, Division 2, Groups A, B, C & D. Use of an optional TYPE Y AIR PURGE improves the rating to Class I, Division 1, Groups A, B, C & D hazardous areas and is not FM approved as such. The flame cell is always rated for sampling from Division 1 or Zone 1 hazardous areas. The rating must meet or exceed the hazardous area rating requirements for its location.

4.3. Location and Mounting

Obtain a representative sample. Position the probe for a reliable and accurate sample and ensure that the analyzer exhaust does not dilute the sample. This requires an understanding of the behavior of the gases and vapors being monitored, and the process itself. Consider the effects of fans and dampers in normal and upset conditions.

Install as close as possible. For good sampling and a fast response, the sample and exhaust tubes must be short.

Keep sensor upright. Do not tilt more than 15° from the vertical axis (at 0° about the X and Z axes).

Maintain the hazardous location rating. Install in an area suitable for the analyzer's rating.

Avoid Pressure Spikes. Keep away from fans, dampers, tees and elbows. For best performance, position the probe where the duct pressure remains near atmospheric pressure.

Provide access for service. Allow for safe and effective routine maintenance. Keep clearance space around the analyzer. If necessary, catwalks, railings or ladders may be required.

Ensure easy probe removal for cleaning. The sample and exhaust lines must be removable for periodic cleaning, especially in processes having high levels of particulate. If a pipe is installed between the oven's or duct's inner and outer walls, this pipe should be large enough to allow installation of the sample and exhaust lines, including all bends in those lines.

Insulate duct-mounted sensors using a gasket. The gasket allows mounting on surfaces up to 75°C. For hotter surface temperatures mount the sensor with a 1 to 2 inch offset. Insulate exposed sample/exhaust tubing to prevent condensation. The sensor can sample from process ducts having much higher interior temperatures, limited mainly by the materials of the sample probe.

Seal duct entry holes. Prevent leaks that cause cooling or dilution of the sample, or allow process gases to leak into the analyzer housing. If used, fully weld flanges or mounting saddles.

Use inlet restrictor fittings. The fuel inlet fitting, and zero and span inlet fittings, contain sintered metal restrictors to control flow rates. These must be used to make the tubing connections. They can only be replaced with identical restrictor fittings.

Remove the shipping bracket. Some shipment methods use a steel bracket to support the flame cell in transit. The bracket is tagged, indicating that it must be removed before the analyzer is operated.

4.4. Sample and Exhaust Tubing

Use suitable materials. Stainless steel is the minimum material for the sample and exhaust tubes. High temperature processes may require other materials. Copper is generally unacceptable.

Keep tubes short. Short tubes give the fastest response and require the least maintenance. Long tubes can greatly delay the sensor response and affect the safety of the gas alarm system. Sample transport time is approximately 6 feet (2 meters) per seconds through standard ¼ inch tubing.

Keep sample-tubing volumes low. Filters and large diameter tubes cause sampling delays. Use ¼ inch 0.035 inch wall thickness tubing. Do not add bowl filters to the sample line.

Keep tubes hot. Prevent condensation and clogging. Never expose sample or exhaust tubing to room temperature, even for a very short distance. All tubes must be kept above the dew point(s) and flash point(s). If the sensor is not mounted directly on the process duct, the sample and exhaust tubes must be heat-traced to a temperature above the flash point(s) and dew point(s) of all substances in the sample stream. In cases where duct mounting is not possible, please contact Control Instruments Corporation for guidance in heat-tracing sample and exhaust lines.

Avoid short radius bends. This can cause clogging and make service difficult.

Keep tube openings perpendicular to the duct flow. Prevent sampling problems from a pitot effect. Do not aim the sample or exhaust tube openings directly into the process flow.

Position the sample probe properly, near the center of the duct where it can obtain a representative sample. Profiling of the duct might be necessary.

Position exhaust tube correctly. The exhaust should be at the same duct pressure as the sample, positioned where it cannot dilute the sample.

Install tube at or below the horizontal. If some condensation of water vapors is unavoidable, the sample and exhaust tubes must slope downward to carry condensate away from the analyzer. Sample tube diameter of 3/8 inch may be needed to avoid plugging by water droplets.

Keep lines leak-tight. Avoid unnecessary joints or couplings. Leaks cause false low readings.

Provide access for service. Cleaning or replacement may be required. If necessary, install a tube union to allow removal of the tubes.

Seal duct openings. Access holes in the duct wall must be sealed to prevent leaks that might dilute the sample, or cool the sample tubing and cause condensation.

Tube installation/removal. There are several mounting options:

4.4.1. Dual Tube Duct Mount

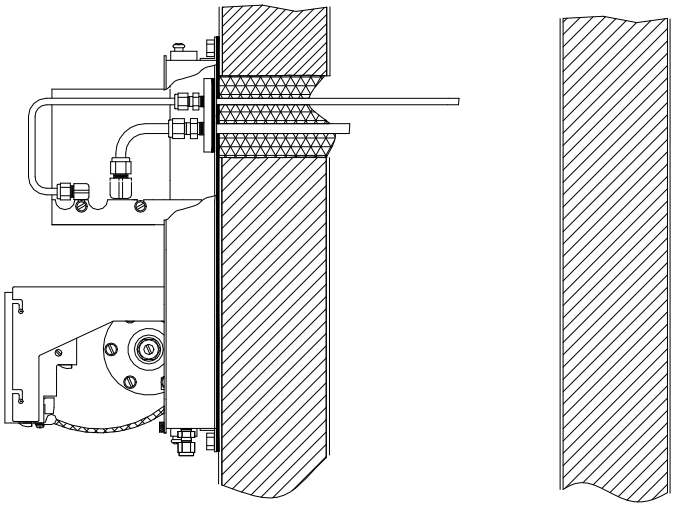
Reference part numbers MTG054 and MTG058

WARNING! Sample should be 6 inches [15.24cm] longer than exhaust to prevent the exhaust from influencing the sample reading.

The sample tube and exhaust tube each pass through their respective bulkhead fittings. The exhaust tube is cut to a length that just penetrates the duct. The sample tube is cut to reach 50-75% into the duct (1/2-3/4ths duct's width).

For MTG054, the sample tubing is ¼ inch OD, for MTG058 it is 3/8 inch OD. All parts of the sample tube must be protected from exposure to cold surfaces or cold air. The analyzer is sealed and insulated with a gasket. The entry into the duct must also be sealed so that a cold spot does not form when room air enters the duct opening adjacent to the probe.

If the maximum temperature inside the duct exceeds 250°C, insulation must be packed into the space between the rear of the analyzer and the inside wall of the duct, as shown.



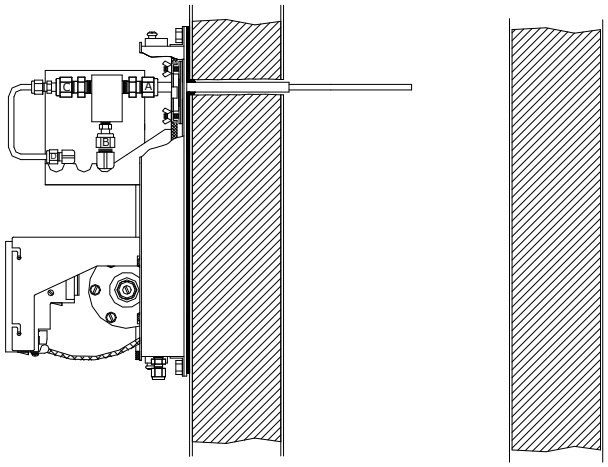
4.4.2. Concentric Sample Probe

Reference part number HTR063 or HTR075

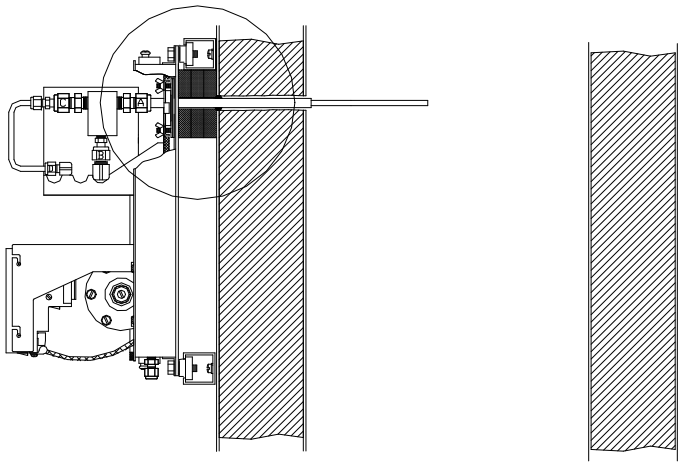
The exhaust tube surrounds the sample tube to protect it from heat losses. The exhaust is cut to length and de-burred. It just penetrates the duct. The sample is positioned to obtain a representative sample, usually between the center of the duct to 3/4ths the width/diameter of the duct. If possible, the analyzer is mounted flush to the duct surface, using a gasket to seal and insulate.

WARNING! Sample should b 6 inches [15.24cm] longer than exhaust to prevent the exhaust from influencing the sample reading.

For the standard probe, part number HTR063, the sample tubing is ¼ inch OD. An optional probe, part number HTR075 is similar, except the sample tubing is 3/8 inch OD. HTR075 is useful where condense water vapors might occasionally form from mists in the duct. Its larger sample tubing diameter is not easily obstructed by a water droplet.



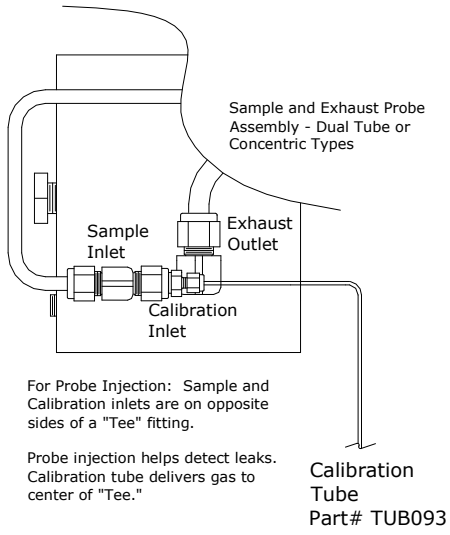
When the duct surface is above 75ºC, or if the duct is circular, the analyzer is not flush with the surface of the duct. The analyzer should be one or two inches (3 to 5 cm) from the duct. The concentric probe is insulated so that no part is exposed. The entry into the duct is sealed so a cold spot does not form at the duct opening adjacent to the probe.



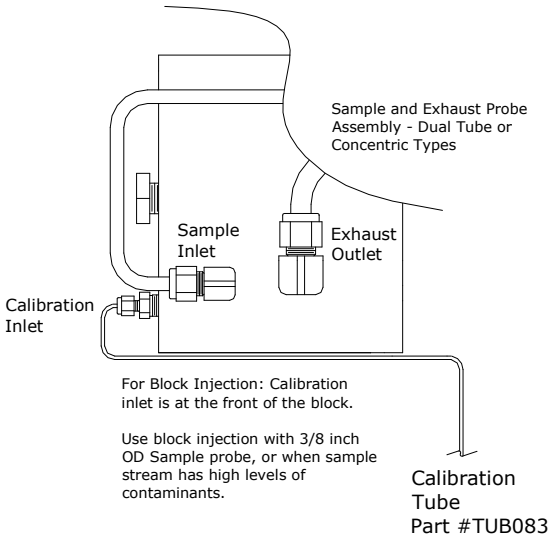
4.4.3. Calibration Injection Methods

The standard method of injecting calibration gas directly opposite the incoming sample stream, shown at left, below, helps detect leaks in the flame cell. Applications with high levels of contaminants use the block injection method, below right, to protect the calibration tube nozzle.

Calibration Probe Injection



Calibration Block Injection

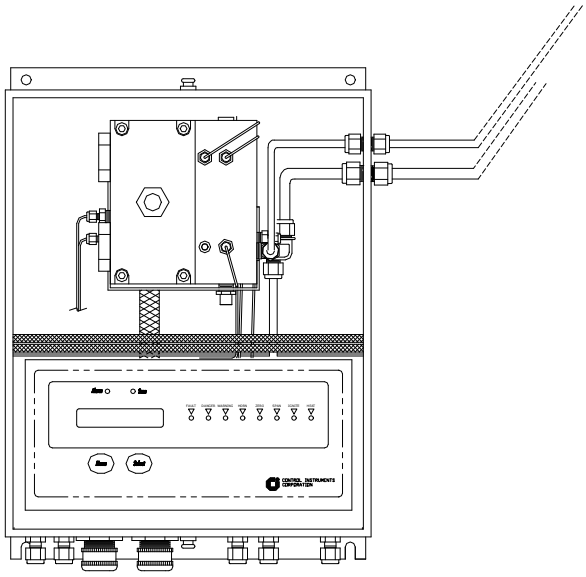


4.4.4. Wall Mount

Reference part number MTG053.
WARNING! Sample should b 6 inches [15.24cm] longer than exhaust to prevent the exhaust from influencing the sample.

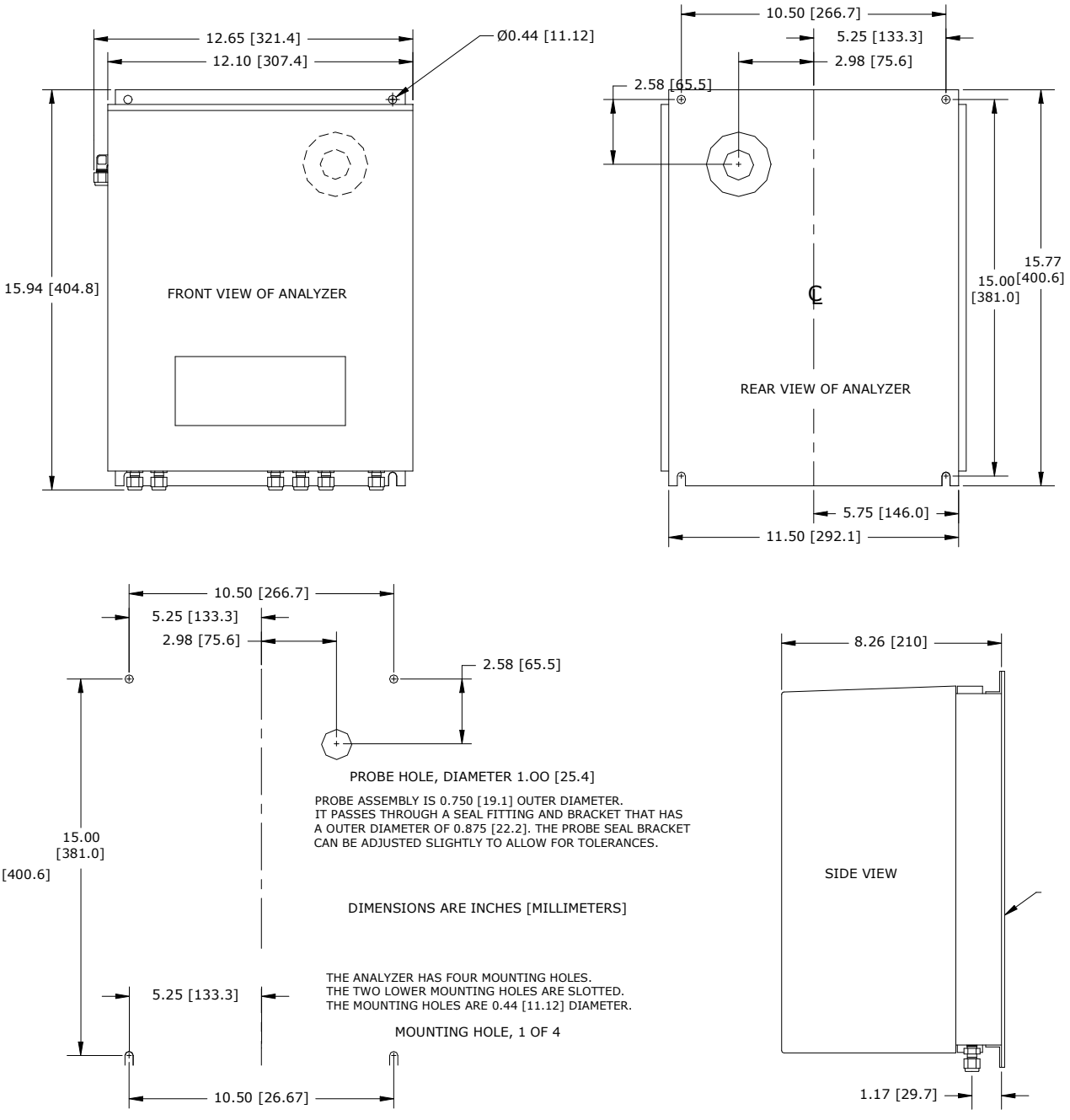
For wall mount make sure that the exhaust tube is downstream of the sample inlet

Both the ¼ inch OD sample tube and 3/8 inch OD exhaust tube connect to fittings on the sidewall of the analyzer. The sample tube is positioned to obtain a representative sample and is cut to length and deburred. The exhaust tube is positioned at a point in the duct slightly downstream and/or at a distance from the sample inlet, so that the exhaust cannot dilute the sample. Cut tubes to length and debur.



4.4.5. Mounting Dimensions

Use four mounting bolts. For the optional Wall Mount type (MTG053) sample and exhaust tubes connect to the side of the analyzer. For all others, sample probes connect through the rear of the analyzer's upper right corner. Leave 6 to 10 inches (15 to 25 cm) clearance on all sides. Some analyzers need room for a purge/pressurization system at the upper left.



4.5. Utilities

Electrical power, fuel and compressed air must be supplied continuously without interruption. Turning off the analyzer at night or on weekends can result in startup delays and operating problems.

Compressed air is used to prevent an accumulation of fuel within the enclosure in case of a leak. If compressed air is temporarily unavailable, turn off the fuel supply as a precaution.

Sampling into an unheated sensor can result in condensation and clogging. During installation, keep sample and exhaust tubing disconnected until startup. If the analyzer is to be left unheated for some time, and if the process being monitored contains substances that might condense when cooled, then the analyzer should be locked out and the sample and exhaust tubing should be disconnected.

4.6. Compressed air supply

Use clean, dry compressed air. The analyzer requires clean, dry "instrument grade[17]" compressed air that is free of oil and water. Water or oil in the air source can cause damage or affect accuracy.

Filter and regulate the compressed air. Install an external air pressure regulator and filter. Set the pressure on this regulator to 20 PSIG (1,4 bar). Allow for a maximum air consumption rate of 1.1 standard cubic foot per minute (32 liters per minute) for each analyzer. A pressure relief valve set at 50 PSIG (3,45 bar) can be added at the analyzer if desired.

Use 3/8" OD copper tubing or better. Refer to flow diagrams for correct tubing diameters, lengths and quantity of analyzers.

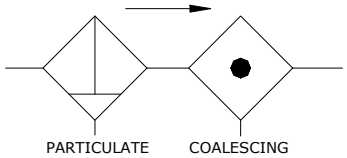
Install a manual "shut off" valve near the Compressed Air inlet, to simplify service and maintenance routines. Use ¼" OD copper tubing to connect the "shut off" valve to the air inlet.

Keep the compressed air dew point well below the coldest expected temperature. Failure to do so may result in frozen compressed air lines. Outdoor compressed air lines require special care.

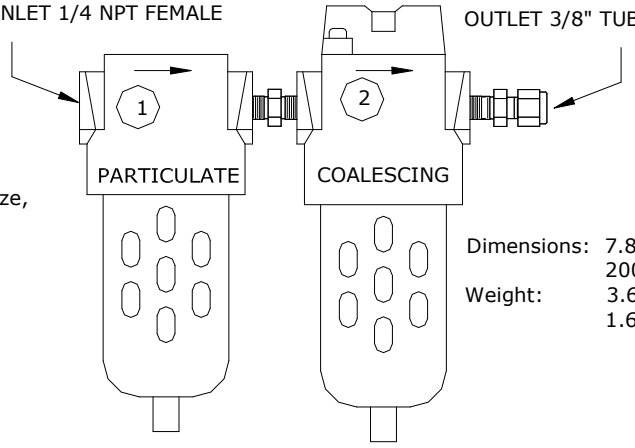
Protect inert processes from air in the exhaust. The analyzer's exhaust contains compressed air and dilution air. 100% nitrogen may be used as the compressed "air" source, but the dilution must still use air. Make certain that this volume of air will not adversely affect an inert process, or take special precautions to plan an alternate exhaust method.

Connect the compressed air to the (optional) air dilution inlet. If this factory-installed option is used, it requires 20 PSIG (140 kPa) at the dilution air inlet.

Air Filter Assembly
Part #FLT070
Maximum working pressure 150 PSIG (10 bar)
Maximum temperature 125°F (52°C)
Suitable for up to 18 analyzers
Maximum flow rate 14 SCFM (400 LPM)
Assembly exceed ISO Class 1 for particulate size, solid contaminates and oil content.

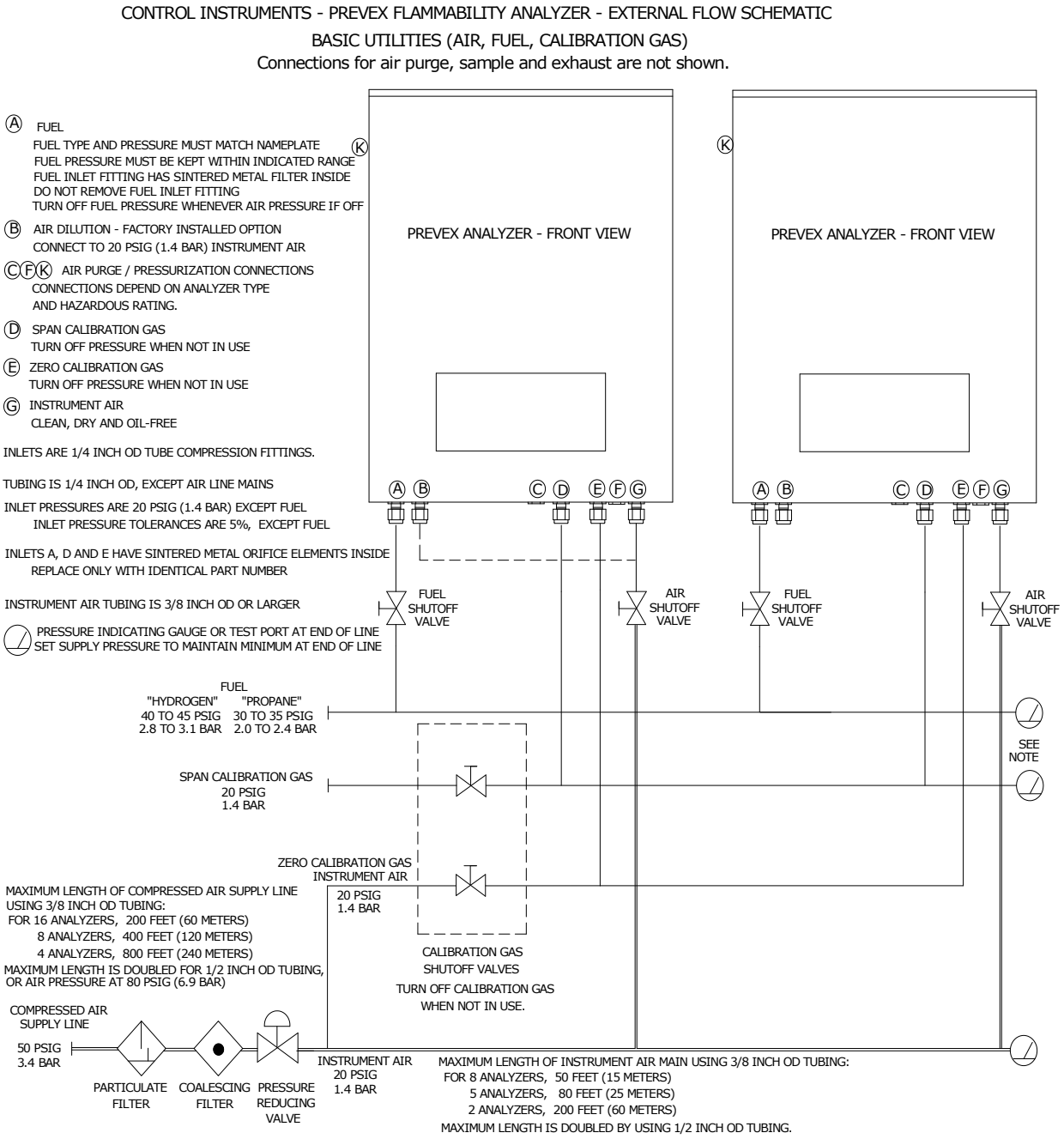


BLOCK DIAGRAM



Dimensions: 7.8 x 7.8 x 3.5 inches
200 x 200 x 90 mm
Weight: 3.6 lb
1.6 kg

[17] Refer to ISA-7.0.01-1996 for specifications of instrument grade air.



Typical installation of pneumatic utilities for a PrevEx analyzer.

4.7. Fuel supply cylinders

Observe precautions and regulations for safe storage and handling. The fuel supply system should conform to all applicable codes[18].

Use only the specified fuel. The sensor is factory built to accept only one type of fuel.

Use inlet restrictor fittings. The fuel inlet requires a special fitting with a factory-installed restrictor. Do not remove or replace this fitting. It is necessary for controlling the flow of fuel.

Separate fuel cylinders from calibration cylinders. Calibration cylinders are equivalent to compressed air and should be separated from fuel cylinders by a safe distance or rated barrier.

Install warning sign and post operating instructions.

Use an approved cylinder rack or mounting system.

If required by code, install a pressure 200 PSIG (13,8 bar) relief device vented to a safe outdoor location.

Limit the number of sensors on a common fuel line, to allow use of a safer excess flow limit.

Install an excess flow valve or flow restrictor. Allow a flow rate of at least 50 cubic centimeters per minute per analyzer for hydrogen fuel, and 10 cc/min per analyzer for propane fuel.

Install a low-pressure switch and alarm. Low fuel pressure causes false low readings. Especially for process control, monitor the fuel line with a pressure switch that alarms if the pressure falls below the minimum.

Install a flexible tubing loop at the cylinder, to facilitate cylinder changes.

Use ¼ inch OD stainless steel or copper tubing. Do not use flexible tubing. Keep tubing clean to prevent contamination. Do not use flux and solder, black pipe and pipe dope, or sealant. Use leak check solutions only on the outside of the tubing. Fuel flow and pressure drops are insignificant as long as leaks are prevented. **¼ inch OD tubing is the correct size**. **Do not use large diameter fuel lines or headers.**

Use compression fittings. Soldered or threaded connections that use flux or sealant will contaminate the fuel line, and cause reading drift or clogging of the fuel inlet restrictor. Fuel purity is important.

Install a shutoff valve at the fuel inlet for use during maintenance.

Test fuel lines for leaks. Eliminate all fuel line leaks.

Set the fuel regulator at the cylinder to deliver the correct pressure. Do not over- or under-pressurize the fuel line.

Set backup cylinder pressure lower than the primary. For systems with two cylinders, the pressure of the primary regulator should be slightly higher than the secondary regulator. In this way, fuel is drawn from the primary cylinder until it is nearly empty. Then the pressure drops slightly to the pressure of the secondary cylinder. This gives an indication that the primary cylinder is empty and needs replacement.

Wait for fuel lines to purge before startup. It may be necessary to wait one hour or more for fuel lines to purge themselves of trapped air. The reading drifts slowly upscale until all air is removed.

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[18] For example, in the USA, these may include: 29CFR chapter XVII, Paragraphs 1910.101(b) and 1910.103(a)(2)(i)(b) *Hydrogen Systems less than 400 cubic feet,* Compressed Gas Association, Pamphlet CGA P-1-1984 *Safe Handling of Compressed Gases in Containers, or* National Fire Protection Association - NFPA 50A *Standard for Gaseous Hydrogen Systems at Consumer Sites - Safe Handling of Compressed Gases in Containers.*

4.8. Zero and Span Test Gases

Use inlet restrictor fittings supplied with analyzer. The zero and span inlet fittings contain sintered metal restrictors that control the calibration flow rate. Do not remove these fittings. Do not replace them with other fitting types.

Supply "Zero Air" at 20 PSIG (140 kPa) to the zero inlet during calibration. If the compressed air supply is used for zero air, it must be clean, dry, oil-free, and regulated to 20 PSIG.

Supply "Span Gas," 1.15% Ethylene in Air, at 20 PSIG (140 kPa) to the span inlet during calibration. Cylinder part number CYL018 or equal. A pressure relief valve set at 50 PSIG (3,45 bar) can be added at the analyzer if desired. See calibration instructions for requirements.

Shut zero and span gas off at the cylinder valve when not in use. This prevents loss of gas from minor leaks and prevents unintentional or accidental injection of test gases. When compressed air is used in place of a zero gas cylinder, **provide a zero gas shutoff valve**, to turn off air pressure to the zero inlet when not in use.

4.9. AC Power

Observe proper precautions for handling AC power. Do not open covers when circuits are live.

Allow 400 Watts (VA) per sensor peak power. Average power consumption is lower.

Supply correct voltage. Voltage is factory configured to either 120VAC or 230VAC—but NOT BOTH. Input voltage is marked on the serial plate on the side of the analyzer.

Supply power continuously. Electric heaters stabilize readings and prevent condensation. Loss of power can cause condensation that requires cleaning before the analyzer can be put into use.

Supply multiple analyzers from same main circuit. Prevent ground loops and electrical problems by properly grounding all analyzers at the same point. This is required for serial communications.

Use proper grounding. Connect the AC main ground wire to the ground lug on the chassis. A green ground wire with yellow stripe connects the controller to the ground lug.

If power is off for some time, turn off the air and fuel supply.

Install a power switch near the sensor. The sensor uses non-incendive circuits, and therefore does not have a power switch. A suitable power switch should be installed so that power can be easily disconnected for servicing the analyzer.

Use conduit seals or cable glands to seal wiring entrances. These are required if an air purge system is connected.

Use size 14 AWG wire (or equal) for power. Do not exceed the maximum recommended length of AC power wires for different wire gauges as indicated in the table below.

Access to wiring terminals is at the rear of the controller. Terminals plug in to the controller. Access terminals by lifting the controller up, and swinging the top down, leaving two bottom pins in place. Unplug terminals to remove controller.

Install a small service loop in wire leads. Prevent strain on the terminals and allow free movement of the controller for access during service.

Maximum Length of AC Power Leads
Maximum length of wire from AC source to single analyzer

AWG	Metric (diameter)	Feet	Meters
12	2,0mm	1000	305
14	1,6mm	600	183
16	1,3mm	350	107

4.10. RS-485 Serial Communications

Serial communications enable remote monitoring for improved service and troubleshooting.

Observe correct grounding and AC wiring. Damage can occur if common-mode voltages are exceeded. Devices, which share the RS-485 connection, must all be grounded to the same common grounding point. If a DC powered master device is used, connect the (-) DC power common to earth ground.

Use shielded 22 AWG (metric 0,6mm) **three-conductor wire** such as “Belden 3106A” or equal. Use foil inner shielding with an overall braided shield. Installation of a spare is recommended.

Keep RS485 wiring separate from AC wires in metallic conduit to minimize RFI and EMI.

Protect RS-485 wires from excessive temperature and humidity, to avoid wire damage, aging, shorting and signal loss.

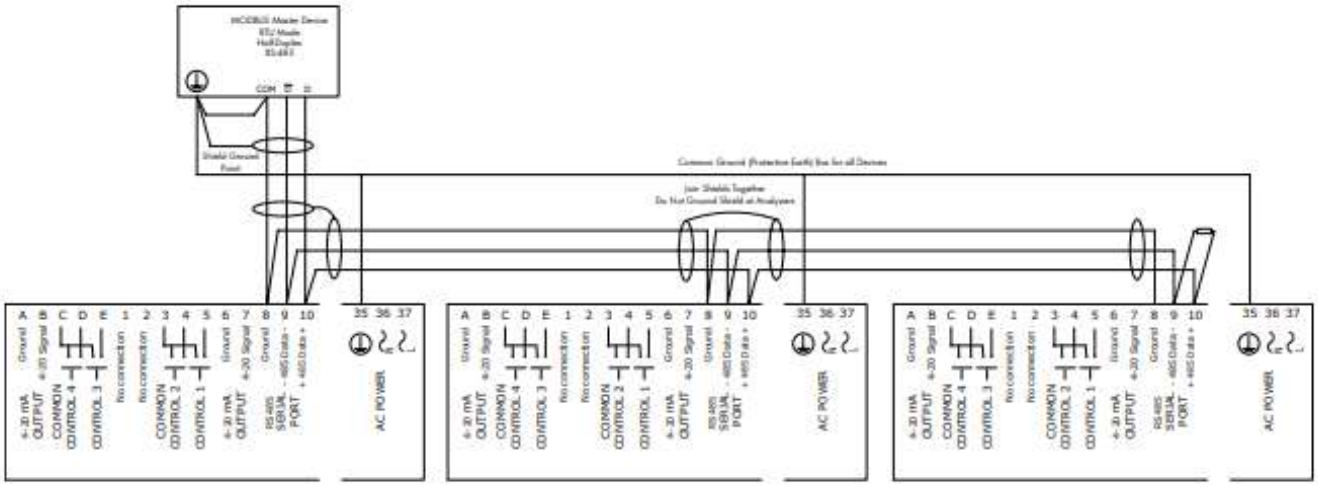
The maximum line length of the RS-485 three-conductor is 4,000 feet. For greater distances a repeater is required.

Connect wires in parallel. For multiple sensors, join all (+) connections, join all (-) connections, and join all common (grounds). Use a “daisy-chain” configuration. The maximum recommended “stub length” (length of branch circuit) is ten feet.

Ground the shield drain wire at a single point. Keep the shield and drain wire insulated from conduit and junction boxes. Interconnect the shields on each segment of cable but attach the shield to ground only at a single ground point.

Install wire with a small service loop to prevent strain on the terminals and to allow access to the terminals for service procedures.

Install termination resistor as needed.



4.11. Relays

1. **Do not exceed the 60-Watt relay contact ratings.** All contacts are rated for 60 Watts, non-inductive loads.

2. **Use the FAULT and DANGER relays.** The fault and danger relays should be used in the safety shutdown system. Refer to the safety manual, H7FTA712 for additional information.

3. **Use failsafe wiring for WARNING, DANGER, FAULT and CAL.** These relays are "normally energized (NE)." They de-energize for an alarm, power loss, or wire break. Form A contacts close when energized, and open for alarms. Form C contacts have both open and closed contacts use the contact that is closed when energized and opens for alarms.

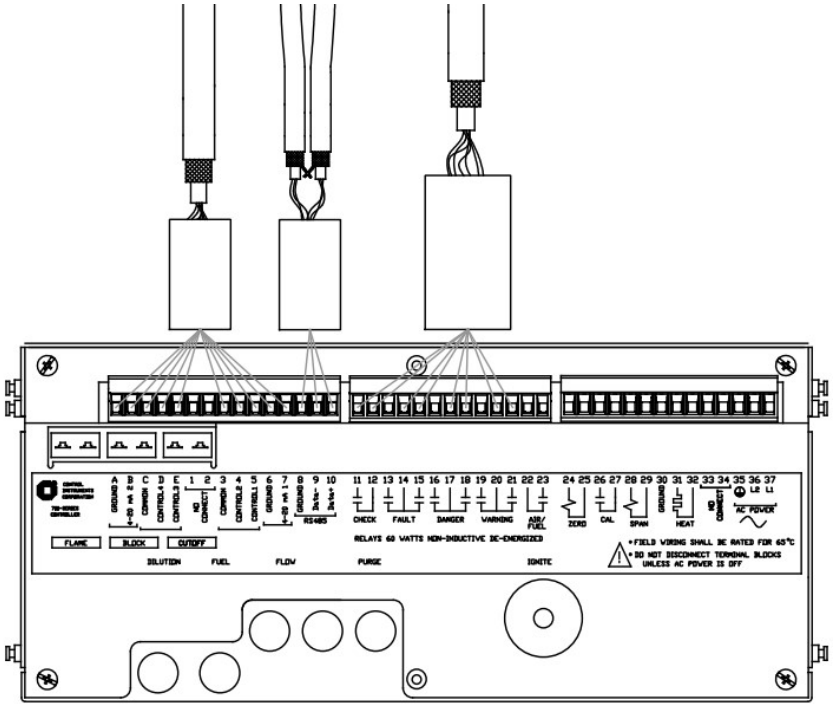
The standard controller's form A relays for warning, danger, and fault are dual form A types. There are three terminals per relay- the center terminal is common and the two outer terminals are redundant form A contacts that can either be used in series or can be used for redundant monitoring.

4. **SERVICE NEEDED "Check" Relay is a normally de-energized relay.** The factory default for the Service Needed "Check" relay is normally de-energized – it energizes when tripped. When enabled, the "CHECK RELAY NE" setting in the CONTROLS menu makes the relay normally energized – it de-energizes when tripped.

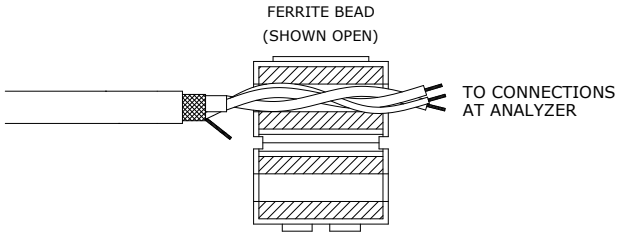
Controllers with form A alarm contacts (standard)			
Name	**Form**	**Normal State**	**Function**
WARNING	A	NE	Reading above warning level, or latched warning alarm.
DANGER	A	NE	Reading above danger level, or latched danger alarm.
FAULT	A	NE	A fault exists and the analyzer cannot make valid readings or alarms.
CALIBRATION (CAL)	A	NE	A calibration or calibration test is in progress and the analyzer cannot make valid process readings or initiate alarms.
SERVICE NEEDED	A	ND	Service is needed soon to prevent a fault. "check relay ne" can be enabled in settings
AIR/FUEL CUTOFF	A	ND	Can be configured using RLY/CNFG setting in the OUTPUTS menu

4.12. Wiring for EMC (Electromagnetic Compatibility)

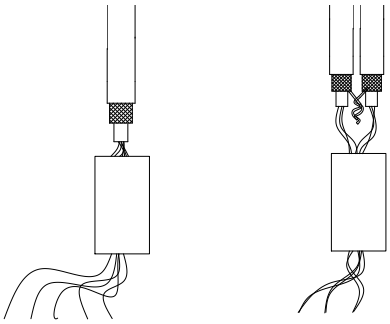
Shielded control cables, surge suppressors, and installation of ferrite beads, are required to suppress electromagnetic interference. Control cables for the 4-20 milliamp signal, RS-485 communications link, relays, and control inputs shall be shielded with an inner foil shield and an overall braided shield. Ferrite beads (250 Ohms at 100 Megahertz) shall be placed near the controller's terminals. Shields and ferrite beads are not required for the internal wiring or the AC power wiring on terminals #24 to #37.



Controller wiring with shielded cables and ferrites on control cables.



Installation of ferrite bead on control wiring. Remove outer insulation and shields. Fit wires into center and snap shut. Keep wires short. A tight fit is best.



"Star" (left), and "Ring" or "Daisy-chain" (right) configurations

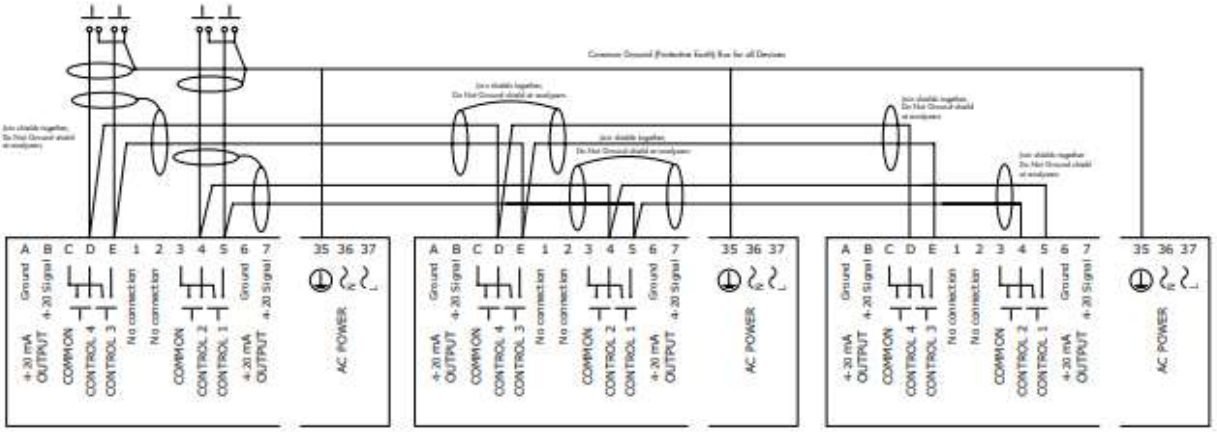
For "star" configuration wiring, cut off the drain wire. For "daisy-chain" wiring, in which a single terminal has both an incoming and outgoing wire, join the drain wires together as shown.

4.13. Remote Control Inputs

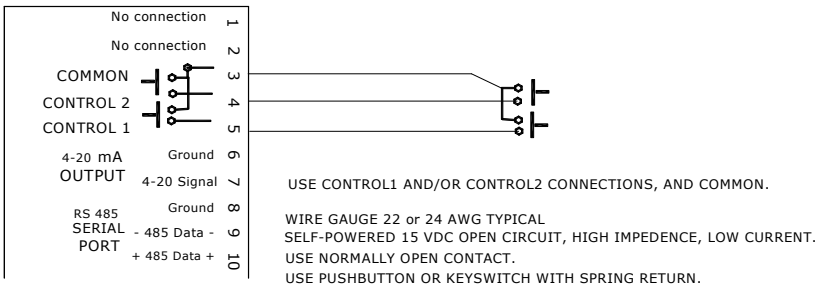
Either two or four inputs are provided. They are programmed in the CONTROLS menu. Each can perform one of the commands from the COMMANDS menu. For example:

- Manual ALARM RESET and ACKNOWLEDGE pushbuttons for alarms.
- Local MENU and SELECT pushbuttons to allow access without removing the cover.
- Switch to initiate calibration.
- Switch to lock out calibrations and tests and/or changes to settings.

1) Terminal block positions with a common (ground) are available for control inputs.
2) Use a normally open, momentary contact, dry circuit, relay or switch.
3) Use 22 or 24 AWG wire (metric 0,5mm to 0,6mm) with a braided shield.
4) Activate with a contact closure between the control input and common**:**
   a) Commands to "lock," "unlock," or "check utilities" are active while the contact closure is sustained.
   b) All other commands initiate with a momentary contact closure of 250 milliseconds or more.
5) Controls are self-powered. **Do not attach AC or DC powered control signals** to controls.



REMOTE CONTROL WIRING, TYPICAL, SINGLE ANALYZER



4.14. 4-20mA Signal Output(s)

1. **The non-isolated 4-20mA output can drive a maximum load of 275 ohms**, non-inductive, including line length.
2. **Use shielded 22 AWG** (metric 0,6mm) **shielded twisted-pair wire** such as Belden 83319E or Alpha 2824/2 or equal.
3. **The 4-20 milliamp signal is self-powered.** It is not loop-powered, and should not be connected to a power source.

4.15. Visual and Audible Alarms and Indicators

Install audible and visual devices to alert personnel in the vicinity of the analyzer. At a safe distance from the process, install a display that gives the reading and alarm status of the analyzer.

The analyzer reading (%LFL) should be displayed at the control panel where adjustments to the process are made that might affect the concentration of flammable gases. The operators should understand the meaning of the analyzer reading, and how it relates to the process under control.

A large, distinctive visual and audible alarm, easily observed and heard from all working locations, should be used to alert personnel to the presence of an alarm.

The danger and fault alarms should be used for an immediate safety shutdown to bring the concentration of flammable gases in the process safely down to 0%LFL. The warning alarm can be used to slow down or stop the process in an orderly manner, so that a danger alarm is avoided. But in any case, the danger alarm must safely bring the process to a halt.

The analyzer is installed close to the atmosphere being sampled. During alarms, the atmosphere may become flammable or explosive. Avoid the area near the analyzer during alarms, until it is known that the concentration of flammable gas has been brought to 0%LFL.

WARNING! During alarms, keep at a safe distance until the cause is found and corrected.

4.16. Security

WARNING! The enclosure should be locked to secure the analyzer against tampering.
Lock the outer cover to prevent access to the MENU and SELECT pushbuttons. The cover requires a tool to open; this tool-operated safety feature should not be removed.

Several methods are available to prevent accidental or unauthorized calibrations or changes to settings. The person responsible for overall process safety must implement an effective design.

- Control inputs can be configured with commands to lock calibrations and tests and/or settings. If a control input is not used, the lock setting in the security menu controls security directly.
- The CALIBRATING (CAL) relay can be used to signal that a calibration or test is in progress so that the machine or process can be put in a safe state.
- The MILLIAMP CAL SET and MILLIAMP FAULT settings can force the milliamp output to a special value so that the machine or process can be put in a safe state.
- Placing a cover over the phototransistors when not being used to prevent unauthorized access and to prevent unwanted activation due to ambient light

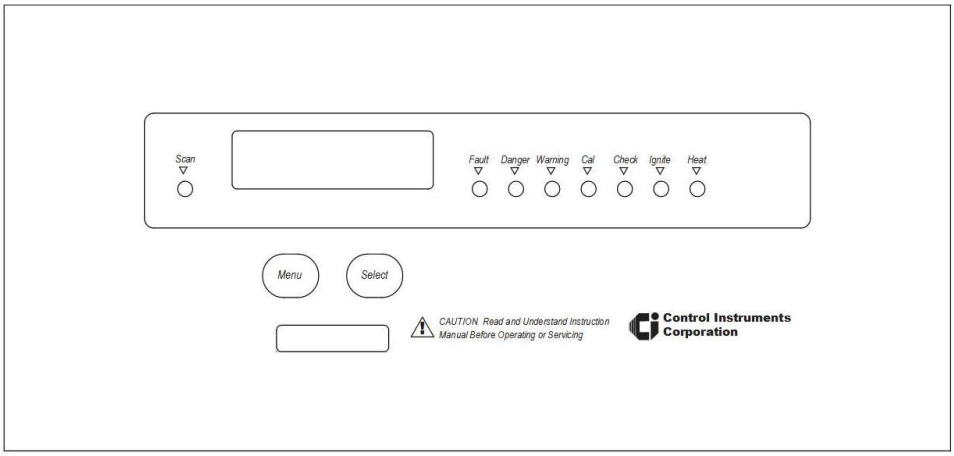
Either a control input, or a register, controls the lock. The LOCK SETTING is the type of security to be used, and the LOCK STATE is the current level of security.

If a CONTROL input is configured to lock or unlock:
If a control input is used, then the control input activates the LOCK SETTING
1. Configure the LOCK SETTING to the desired type of lock.
2. Configure the CONTROL input:
   a. For failsafe wiring, use a contact closure of the CONTROL input to UNLOCK, so that a wire break or open circuit results in a lock.
   b. For non-failsafe wiring, use a contact closure on the CONTROL input to LOCK, so that a wire break or open circuit unlocks.
3. Verify that activating the control input causes the LOCK STATE to match the LOCK SETTING and has the desired effect.

If the LOCK SETTING is used to lock or unlock:
If a control input is not used, then the LOCK SETTING *on the security menu can be used to set the* LOCK STATE.

5. Display, Indicators, Controls

The controller has an alphanumeric display with status indicators. The display is controlled by MENU and SELECT pushbuttons, or by photo transducers that allow access with the cover in place. When the menus are not in use, the flammability reading and status are displayed.



5.1. Indicators

Indicator	Color	Meaning
SCAN	Green	Flashes at least twice per second during normal operation. "Winks" briefly after completing a serial communication reply.
FAULT	Yellow	On when a fault is present. Flashes once each ten seconds if "Service Needed."
DANGER	Red	Indicates a high high level alarm.
WARNING	Red	Indicates a high level alarm.
CAL	Yellow	On when the zero or span gas solenoid valve is energized during calibration.
CHECK	Yellow	On when a "Service Needed" check is active
IGNITE	Green	On during two-second ignition pulse of the high voltage spark igniter.
HEAT	Green	Turns on when the flame cell heater is on. Once the operating temperature is reached, the heater and LED slowly cycles on and off.

5.2. Menu and Select Pushbuttons

The MENU pushbutton displays a list of menus. SELECT a menu to display its registers. Within a menu, all register values will be displayed underneath the register's name. Pressing SELECT will allow you to change the value of any register whose value can be changed. Pressing the MENU button will go to the next register. When finished viewing registers, press SELECT when EXIT is displayed to leave the menu; or, simply wait: menus are displayed for several seconds, and register values for three minutes, after which, without pressing any pushbutton, the display will time out and exit automatically. In VERSION 7.01 and above, exiting out of a register menu will not return you to the scan and readings page, but will just exit out to the menu name. The user can use the MENU button to scroll through the menus until EXIT appears. Using the SELECT button here will exit out of the menus and return the user to home screen with the flammability readings.

To test the display, press and hold both the Menu and Select pushbuttons at the same time to turn on all the segments. When power is first turned on, the display automatically tests all segments.

Some registers can be changed. Pressing SELECT while its value is displayed increments the value to the next available setting. Once the desired value is displayed using SELECT, wait a few seconds until the display flashes. Continue to wait until the flashing stops. As the flashing stops, the new setting is entered and saved. To abort the change without saving it, press MENU before the flashing stops. The old value will be restored.

5.3.

5.3. MENUS AND REGISTERS

COMMANDS MENU	
COMMANDS ALWAYS ACCESSIBLE TO THE OPERATOR	
Alarm Reset	Clear alarms
Acknowledge	Clear "Service Needed - Check"
NOTE: Calibration commands accessible to the operator when enabled by security. These commands temporarily bypass new warning and danger alarms and trip the calibration in progress relay	
Zero adjust	Adjust flammability readings within +/-3%LFL to 0 %LFL
Zero calibration	Zero calibration with gas, make adjustment to 0%LFL
Span calibration	Span calibration with gas, make adjustment to the calibration reading
Full calibration	Perform zero and span calibration with gases, make adjustments
Zero test	Test using zero gas without adjustment
Span test	Test using span gas, without adjustment
Full test	Launch zero and then span tests, no adjustments
GAUGES MENU	
PEAK HIGH %LFL	Highest flammability reading since the last "reset" command in %LFL
PEAK LOW %LFL	Lowest flammability reading since the last "reset" command in %LFL
BLOCK TEMP	Temperature of the flame cell block in Degrees C
BLOCK SETTING	Temperature setting for the flame cell block in Degrees C
SAMPLE FLOW	Sample flow rate in Standard Liters per Minute
FLAME TEMP	Current flame temperature (Thermocouple TC1 reading) in Degrees C
ZERO FLAME TEMP	Flame temperature measured during the last successful zero calibration
SPAN FLAME TEMP	Increase in flame temperature measured during the last successful span calibration
FLOW SYSTEM MENU	
SAMPLE FLOW	Current sample flow rate
CAL SAMPLE FLOW	Sample flow rate measured during the last successful calibration
LOW FLOW LEVEL	Flow rate at which a low sample flow fault trips
CHECK FLOW SET	Deviation in flow rate at which a low sample flow "Service Needed" "check" activates
ORIFICE PRESSURE	Current pressure measured across the sample flow orifice
ORF PRESS ADJ	Adjustment to the sample flow orifice pressure sensor
DILUTION MONITOR MENU	
WHEN ENABLED BY DEVICE SETTTINGS	
DILUTION PRESS	Current dilution pressure
DILUTION SETTING	Dilution pressure setting
DILUTION LIMIT	Setting for the allowable deviation between the dilution pressure and setting
DILN PRESS ADJ	Adjustment to the dilution pressure sensor
FUEL MONITOR MENU	
WHEN ENABLED BY DEVICE SETTTINGS	
FUEL PRESSURE	Current fuel pressure
FUEL SETTING	Fuel pressure setting
FUEL LIMIT	Setting for the allowable deviation between the fuel pressure and setting
FUEL PRESS ADJ	Adjustment to the fuel pressure sensor
PURGE MONITOR MENU	
WHEN ENABLED BY DEVICE SETTTINGS	
ENCLOSURE PRESS	Current enclosure pressure
ENCL PRESS ADJ	Adjustment to the enclosure pressure sensor

CALIBRATION MENU	
CAL READING	Setting for the span calibration
ZERO AS FOUND	Flammability reading recorded during zero calibration prior to adjustment
SPAN AS FOUND	Flammability reading recorded during span calibration prior to adjustment
RESPONSE TIME	Response time recorded during last full calibration
SPEED SETTING	Speed of response and noise filter
ZERO ADJUSTMENT	Amount of adjustment that has been made to the zero-flammability reading through the "Zero Adjust" command
CALIBRATION TIME	Duration of each phase of calibration in minutes
SPAN TEMP SET	Setting for target flame temperature rise for span calibration
FLAME TEMP	Current flame temperature (Thermocouple TC1 reading) in Degrees C
ZERO FLAME TEMP	Flame temperature recorded during last successful zero calibration
SPAN FLAME TEMP	Rise in flame temperature recorded during last successful span calibration
RDNG LAST CAL	Flammability reading recorded during last successful span calibration
ALARMS MENU	
WARNING LEVEL	Setting for "high flammability" warning alarm, in %LFL
DANGER LEVEL	Setting for "high high flammability" danger alarm, in %LFL
RATE ALARM	Setting to enable or disable the rate-of-rise alarm, 0 = disabled,1 = enabled
WARNING LATCH	Setting to latch warning alarm and relay until manual reset, 0 = disabled (autoreset), 1 = latched (enabled)
DANGER LATCH	Setting to latch danger alarm and relay until manual reset, 0 = autoreset (disabled), 1 = latched (enabled)
FAULT LATCH	Setting to latch fault and fault relay until manual reset, 0 = autoreset (disabled), 1 = latched (enabled)
RDNG FAULT LEVEL	Setting to trip a fault on low "downscale" flammability readings, in %LFL, between -10%LFL and -20%LFL
AUTO IGNITION	Setting for number of automatic re-ignition attempts, from 0 to 120, at fifteen second intervals
ALARM TESTS MENU	
	WHEN ENABLED BY SECURITY, THESE TESTS TRIP RELAYS OR ACTIVATE THE SPAN VALVE
	TEST LAST FOR UP TO ONE HOUR - IT IS ENDED BY AN ALARM RESET OR CALIBRATION COMMAND
COMMANDS	Execute a command to control the device (see Commands menu for full list):
	Alarm reset — Reset command halts testing
	Test warning rly — Turn on warning relay for up to one hour or until an alarm reset command
	Test danger rly — Turn on danger relay for up to one hour or until an alarm reset command
	Test fault relay — Turn on fault relay for up to one hour or until an alarm reset command
	Test cal relay — Turn on calibrating relay for up to one hour or until an alarm reset command
	Test check relay — Turn on "Service Needed - Check" relay for up to one hour or until an alarm reset command
	Gas alarm test — Open span valve for up to 60 minutes or until an alarm reset command
4-20MA OUTPUTS MENU	
MILLIAMP OUTPUT	Flammability reading currently being output, including the effect of the calibration and fault overrides and milliamp test, in %LFL
4MILLIAMP ADJUST	Adjustment to the 4-milliamp output for the 0% LFL flammability reading, Default = 0. Adjustable from -2.0 milliamps (-20) to +2.0 milliamps (20) in tenths of milliamps. Does not normally require adjustment.
MILLIAMP CAL SET	Setting to signal that a calibration is in progress. It sends this % LFL flammability reading to the milliamp output. Adjustable from -10 %LFL to 117 %LFL. Defaults to 113 %LFL. When set to 0 %LFL the override is disabled and has no effect. NOTE: In earlier versions below 7.01 this register was displayed as milliamps.
MILLIAMP FAULT	A setting to signal that a fault exists. It sends this % LFL flammability reading to the milliamp output. Adjustable from -10% LFL to 117 %LFL. Defaults to 106 %LFL. When set to 0%LFL the override is disabled and has no effect. NOTE: In earlier versions below 7.01 this register was displayed as milliamps.
MIN MILLIAMP	Setting for minimum "floor" of the milliamp output. Can be used to prevent outputs below 4 milliamp or to test the output by increasing the output to higher levels.
CONTROLS MENU	
CONTROL 1	Command that executes when the Control 1 input is activated (see Commands list for codes and their meanings)
CONTROL 2	Command that executes when the Control 2 input is activated
CONTROL 3	Command that executes when the Control 3 input is activated NOTE: Controllers without Control3 & Control4 hardware still display these registers on the menu
CONTROL 4	Command that executes when the Control 4 input is activated
RELAY SETTING	Setting to control the "Air/Fuel Cutoff" relay NOTE: "Horn" function has been removed, replaced with the "Off" setting
CHECK RELAY NE	Setting to "Air/Fuel Cutoff" relay state "normally energized." Setting 1 = enabled makes the relay normally energized, 0 = disabled makes the relay normally de-energized

COM PORT MENU	
CURRENT BAUD	Code for the current Baud rate., can differ from the default setting if autobaud is enabled
MODBUS ADDRESS	Setting for MODbus address from 1 to 32, default =1
DEFAULT BAUD	Setting for Baud rate used when automatic baud rate adjustment is disabled
BAUD AUTO ADJUST	Setting to automatically search & set Baud rate, overriding the default Baud setting, 0 = disabled, 1 = enabled, default = enabled
ID NUMBER	Setting for user-defined identification, can be any number from 0 to 99, Default = 0

SETTINGS MENU	
FACTORY SETTINGS ARE USED FOR DEVICE CONFIGURATION – THEY ARE ACCESSIBLE ONLY WHILE IN A "COLD START" CONDITION	
DEVICE TYPE	Analyzer type - This register is written once at factory, from 1 to 23. Defaults to 0 to indicate "Data Lost." The device type setting should match the analyzer nameplate.
SOFTWARE VERSION	Version number, for example 701, which is displayed as "7.01"
SERIAL NUMBER	Last three digits of the serial number. Example: Serial number 101 for year built 2022 displays as "22-101." The serial number adjusts from 100 to 9999, year built adjusts from 2001 to 2100.
YEAR BUILT	
LANGUAGE	Display language: 0 = English, 1 = German
FLAME TC GAIN	Percent gain of the TC1 flame thermocouple from 95 to 115, default = 100
TC TIME CONST	Time constant of the flame thermocouple from 9 to 12 seconds in tenths of seconds (90 to 120). Defaults to 10.7 seconds
MINIMUM SPEED	Minimum speed setting, when enabled allow SPEED SETTING (Reg 50) to be turned completely “OFF”
DILUTION MONITOR	Setting for the dilution monitor, 1 = enabled, 0 = disabled
FUEL MONITOR	Setting for the fuel monitor, 1 = enabled, 0 = disabled
PURGE MONITOR	Setting for the purge monitor, 1 = enabled, 0 = disabled

SECURITY MENU	
LOCK SETTING	Setting to lock out calibrations and tests and/or to lock parameters - see "Security" section of manual.
LOCK STATE	Current state of the lock(s), 0 = Unlocked, 1= LockCal&Tests, 2 = LockSettings, 3 = LockAll

5.4. COMMANDS

ALARM RESET and ACKNOWLEDGE are the only commands used by the operator. They are always accessible.

ALARM RESET	Clears the latch for warning, danger, or faults. It initiates re-ignition of the sensing flame. It clears and restarts timing for the air/fuel cutoff.
ACKNOWLEDGE	Deactivates the “Service Needed” relay and message.

All other commands are for calibration and maintenance, and should be secured against unauthorized or unintentional activation with a software or hardware lock.

ZERO ADJUST	Immediately adjusts the flammability reading to 0%LFL without using zero gas. The total cumulative amount of adjustment is +/-3%LFL

Calibration commands initiate calibration. Solenoid valves inject calibration gases to the analyzer. If the response is acceptable, the readings are adjusted; else a calibration fault is activated. Hardware faults, which might make calibration inaccurate, cause the calibration to abort.

ZERO CALIBRATION	Injects Zero Gas and updates ZERO FLAME TEMP to give a 0%LFL reading. If the response is bad, a zero fault occurs.
SPAN CALIBRATION	Injects Span Gas and updates SPAN FLAME TEMP to give a %LFL reading equal to CAL READING. If the response is bad, a span fault occurs.
FULL CALIBRATION	Performs a zero calibration, followed by a span calibration.

TEST commands do not calibrate, they only inject calibration gases. They do not adjust readings, and do not generate faults. They will attempt to execute even when a hardware fault exists. Use test commands for testing and troubleshooting only.

ZERO TEST	Injects Zero Gas, so the response can be observed.
SPAN TEST	Injects Span Gas, so the response can be observed.
FULL TEST	A combined zero and span test.

5.5. GAUGES MENU

The GAUGES menu has registers for the reading, status and operation. All are read-only, except for SET TEMP and LOW FLOW, which can be adjusted, but normally require none.

PEAK HIGH %LFL	The highest reading since last alarm reset.
PEAK LOW %LFL	The lowest reading since last alarm reset.
BLOCK TEMP	The flame cell temperature in °C. It should be equal to BLOCK SETTING.
BLOCK SETTING	Sets the flame cell temperature in °C. The controller cycles the heater to maintain the block temperature, typically, within ±2°C.
SAMPLE FLOW	The sample flow rate in standard liters per minute.
FLAME TEMP	The current flame temperature (Thermocouple TC1 reading) in °C.
ZERO FLAME TEMP	The flame temperature from the last successful zero calibration.
SPAN FLAME TEMP	The increase in flame temperature from the last successful span calibration.

5.6. FLOW SYSTEM

The FLOW SYSTEM menu has registers related to the sample flow.

SAMPLE FLOW	The sample flow rate in standard liters per minute.
CAL SAMPLE FLOW	The sample flow from the last successful calibration. Deviation from the calibration sample flow activates “Service Needed – Check Flow,” and the high and low flow faults.
LOW FLOW LEVEL	The sample flow rate below which a Low Flow Fault trips.
CHECK FLOW SET	The change in sample flow rate from the last successful calibration at which “Service Needed – Check Flow” activates.
ORIFICE PRESSURE	The current pressure across the sample orifice.
ORF PRESS ADJ	The amount of adjustment made to the sample flow pressure transducer.

5.7. DILUTION MONITOR

This menu appears only when the dilution monitor is enabled. The dilution pressure transducer hardware must be installed. These registers are related to monitoring the pressure of dilution air into the sample stream of an air dilution type analyzer to produce “Service Needed – Check Dilution” or a high or low dilution fault when the pressure deviates from the setting.

DILUTION PRESS	The current dilution pressure
DILUTION SETTING	The dilution pressure setting
DILUTION LIMIT	Setting for the allowable deviation between the dilution pressure and setting
DILN PRESS ADJ	The amount of adjustment to the dilution pressure sensor

5.8. FUEL MONITOR

This menu appears only when the fuel monitor is enabled. The fuel pressure transducer hardware must be installed. These registers are related to monitoring the fuel pressure to produce “Service Needed – Check Fuel” or a high or low fuel fault when the fuel pressure deviates from the setting.

FUEL PRESSURE	The current fuel pressure
FUEL SETTING	The fuel pressure setting
FUEL LIMIT	Setting for the allowable deviation between the fuel pressure and setting
FUEL PRESS ADJ	The amount of adjustment to the fuel pressure sensor

5.9. PURGE MONITOR

This menu appears only when the purge monitor is enabled. The enclosure pressure transducer hardware must be installed. These registers are related to monitoring the enclosure pressure to ensure that the purge system is working. It will produce a “Service Needed – Check Enclosure” enclosure pressure deviates from the acceptable limits.

ENCLOSURE PRESS	The current enclosure pressure
ENCL PRESS ADJ	An adjustment to the enclosure pressure sensor

5.10. CALIBRATION MENU

This menu contains registers for controlling calibration and recording the results.

CAL READING	Setting for the span calibration.
ZERO AS FOUND	Flammability reading recorded during zero calibration prior to adjustment.
SPAN AS FOUND	Flammability reading recorded during span calibration prior to adjustment.
RESPONSE TIME	Approximate T60 response time recorded during last full calibration.
SPEED SETTING	Speed of response and noise filter. Sets the speed of response for the sensor. The HIGH setting gives the fastest response; the LOW setting is slower but has better noise immunity.
ZERO ADJUSTMENT	The amount of adjustment that has been made to the zero flammability reading through the "Zero Adjust" command
CALIBRATION TIME	Duration of each phase of calibration in minutes. 1 – 4 minutes adjustable.
SPAN TEMP SET	Setting for target flame temperature rise for span calibration. If the span flame temperature is more than 20°C from this setting a span fail fault trips. A deviation of 10°C activates “Service Needed – Check Span.
FLAME TEMP	Current flame temperature (Thermocouple TC1 reading) in °C.
ZERO FLAME TEMP	Flame temperature recorded during last successful zero calibration
SPAN FLAME TEMP	Rise in flame temperature recorded during last successful span calibration
RDNG LAST CAL	Flammability reading recorded during last successful span calibration. If it does not match the CAL READING a “Service Needed – Check Span” activates.

5.11. ALARMS MENU

The ALARMS menu control the warning, danger and fault alarms. These registers are for configuration only, and do not normally require adjustment.

WARNING LEVEL	Setting for "high flammability" warning alarm, in %LFL. When the flammability reading exceeds the warning level, the warning LED indicator and relay are activated. If configured to be a latching alarm, the warning remains active until the flammability reading goes back below the warning level and the reset command is given. Otherwise, the warning LED indicator and relay are deactivated once the flammability reading goes back below the warning level. The WARNING alarm is automatically activated whenever the DANGER alarm is active.
DANGER LEVEL	Setting for "high high flammability" danger alarm, in %LFL. When the flammability reading exceeds the danger level, the warning and danger indicators and relays are activated. These remain active until the flammability reading goes back below the danger level and the reset command is given, or an auto reset occurs.
RATE ALARM	Setting to enable or disable the rate-of-rise alarm, 0 = disabled,1 = enabled. If the rate of rise of the flammability reading is such that it will soon exceed the danger alarm level, the danger alarm is activated.
WARNING LATCH	Setting to latch warning alarm and relay until manual reset, 0 = disabled (autoreset), 1 = latched (enabled). A latching alarm is kept active, even if the condition that caused the alarm is corrected, until it is cleared by the alarm reset command.
DANGER LATCH	Setting to latch danger alarm and relay until manual reset, 0 = autoreset (disabled), 1 = latched (enabled). When the danger latch is disabled, an external device must be provided to perform the latching function.
FAULT LATCH	Setting to latch fault and fault relay until manual reset, 0 = autoreset (disabled), 1 = latched (enabled). An auto ign setting of 1 or more can override the latches setting for flameout faults that successfully re-ignite.
RDNG FAULT LEVEL	Setting to trip a fault on low "downscale" flammability readings, in %LFL, between -10%LFL and -20%LFL.
AUTO IGNITION	Setting for number of automatic re-ignition attempts, from 0 to 120, at fifteen second intervals. When set to 0, this feature is turned off, and there is a single attempt to ignite the sensing flame. When 1 or more, up to a limit of 120, there are additional re-ignition attempts at 15-second intervals. If re-ignition is successful, and there are no fault conditions, normal operation resumes as if an alarm reset had occurred, regardless of the fault latch.

5.12. ALARM TESTS MENU

When enabled by security, this menu can be used to send commands that trip relays or activate the span valve. The maximum duration of the test is one hour. The test ends with an alarm reset or the start of a calibration. The commands are:

ALARM RESET	The alarm reset command halts testing. The tester should halt the test promptly after completion.
TEST WARNING RLY	Turn on warning relay for up to one hour, or until an alarm reset command.
TEST DANGER RLY	Turn on danger relay for up to one hour, or until an alarm reset command.
TEST FAULT RELAY	Turn on fault relay for up to one hour, or until an alarm reset command.
TEST CAL RELAY	Turn on calibrating relay for up to one hour, or until an alarm reset command.
TEST CHECK RELAY	Turn on "Service Needed - Check" relay for up to one hour, or until an alarm reset command.
GAS ALARM TEST	Open span valve for up to 60 minutes, or until an alarm reset command. The gas alarm test does not disable the warning and danger alarms. If the test gas concentration is above the warning or danger level, these alarm will trip during the gas alarm test.

5.13. 4-20mA OUTPUTS

This menu contains registers for configuring and testing the milliamp output(s).

MILLIAMP OUTPUT	The flammability reading, in %LFL, currently being output, including the effect of the calibration and fault overrides and the minimum milliamp setting.
4MILLIAMP ADJUST	Adjustment to the 4 milliamp output at the 0% LFL flammability reading. Default = 0. Adjustable from -2.0 milliamps (-20) to +2.0 milliamps (20) in tenths of milliamps. Does not normally require adjustment.
MILLIAMP CAL SET	A setting to signal that a calibration is in progress. It sends this %LFL flammability reading to the milliamp output. Adjustable from -10%LFL to 117%LFL. Defaults to 113%LFL. When set to 0%LFL the override is disabled and has no effect. Note - in earlier versions below 7.01 this register was displayed as milliamps.
MILLIAMP FAULT	A setting to signal that a fault exists. It sends this %LFL flammability reading to the milliamp output. Adjustable from -10%LFL to 117%LFL. Defaults to 106%LFL. When set to 0%LFL the override is disabled and has no effect. Note - in earlier versions below 7.01 this register was displayed as milliamps.
MIN MILLIAMP	Setting for minimum "floor" of the milliamp output. Can be used to prevent outputs below 4 milliamps or to test the output by increasing the output to higher levels.

5.14. CONTROLS

This menu sets the commands for the four wired control inputs, configures the AIR/FUEL relay and sets the state of the CHECK relay.

CONTROL 1	Set the command that will execute when the Control 1 input is activated.
CONTROL 2	Set the command that will execute when the Control 2 input is activated
CONTROL 3	Set the command that will execute when the Control 3 input is activated.
CONTROL 4	Set the command that will execute when the Control 4 input is activated.
RELAY SETTING	Setting for AIR/FUEL cutoff relay. 0 = de-energized, 1 = air/fuel cutoff, 2 = energized.
CHECK RELAY NE	Normally energized (fail safe) state of the Check relay, Enabled or disabled..

5.15. COM PORT

This menu configures the serial communications port. The character length is always 8 bits with one stop bit and no parity. The baud rate is adjustable, and can be made automatic.

CURRENT BAUD	Code for the current Baud rate. Can differ from the default setting if Baud Rate automatic adjustment is enabled.
MODBUS ADDRESS	Setting for Modbus address from 1 to 32, default =1.
DEFAULT BAUD	Setting for Baud Rate used when automatic baud rate adjustment is disabled.
BAUD AUTO ADJUST	Setting to automatically search and set the Baud Rate, overriding the Default Baud rate setting, 0 = disabled, 1 = enabled, default = enabled.
ID NUMBER	Setting for user-defined identification, Can be any number from 0 to 99. Default = 0.

5.16. SETTINGS MENU

Factory settings are used for device configuration – they are accessible only while in a "cold start" condition

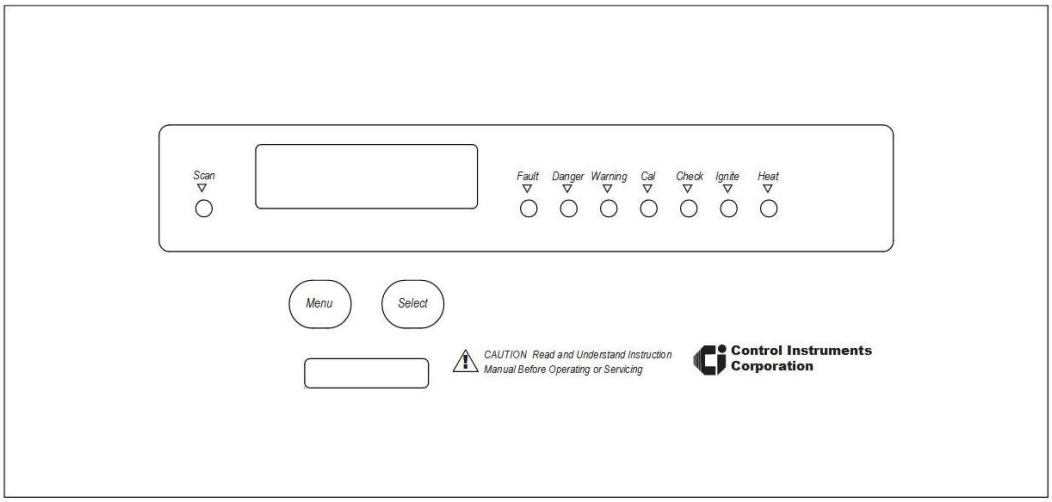
DEVICE TYPE	Analyzer type – This register is written once at factory, from 1 to 233. Defaults to 0 to indicate "Data Lost". The device type setting should match the analyzer nameplate.
SOFTWARE VERSION	Version number, for example 701, which is displayed as "7.01".
SERIAL NUMBER	Last three digits of the serial number. Example: Serial number 101 for year built 2022 displays as "22-101." The serial number adjusts from 100 to 9999, year built adjusts from 2001 to 2100
YEAR BUILT	
LANGUAGE	Display language: 0 = English, 1 = German
FLAME TC GAIN	Percent gain of the TC1 flame thermocouple from 95 to 115, default = 100.
TC TIME CONST	Time constant of the flame thermocouple from 9 to 12 seconds in tenths of seconds (90 to 120). Defaults to 10.7 seconds
MINIMUM SPEED	Minimum speed setting, when enabled allow SPEED SETTING (Reg 50) to be turned completely "OFF".
DILUTON MONITOR	Setting for the dilution monitor, 1 = enabled, 0 = disabled.
FUEL MONITOR	Setting for the fuel monitor, 1 = enabled, 0 = disabled.
PURGE MONITOR	Setting for the purge monitor, 1 = enabled, 0 = disabled.

5.17. SECURITY MENU

LOCK SETTING	Setting to lock out calibrations and tests and/rs – see "Security" section of manual.
LOCK STATE	Current state of the lock(s), 0 = Unlocked, 1 = LockCal&Tests, 2 = LockSettings, 3 = LockAll.

5.18. Flashlight-activated Commands

Phototransducers located just below the MENU and SELECT pushbuttons allow access to COMMANDS when the enclosure cover is on.



1. Cover the left and right phototransducers with fingertips.
2. Shine a flashlight on the center phototransducer.
   Note: an incandescent light works best, some LED lights do not.
3. Commands will appear in the display, one at a time, every few seconds. Only RESET and ACKNOWLEDGE commands appear when the CAL LOCK is active.
4. When the desired command appears, remove the flashlight and uncover the left and right phototransducers. To exit without executing a command, use the EXIT command.
5. The selected command flashes for several seconds before activating.
6. Once the command activates, flashing stops, and the reading and status is displayed.

   Note: the “Flashlight” command is no longer accessible through Modbus.

6. Startup Procedure

To be performed by qualified persons only.

6.1. Initial conditions

Before starting, AC power, fuel pressure, air pressure and calibration pressure are all OFF. To be safe, turn utilities on in the correct sequence as shown below.

6.2. Air

Apply compressed air to the analyzers. Remember that the compressed air allows the analyzer to use fuel safely. Keep the fuel and power off until the air is on and checked.

1. Temporarily disconnect air connections from analyzer(s)
2. Turn air supply pressure to a minimum value, (a few PSIG).
3. Turn air on to blow air lines clean of anything that might have accumulated.
4. Reattach air lines to each analyzer
5. Set air pressure to deliver 20 PSIG to each and every analyzer.
6. Check pressure at each analyzer.
7. Verify typical flow rate and LOW FLOW or HIGH FLOW fault alarm is not tripped.

6.3. Fuel

Once the compressed air is on, the fuel supply can be turned on. It will take some time for the fuel lines to purge before the fuel reaches the analyzer.

1. Visually check fuel connection.
2. Turn fuel supply regulator to low pressure.
3. Turn on fuel supply main valve.
4. Turn fuel supply regulator up to pressure indicated on serial plate.
5. Check for leaks: Shut off fuel cylinder valve and observe outlet pressure. If pressure holds steady more than 5 minutes, there are no significant leaks. If pressure drops suddenly there are leaks. If pressure does not hold for at least 30 seconds, the main valve must be left off until all connections are checked and the leak is found and fixed, or else the fuel cylinder will empty itself.
6. Turn on main fuel valve.

Depending on the volume of the fuel system piping, it may take hours or more for the fuel lines to fully purge.

6.4. Power

Carefully check the wiring before turning power on.

1. Check wiring for misconnections and short.
2. Look for loose strands on terminals.
3. Measure from AC input to ground, should be open circuit
4. Check relay connections for compatibility. If in doubt unplug relay connections until they can be proven.
5. Turn power on.

Analyzer has "BLOCK TEMP LOW" fault. Heater is on. Fault indicator on. Alarm on, or flashing.

Analyzer begins to heat up. Depending on the starting temperature and operating temperature, after ½ hour to 1½ hour, the analyzer reaches operating temperature, and the heater cycles on/off. Analyzer may still be in fault condition "FLAMEOUT."

6.5. Ignition

Once fuel lines have purged so fuel reaches the analyzers, and the analyzer is up to its operating temperature, and there are no other faults except "FLAMEOUT," the sensing flame can be lit.

1. The analyzer may automatically ignite itself. If not, use ALARM RESET to ignite. It may be necessary to reset up to three times.
2. Use the FLAME TEMP register to see if the flame is lit.
3. Check the settings and reading on the GAUGES menu.
4. To repeat the ignition attempt, use ALARM RESET on the COMMANDS menu again.

AUTO IGNITION on the ALARMS menu can be configured to automatically make re-ignition attempts.

REGISTER	Description
FLAMMABILITY	Until flame is lit, reading is negative, usually below -35% LFL, typically -99 %LFL Once flame is lit, reading is higher, usually above -35% LFL, and climbs up slowly as fuel lines purge and analyzer stabilizes. It finally settles out near 0%LFL.
STATUS	When power is first applied, STATUS is BLOCK TEMP LOW until analyzer heats up. With air on and the analyzer up to temperature, STATUS will be FLAMEOUT.
BLOCK TEMP	When up to temperature, is equal to BLOCK TEMP SET
BLOCK TEMP SET	Depends on the device type.
SAMPLE FLOW	Should be higher than LOW FLOW LEVEL or else a fault exists.
LOW FLOW	Factory set, usually 1.4 LPM.
FLAME	Until the flame it lit, the reading is about the same as BLOCK TEMP. Once the flame is lit, FLAME is much higher than BLOCK TEMP, and it continues to rise as the fuel lines purge and the analyzer's temperature stabilizes. The controller indicates FLAMEOUT until the FLAME TEMP is above 425°C. It eventually settles near 500°C.

Avoid adjusting the fuel regulator to try to bring the flame up. Wait for fuel lines to purge completely, until then the flame will be smaller than normal. Wait for the fuel system and analyzer to stabilize before making any adjustments.

6.6. Preliminary Calibration

Check calibration line for leaks. Perform a preliminary calibration and check results.

1. Briefly turn calibration cylinder main valve on and then immediately off to pressurize the calibration lines.
2. Observe the calibration cylinder tank pressure. It should hold constant for at least 5 minutes. If not there is a leak that must be eliminated or else the calibration cylinder will empty itself. Check all connections and repair any leaks before proceeding.
3. Turn calibration cylinder(s) on.
4. Use FULL CALIBRATION on COMMANDS menu to initiate the automatic calibration sequence.
5. For each analyzer, observe the reading during the calibration process.
   - Cal LED turns on. Zero indicator is on the display.
   - After about one minute reading is near 0%LFL.
   - Zero is off. Span comes on.

- Reading climbs up to near CAL READING, typically about 60%LFL.
- Within one minute reading is very near CAL READING.
- Span ends. Cal LED stays on for 60 seconds to recover (display is CALIBRATION ENDS) . Reading returns to 0%LFL.

6.7. Inputs and outputs

Check the 4 - 20 milliamp output and set up its "special states." Configure the controls. Check the alarm levels, Test the relays. Verify the action of the outputs for Safety Shutdown or E-Stop.

6.7.1. 4 - 20 milliamp output

Check the milliamp output. The MIN MILLIAMP *register can increase the output for test, or to prevent outputs that are too low for the control device that reads them. The* MIN MILLIAMP *setting is always active. It is saved in non-volatile memory through power cycles.*

1. Use MIN MILLIAMP on the 4-20MA OUTPUTS menu to increase the output.
2. Output several different readings and check for the desired effect.
   For example, 0%LFL=4mA, 50%LFL=12mA, and 100%LFL=20mA.
3. Set MILLIAMP FAULT, the output during faults, for desired effect.
4. Set MILLIAMP CAL SET, the output during calibrations, for desired effect.
5. Set MIN MILLIAMP to the lowest acceptable output.
   For example, -10%LFL typically, or 0%LFL if the control device cannot read below 4mA.

Note: The 4MILLIAMP ADJ settings does not normally require adjustment.

6.7.2. Alarms and Relays

1. Check relay wiring for misconnections and shorts.
2. Check WARNING and DANGER levels in ALARMS menu.
3. Use ALARM TESTS menu to activate relays.
4. Verify that actions of each relay produce the desired result.
5. Verify that the fault and danger activate the E-Stop (safety shutdown).

6.7.3. Controls

1. Configure CONTROL1 and CONTROL2 for desired effect.
2. If installed, Configure CONTROL3 and CONTROL4.

6.7.4. Security

Either a control input, or a register, controls the lock. The LOCK SETTING *is the type of security to be used, and the* LOCK STATE is the current level of security. See "security" in the Installation section.

1. If a control input is used, check that it activates the correct LOCK SETTING.
2. If a control input is not used, check the LOCK SETTING

6.8. Records

For future reference, create a permanent written record of the settings.

6.9. Stability test

After the first 24 to 72 hours of operation the analyzer and all its utilities should be stable. Test accuracy with calibration gases and check results. If the zero and span values have not drifted significantly, the analyzer and its utilities have stabilized, and are ready for a complete calibration.

FLAME TEMP on the GAUGES menu should be between 495°C and 505°C. The zero and span should be within +/-3%LFL of the correct values. If the drift is greater than 3%, repeat the stability test.

1. Turn calibration cylinder(s) on.
2. Use FULL CALIBRATION on COMMANDS menu.
3. For each analyzer, observe the reading during the calibration process.
   - Cal LED is on. Zero indicator is on display.
   - After about one minute reading is near 0%LFL.
   - Zero valve is off. Span valve comes on.
   - Reading climbs up to near CAL READING, typically about 60%LFL.
   - After about one minute reading is near CAL READING.
4. When calibration ends, check the results in the CALIBRATION menu.
   - ZERO AS FOUND should be with +/-3%LFL
   - SPAN AS FOUND should equal cal reading within +/-3%LFL
   - RESPONSE TIME should be less than three seconds.
   - ZERO ADJUSTMENT should be 0%LFL
   - RDNG LAST CAL should equal CAL READING

6.10. Response time verification

An analyzer speed of response test should be performed. The addition of sample transport time and system response time (time for corrective action) will determine the overall response time of the safety system. This combined time needs to be fast enough to prevent a hazardous condition.

6.10.1. Testing analyzer speed of response

The approximate speed of response of the analyzer is captured during a full calibration and is recorded in the RESPONSE TIME *register. The* RESPONSE TIME *includes the time for test gas to enter the analyzer plus the time for the analyzer to make a reading.*

6.10.2. Adding in sample transport time

If sample tubing is added between the analyzer and the measuring point, the "transport time" increases. The sample transport times below are based on a 2.0 LPM *flow rate. Higher flow rates are proportionally faster. Lower flow rates are slower.*

- For ¼ inch diameter tubing in a standard analyzer: 0.5 seconds per meter
- For ¼ inch diameter tubing in an air dilution analyzer: 1.0 seconds per meter.
- For 3/8 inch diameter tubing in a standard analyzer: 1.4 seconds per meter.
- For 3/8 inch diameter tubing in an air dilution analyzer: 2.8 seconds per meter.

6.11. Verification of Readings

It is important to verify that the readings from the analyzer match the readings expected from the process.

7. Calibration

To be performed by qualified persons only.

WARNING! The accuracy of the analyzer, and its ability to make alarms, depends completely upon proper calibration. Follow all calibration instructions carefully.

Calibration adjusts the analyzer for accurate %LFL readings. First, an initial calibration verifies important aspects of the calibration and its configuration, and determines the correct calibration reading[19] - CAL READING. Thereafter, routine recalibrations are performed on a regular basis to maintain accuracy and to verify the response. If any fact affecting the initial calibration changes, the initial calibration process should be repeated.

7.1. Initial calibration

Perform the initial calibration before the analyzer is put into service, and whenever process changes are made, especially changes to the types of flammable gases being measured, the span calibration gas, or the process temperature. Maintain a permanent record of all calibration information.

Review of limitations

Prior to putting the analyzer into service, review the "Theory of Operation" section for limitations to its effective operation. Include all foreseeable changes. Any limitation that might adversely affect the reliability of the analyzer should preclude its use until a correction can be found.

Identification of all flammable gases to be measured

List all flammable gases to be measured, now or in the future, in the record of initial calibration. Include the name, the LFL value, the Flash Point, and the response factor.

Response factors for different flammable gases

A response factor compares a particular gas to the standard calibration[20]. If the standard calibration is completely accurate for the gas of interest its response factor is 1.0. A low response factor means low readings: the calibration reading should be increased to make readings accurate. A high response factor means a gas has an exaggerated response compared to the standard - a "safe side" error[21]. The sensing flame has uniform response factors for many gases. Most solvent vapors have response factors between 0.9 and 1.1. They can be measured within the analyzer's specified accuracy using the standard calibration, so correction for different solvents is often unnecessary.

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[19] The calibration reading is determined according to code requirements. There is one method for FMc and FM approved types, and another for ATEX/CENELEC types. The results are similar but not identical.

[20] Response factors are not based on the LFL value of the calibration gas itself. The calibration gas is only used to create a known, predictable increase in the flame temperature. Do not change the calibration reading when the local authority publishes a different LFL value of the calibration gas itself.

[21] The error due to response factor variation is the actual %LFL reading times (R_{gas} - R_{cal}), where R_{cal} is the response factor used for the calibration (typically 1.0 for the standard calibration), and R_{gas} is the response factor of the gas that is actually present. If more than one gas is actually present, R_{gas} is the weighted average of all response factors, where the weighing factor is each gas's percentage of the total %LFL reading.

Flash Point verification

Make certain the Flash Points of any and all flammable gases to be monitored are less than the operating temperature of the analyzer and all parts of the sampling system. The Flash Point of each individual flammable gas is important, not only the Flash Point for the complete mixture.

7.1.1. Calibration reading for FM and FMc type analyzers

Standard calibrations

approved for FM and FMc type analyzers
with 7.84% per 100°C temperature correction factor per NFPA-86, suitable for the standard solvents

FM and FMc approved models are calibrated with *1.15% by volume ethylene in air* within a 2% tolerance. Ethylene in air between 1.13% and 1.17% by volume is an acceptable calibration span gas.

The calibration reading is adjusted for higher temperatures per the NFPA-86 code, which requires a 7.84% increase in **CAL RDNG** for each 100°C increase in process temperature.

CAL RDNG	**Process temperature**
1.15% Ethylene in Air = **50%**	LFL of standard solvent vapors at **77°F** (25°C)
1.15% Ethylene in Air = **56%**	LFL of standard solvent vapors at **250°F** (120°C) per NFPA-86
1.15% Ethylene in Air = **60%**	LFL of standard solvent vapors at **400°F** (200°C) per NFPA-86 (factory default)
1.15% Ethylene in Air = **64%**	LFL of standard solvent vapors at **600°F** (315°C) per NFPA-86

Adjusting FMc and FM type analyzers for other substances

The standard calibration is accurate for flammable gases with response factors between 0.9 and 1.1. Response factors above 1.1 produce "safe-side" error and can be ignored. Response factors below 0.9 may require correction of the calibration reading.

For mixtures of several solvents, where one has a much lower response factor than the others, $\mathbf{R_{min}}$, it is safest to divide the standard calibration reading by $\mathbf{R_{min}}$. The higher calibration reading will give readings for all gases in the mixture at or above their actual flammability[22].

To remove some of the "safe-side error," the response factors for "minor components" in the mixture may be ignored. A minor component is a flammable gas with a low response factor, that nevertheless is present in such small amounts in the mixture that the resulting error[23] can be ignored.

If all gases being monitored have response factors greater than 1.0, it is permitted, but not necessary, to divide the calibration reading by $\mathbf{R_{min}}$, the lowest response factor of all the gases, to lower the calibration reading and remove the "safe side" error.

[22] Refer to National Fire Protection Association, NFPA-86 "Ovens and Furnaces" for this requirement.

[23] NFPA-86 indicates an error of not more than 3% of the reading. This means flammable gases with response factors above 0.7 that are always less than 10% of the total flammable gas mixture are minor constituents. The rated accuracy of the analyzer is 10% of the reading, which corresponds to a response factor of 0.7 that is less than one-third of the mixture.

7.1.2. Calibration reading for ATEX type analyzers

Standard calibration
approved for ATEX types

The standard ATEX calibration is "1.15% Ethylene in Air = 50% LFL solvent vapors in air at 20°C." Then correct for response factor(s), calibration gas type and concentration, alternate LFL values (if any), and temperature correction (if any).

CAL RDNG

1.15% Ethylene in Air = **50%** LFL of standard solvent vapors at **20°C**

Adjusting ATEX / CENELEC type analyzers for other substances

To measure a particular flammable gas with maximum accuracy, divide the calibration reading by the response factor **R** from the table in section 7.1.6.

For mixtures of several solvents, where one has a much lower response factor than the others, $\mathbf{R_{min}}$, it is safest to divide the standard calibration reading by $\mathbf{R_{min}}$. The higher calibration reading will give readings for all gases in the mixture at or above their actual flammability.

To remove some of the "safe-side error," the response factors for "minor components" in the mixture may be ignored. A minor component is a flammable gas with a low response factor, that nevertheless is present in such small amounts in the mixture that the resulting error is insignificant. For example, a substance with **R** = 0.7 at a concentration below 10% LFL causes an error less than 3% LFL using the standard calibration.

If all gases being monitored have response factors greater than 1.0, it is permitted, but not necessary, to divide the calibration reading by $\mathbf{R_{min}}$, the lowest response factor of all the gases, to lower the calibration reading and remove the "safe side" error.

Alternate calibration gases and concentration for ATEX / CENELEC type analyzers

If calibration gas is not available in concentration C^0, a different concentration C_C may be used within an acceptable range. The calibration reading is multiplied by $(P * C_C) / C^0$.

For ethylene, C^0 is 1.15% by volume, C_C must be between 1.0% and 1.2%, and P =1.
For propane, C^0 is 0.85% by volume, C_C may be between 0.75% and 0.9%, and P = 0.93.

Alternate Lower Flammable Limit values

Each flammable gas has its own LFL value, the percent by volume concentration in air at which that particular gas becomes flammable. The response factors in section 7.1.6 use LFL values from several sources. If traceability to a particular code or authority is required, compare the LFL value in section 7.1.6 with the value published by the local authority. If the values are the same, the response factor remains valid. If the $\mathbf{LFL_A}$ value recognized by the local authority is different than the $\mathbf{LFL_R}$ value in the response factor table, multiply the response factor by the ratio ($\mathbf{LFL_A / LFL_R}$).

Temperature correction for ATEX type analyzers

To reflect the increase in flammability at higher temperatures, there are two methods.

Standard ATEX method regarding temperature correction:

The calibration reading is not corrected for temperature. The calibration means "%LFL at 20°C." The alarm levels are set to prevent operation above the maximum %LFL concentrations permitted by code for the process temperature.

Alternate ATEX method of temperature correction:
To correct the calibration reading for temperature, multiply CAL READING by [1/ (1-$\mathbf{K_T}$*($\mathbf{T_A}$ – $\mathbf{T^0}$))]. Once corrected, the analyzer readings will be accurate for the temperature $\mathbf{T_A}$.
$\mathbf{T_A}$ is the temperature of mixture to be monitored, in °C.
$\mathbf{T^0}$ is the temperature at which the LFL value was determined, in °C, usually 20°C.
$\mathbf{K_T}$ is a (gas-specific) factor for the temperature dependency, usually from 0,000784 to 0,00140. A typical value of $\mathbf{K_T}$ is 0,0014, which corresponds to 14% per 100°C. $\mathbf{K_T}$ should be obtained from the relevant standards or the local authority having jurisdiction.

7.1.3. Calibration reading formula for ATEX type analyzers

A formula for the calibration reading for ATEX type analyzers combines all the corrections described above:

CAL READING = 50%LFL * [C_C/C^0] * [1/R] * [P] * [LFL_R / LFL_A] * [1/ (1-K_T*(T_A – T^0))]

7.1.4. Calibration – Limits to Adjustment

SPAN TEMP SET - the expected span temperature
SPAN TEMP SET is the expected increase in the flame temperature during the span phase of calibration.

For standard analyzers, SPAN TEMP SET must be 100°C or more.

For air dilution type analyzers, SPAN TEMP SET must be 70°C or below.

An attempt to calibrate with an incorrect SPAN TEMP SET will cause the calibration to fail.

An attempt to calibrate when the flame temperature during the span phase of calibration deviates from SPAN TEMP SET by more than 20°C will cause the calibration to fail.

A calibration that succeeds, but with SPAN FLAME TEMP that deviates more than +/-10°C from SPAN TEMP SET, will cause a "Service Needed – Check Span" message.

Before adjusting SPAN FLAME TEMP, find and fix the cause of the deviation. Look for incorrect calibration gas concentration or pressure, incorrect alignment between the burner and thermocouple, accumulation of debris on the thermocouple, or a leak in the sampling system.

CAL READING is adjustable from 40%LFL to 80%LFL.

ZERO FLAME TEMP is valid between 460°C and 525°C. "Service Needed – Check Zero" activates below 465°C and above 520°C.

SPAN FLAME TEMP is valid from 37°C to 175°C and must be within 20°C of SPAN TEMP SET.

7.1.5. Calibration – Aborting Calibration

The calibration is self-timed and automatic. It typically lasts less than 3 minutes. If the response is unacceptable the calibration aborts automatically and the user then uses the "alarm reset" command to return to normal monitoring. If a hardware fault is detected the calibration aborts automatically. To manually abort the calibration, the user may: 1) wait for an unsuccessful calibration and perform an "alarm reset" command to return to normal measurement; 2) interrupt the calibration by cycling power so that the calibration is aborted and the analyzer returns to normal measurement after a few seconds; 3) force a flow fault by momentarily reducing air pressure so that the calibration aborts.

7.1.6. **Response Factor Table**

Response factors **R** marked (*) were determined by DMT - Deutsche Montan Technologie GmbH - using the **LFL_R** value given in the table and temperature basis T0 = 20°C and are the only substances within the scope of the EC-type examination certificate DMT_03_ATEX_G_001_X by DEKRA Testing and Certification GmbH. Substances marked with (*) are covered by certificate DMT_03_ATEX_G_001_X in the measuring range 0 to 80%LFL with exception of n-heptane and o-xylene, which are limited to 0-70% LEL. For Shellsol, there is also the limitation for SNR701.

All other response factors are by Control Instruments Corporation with temperature basis T0 = 25°C. Substances marked " >" are standard.

FP = °C Flash Point. Response factors **R** have less than 10% tolerance over all analyzer types, operating conditions or time based on successful calibrations.

LFL_R	R	Substance	CAS#	FP °C
4.00	1.00	acetaldehyde	75-07-0	-27
4.00	0.82	acetic acid	64-19-7	39
2.70	1.06	acetic anhydride	108-24-7	52
2.50	1.01	acetone (*)	67-64-1	-20
3.00	0.83	acetonitrile	75-05-8	2
2.30	0.95	acetylene	74-86-2	-17
2.40	0.80	Acrylic acid	79-10-7	49
16.00	1.74	ammonia	7664-41-7	-78
1.10	1.14	amyl acetate	628-63-7	23
1.20	0.90	amyl alcohol	71-41-0	33
1.20	0.94	benzol (*) / benzene	71-43-2	-11
1.30	0.94	benzyl alcohol	100-51-6	93
1.50	1.10	butadiene	106-99-0	-
1.80	1.01	butanone (*)	78-93-3	-7
1.40	1.00	n-butane	106-97-8	-60
1.65	0.97	2-Butanol	78-92-2	27
1.60	1.06	butene	25167-67-3	-
1.70	1.00	butyl acetate	123-86-4	22
1.50	0.88	n-butyl acrylate	141-32-2	29
1.70	1.00	n-butyl alcohol	71-36-3	37
1.10	0.86	butyl cellosolve	111-76-2	67
1.40	0.76	butyraldehyde	123-72-8	-12
1.40	1.00	y-butyrolactone	96-48-0	98
1.30	0.83	carbon disulfide	75-15-0	-30
12.50	1.22	carbon monoxide	630-08-0	-205
1.30	0.81	chlorobenzene	108-90-7	30
14.00	1.00	chloroform / trichloromethane	67-66-3	-
8.10	1.41	chloromethane / monochloromethane	74-87-3	0
0.80	0.96	Cumene	98-82-8	31
1.33	1.00	cyclohexane	110-82-7	-20
1.10	0.84	cyclohexanone	108-94-1	43
1.00	0.93	cyclohexene	110-83-8	-12
0.70	0.77	decalin	91-17-8	57
0.80	1.02	decane	124-18-5	46
0.70	0.91	1-decene	872-05-9	49
1.80	0.91	diacetone alcohol	123-42-2	56
2.20	1.17	1,2-dichlorobenzene	95-50-1	65
1.40	0.78	Diethyl carbonate	105-58-8	25
0.90	1.01	Diethylene glycol monobutyl ether	112-34-5	99
1.20	0.93	Diethylene glycol monoethyl ether	111-90-0	96
0.80	0.96	diisobutyl ketone / DIBK	108-83-8	49
1.00	0.60	(dimethylamino) acetonitrile	926-64-7	36
2.00	1.14	dimethyl acetamide	127-19-5	70
0.85	0.88	dimethyl adipate	627-93-0	107

LFL_R	R	Substance	CAS#	FP °C
4.20	1.15	Dimethyl carbonate	616-38-6	17
3.00	1.18	dimethyl ether	115-10-6	-41
2.20	0.95	dimethyl formamide	68-12-2	58
1.20	1.02	dimethyl glutarate	1119-40-0	103
1.35	0.96	dimethyl succinate	106-65-0	85
2.20	1.19	dimethyl sulfide	75-18-3	-
2.60	1.44	dimethyl sulfoxide	67-68-5	95
1.10	1.08	dipropylene glycol methyl ether acetate	88917-22-0	86
2.00	1.11	1,4-dioxane	123-91-1	12
2.30	0.88	1,3-dioxolane	646-06-6	2
0.60	1.00	dodecane	112-40-3	71
3.00	1.28	ethane	74-84-0	-135
3.30	1.08	ethanol / ethyl alcohol (*)	64-17-5	17
0.95	0.98	1-ethoxy -2-propyl acetate	54839-24-6	42
1.30	0.82	1-ethoxy-2-propanol	1569-024	42
1.70	1.14	2-ethoxyethylacetate	111-15-9	52
2.00	1.00	ethyl acetate (*)	141-78-6	-4
1.00	0.87	Ethyl acetoacetate	141-97-9	84
1.80	1.04	ethyl acrylate	140-88-5	9
1.00	1.08	ethyl benzene	100-41-4	27
3.70	0.93	ethyl chloroformate	541-41-3	16
1.90	1.10	ethyl ether	60-29-7	-45
1.05	0.92	ethyl-3-ethoxypropionate	763-69-9	58
0.88	0.90	2-ethyl hexanol	104-76-7	81
1.50	0.71	ethyl L-lactate	687-47-8	49
1.19	0.88	1-ethyl-2-pyrrolidinone (NEP)	2687-91-4	76
2.30	1.00	ethylene (*)	74-585-1	-135
6.20	1.16	ethylene dichloride	107-06-2	13
2.50	1.07	Ethylene glycol monomethyl ether	109-86-4	40
1.10	0.86	ethylene glycol monobutyl ether 2-butoxyethanol	111-76-2	67
1.75	0.90	ethylene glycol monoethyl ether ethyl cellosolve	110-80-5	40
1.40	0.93	Ethylene glycol mono isopropyl ether	109-59-1	45
3.00	1.06	Ethylene Oxide	75-21-8	-
0.60	0.94	Exxsol D95	64742-47-8	96
0.50	0.90	Exxsol D100	869062-45-3	104
n/a	0.90	formaldehyde and methanol aqueous	-	60
18.00	0.89	formic acid	64-18-6	69
2.30	1.37	furan	110-00-9	-35
1.80	0.96	furfural alcohol	98-00-0	75
1.00	1.02	heptane (*)	142-82-5	-4
1.10	1.00	hexane (*)	110-54-3	-22
1.20	1.11	1-hexene	592-41-6	-25
4.00	0.72	hydrogen	1333-74-0	-
4.30	1.02	hydrogen sulfide	7783-06-4	-
1.10	0.91	isoamyl acetate	123-92-2	25
1.80	1.14	isobutane	75-28-5	-88
1.80	1.08	isobutylene	115-11-7	-76
1.00	1.00	isooctane	540-84-1	-12
0.70	0.80	Isopar G	64742-48-9	40
0.70	0.81	Isopar H	90622-57-4	49
0.60	1.07	Isopar K	64742-48-9	50
0.70	0.90	Isopar L	64742-48-9	62
0.60	0.94	Isopar M	64742-47-8	96
0.80	0.94	isophorone	78-59-1	84
1.50	1.04	isoprene	78-79-5	-53
1.80	0.98	isopropyl acetate	108-21-4	2
2.20	0.99	Isopropanol, 2-propanol(*)	67-63-0	12
2.40	1.03	Isopropyl chlorine	75-29-6	-21
1.40	1.02	isopropyl ether	108-20-3	-18
0.88	0.97	mesitylene	108-67-8	53
4.40	1.20	methane	74-82-8	-188

LFL_R	R	Substance	CAS#	FP °C
6.00	1.06	methanol	67-56-1	12
1.30	0.88	1-methoxy-2-propylacetate	108-65-6	43
1.20	0.98	2-methoxyethylamine	109-85-3	12
3.10	1.04	Methyl acetate	79-20-9	-13
2.7	1.02	methyl acrylate	96-33-3	-3
1.00	1.10	methyl amyl ketone	110-43-0	39
1.20	0.82	2-methyl-2-butanol	75-85-4	21
8.10	1.41	methyl chloride / chloromethane	74-87-3	0
12.00	1.10	methylene chloride / dichloromethane	75-09-2	-
1.00	1.19	3-methylheptane, 95%	589-81-1	7
1.80	1.01	methyl ethyl ketone / MEK / 2-butanone (*)	78-93-3	-7
4.00	0.81	methyl formate	107-31-3	-19
1.20	0.98	methyl isobutyl ketone / MIBK	108-10-1	17
1.20	1.09	2-methylpentane	107-83-5	-12
2.30	0.98	methyl methacrylate	80-62-6	13
2.50	1.39	methyl proprionate	554-12-1	
1.60	0.90	methyl tert-butyl ether MTBE	1634-04-4	-40
0.80	0.94	mineral spirits RM-19	-	
0.80	0.92	mineral spirits Shell 140	-	60
0.90	1.00	mineral spirits Varsol	8052-41-3	40
4.90	1.11	monomethylamine	74-89-5	-18
0.90	1.07	naptha, vm&p regular	8032-32-4	20
2.60	1.04	nitropropane	79-46-9	33
0.80	1.00	Nonane	111-84-2	31
3.00	1.10	Novec 7300	132182-92-4	-
1.30	0.93	n-methyl pyrrolidone NMP	872-50-4	93
0.70	0.71	octene	25377-83-7	21
2.40	1.17	2,4-pentanedione	123-54-6	34
1.60	1.22	3-pentanone, 99%	96-22-0	13
1.30	0.87	Phenol	108-95-2	82

LFL_R	R	Substance	CAS#	FP °C
0.80	1.00	pinene	-	32
1.70	0.93	Propane (*)	74-98-6	-104
2.30	1.09	propionaldahyde	123-38-6	-40
2.10	0.92	Propionic acid	79-09-4	51
1.30	0.96	Proprionic anhydride	123-62-6	63
1.80	0.96	propyl acetate	109-60-4	14
2.20	0.99	propyl alcohol / propanol (*)	71-23-8	15
2.00	1.04	propylene	115-07-1	-108
2.40	0.93	propylene carbonate	108-32-7	135
2.20	0.97	propylene glycol	57-55-6	99
1.60	0.90	propylene glycol mono methyl ether / PGME	107-98-2	32
1.30	0.89	propylene glycol mono methyl ether acetate	108-65-6	43
0.90	0.89	Propylene glycol 1-phenyl ether	770-35-4	113
1.70	0.94	pyridine	110-86-1	17
0.60	0.93	Shellsol 60/95	64741-65-7	61
1.20	0.80	styrene	100-42-5	31
1.50	1.09	tert-butyl acetate	540-88-5	22
2.35	1.19	tert-butyl alcohol	75-65-0	4
0.43	0.95	tetradecane	629-59-4	100
2.00	0.94	tetrahydrofuran	109-99-9	-14
1.20	1.03	toluene / toluol (*)	108-88-3	4
7.00	2.09	trichloroethane	71-55-6	-
8.00	1.66	trichloroethylene	79-01-6	-
0.47	1.03	tridecane	629-50-5	79
1.20	1.00	triethylamine	121-44-8	16
1.40	0.79	trimethyl orthoformate	149-73-5	15
2.00	2.29	trimethylamine	75-50-3	-7
0.80	1.00	turpentine	8006-64-2	95
0.58	1.05	undecane	1120-21-4	61
3.00	1.01	vinyl acetate	108-05-4	-8
1.10	1.04	xylene / xylol (*)	95-47-6	27

7.2. Recalibration Procedure

To be performed by qualified persons only. Perform at routine intervals as part of normal maintenance at least once per month or more frequently if necessary. If changes to the types of flammable gases, the span calibration gas, the process temperature, or process controls have been made, the initial calibration procedure in section 7.1 should be repeated.

WARNING! During calibration, the analyzer alarms are bypassed. Calibration should only be performed when it cannot interfere with the safe operation of the process being monitored.

Verify normal operation. The calibration will not proceed if any fault except DOWNSCALE exists. If a fault occurs during calibration, the calibration process will terminate. Make sure the temperature is stable, with the insulating covers in place, for at least twenty minutes before starting. Accuracy is maintained over a range of temperatures spanning not more than 50°C (122°F). Therefore, where the analyzer is to be exposed to changing temperatures in excess of this range, the accuracy must be restored by recalibration.

Unlock calibrations and tests.

Record the ZERO FLAME TEMP, SPAN FLAME TEMP and CAL READING registers in the CALIBRATION MENU.

Open the Zero and Span Gas[24] supply valves. Verify that the pressure is 20 ±1 PSIG (1.4 Bar).

Use the FULL CALIBRATION command. **Then observe:**

- *The zero solenoid valve energizes. The* CAL *indicator and relay activate. Zero air enters the flame cell from the zero inlet and remains flowing for a minute or more[25].*
- *The zero phase ends. The reading before any adjustment is recorded in the* ZERO AS FOUND *register.*
- *If the* FLAME TEMP *is below 460°C or above 525°C, a zero fault trips and calibration terminates.*
- *If the flame temperature is acceptable,* ZERO FLAME TEMP *updates. The reading is 0%LFL for a few seconds.*
- *Zero gas stops, and span calibration gas enters the flame cell from the span inlet.*
- *The reading during the span phase should quickly come close to the* CAL READING *setting.*
- *The span phase ends after a minute or so. The* RESPONSE TIME *is automatically recorded.*
- *The reading before any adjustment is recorded in the* SPAN AS FOUND *register.*
- *At the end of the span phase, if the flame temperature is unacceptable, a span fail fault trips and the calibration process terminates.*
- *The* CAL SAMPLE FLOW, RESPONSE TIME, RDNG LAST CAL *and* SPAN FLAME TEMP *automatically update. The* ZERO ADJUSTMENT *is cleared to 0%LFL. The flammability reading displays the* CAL READING *for a few seconds, then the span valve turns off.*
- *The reading is allowed to recover for sixty[26] seconds.*
- *The calibration ends, the* CAL *indicator and relay deactivate, and the alarms are re-enabled.*
- *If the calibration has drifted, "Service Needed – Check Zero" or "Check Span" activate.*

Review the results. Investigate any significant changes

In the CALIBRATION menu: Check the zero and span "as found" that show the readings prior to adjustment. Verify that the response time remains suitable. Check that the RDNG LAST CAL is correct.

In the FLOW SYSTEM menu: If there is a large difference between CAL SAMPLE FLOW and SAMPLE FLOW, service the flow system and repeat the calibration.

Turn off the Zero Gas and Span Gas supply valves. Restore the lock to secure from unauthorized or unintentional calibrations.

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[24] Before using a new cylinder of calibration gas, its contents should be analyzed to ensure the correct gas type and concentration. Observe the first calibration with a new calibration cylinder carefully to verify the gas type and concentration.

[25] The timing of the zero and span solenoids is controlled by the CALIBRATION TIME register.

[26] The recovery time was increased from fifteen seconds as of July 2025 as V7.01.65

8. Operation

Operators must read and understand this instruction manual. They must understand that the device is for early warning of a potential fire or explosion. They must understand the meaning of the readings and alarms. Alarm relays should be hard wired to the safety shutdown system as part of the overall system design and in conformance to all applicable safety codes. They must not tamper with, or otherwise defeat the intended protection of the alarms and relays.

WARNING! Calibration, testing, maintenance and repair are beyond the scope of the operating instructions and are to be performed by qualified persons only.

Only the ACKNOWLEDGE and ALARM RESET controls should be accessible during operation. See "security" in the Installation Section.

8.1. Reading and status.

Operators should be alert to abnormal readings or changes in the alarm indicators. Familiarity with the normal range of readings under various process conditions will help operators detect abnormally high or low readings. Unusually low readings might indicate analyzer failure that must be corrected immediately. Unusually high readings might indicate the beginning of a serious problem with the atmosphere or process. All alarms, Service Needed messages, and abnormal readings should be recorded in a log and reported for correction.

WARNING! During alarms, the area near the analyzer may be explosive: keep a safe distance until the cause has been found and corrected.

Install remote indicators, and acknowledge and reset controls, in a safe location where operators can easily view the %LFL reading and the alarm status, preferably near a main control panel.

8.2. Acknowledge Command.

The acknowledge command deactivates the Service Needed message and relay. Before acknowledging, record the "Service Needed" message and investigate it to prevent future faults.

8.2.1. How to acknowledge the SERVICE NEEDED message

Use either method below:

- Press the MENU pushbutton until the word COMMANDS appears. Press SELECT. Press the MENU button until ACKNOWLEDGE appears. With ACKNOWLEDGE in the display, press the SELECT button. The display will flash for a few moments as the command executes. Wait for the flashing to stop.
- Or, shine a flashlight on the center phototransducer while covering the two outer phototransducers. Remove the light when the ACKNOWLEDGE command appears on the display. The display will flash for a few seconds, and then stop as the command executes.

Some "Service Needed" message are the result of one-time events - acknowledging them does not correct the problem. CHECK ZERO and CHECK SPAN update only during calibration and test. CHECK DATA means that all settings should be checked

8.3. Alarm Reset Command

Alarms typically latch. They stay active until a reset command is performed. The RESET command clears latched alarms if the condition that caused the alarm is no longer valid.

8.3.1. How to reset alarms

Before resetting, make a record of the alarm, record the PEAK HIGH and PEAK LOW readings, and investigate the cause of the alarm.

- Press the MENU pushbutton until COMMANDS appears in the display. Press SELECT. Press the MENU button until ALARM RESET appears. With ALARM RESET in the display, press the SELECT button. The display will flash for a few moments as the command executes. Wait for the flashing to stop.
- Or, shine a light on the center phototransducer while covering the two outer phototransducers. Remove the light when ALARM RESET command appears on the display. The display will flash for a few seconds, and then stop as the command executes.

If the condition that caused the alarm is no longer valid, the alarm indicator and relay return to normal.

8.3.2. Reset the danger alarm.

Danger alarms should only be reset after all corrective actions have been successfully completed. Flooding of the flame cell with explosive concentrations can extinguish the sensing flame. If reset of a Danger alarm results in a flameout fault, a flammable concentration might exist, especially if the sensing flame cannot be re-ignited.

8.3.3. Reset the flameout fault – Re-ignition.

A flameout may require up to three resets – to clears the latch, to re-ignite the sensing flame, and after a short delay, to clears the flameout fault after successful ignition.

WARNING! Flameout may indicate the presence of an explosive atmosphere.

9. Maintenance

In addition to the operator instructions and calibration, the following maintenance procedures are to be performed semi-annually or as experience indicates they are required.

9.1. General information and precautions

WARNING! The following general precautions, and all warnings noted in this manual, must be observed during all operation and maintenance procedures.

WARNING! Do not perform service when flammable gas might be present, or when servicing might interfere with the safety and operation of the process or atmosphere being monitored.

WARNING! The analyzer uses hydrogen or propane fuel. Observe all proper precautions for handling compressed gases and flammable gases, including:

1. Turn fuel supply off whenever the air pressure is off. If necessary, an external automatic fuel shutoff should be installed.
2. Eliminate all fuel fitting leaks. Use a leak-check (soap) solution, or a leak detector that is sensitive to the particular fuel in use.
3. The sensor enclosure is sealed for purging. Air pressure must be maintained whenever fuel is present to prevent an accumulation of flammable gas.
4. All covers must be kept tight whenever circuits are energized.
5. The fuel inlet fitting has a sintered metal restrictor that limits incoming fuel flow. Do not remove the restrictor. Replace with a new restrictor fitting if clogged or damaged.
6. Keep sampling system clean and leak-tight. Check for leaks after all maintenance work.
7. Non-incendive circuits are used to prevent ignition of fuel leaks, sample leaks, or of the external atmosphere. Do not modify the analyzer circuits, or add ignition-capable devices to the analyzer.

WARNING! The analyzer is electrically heated. Observe proper precautions for working with electricity and hot surfaces, including:

8. Prevent electric shock. Turn off electrical power before performing maintenance.
9. Prevent burns from contact with the heated flame cell components. Use insulated gloves. If necessary allow parts to cool just enough to allow maintenance.
10. Prevent overheating of the flame cell. Always keep the insulated flame cell cover in place during operation.

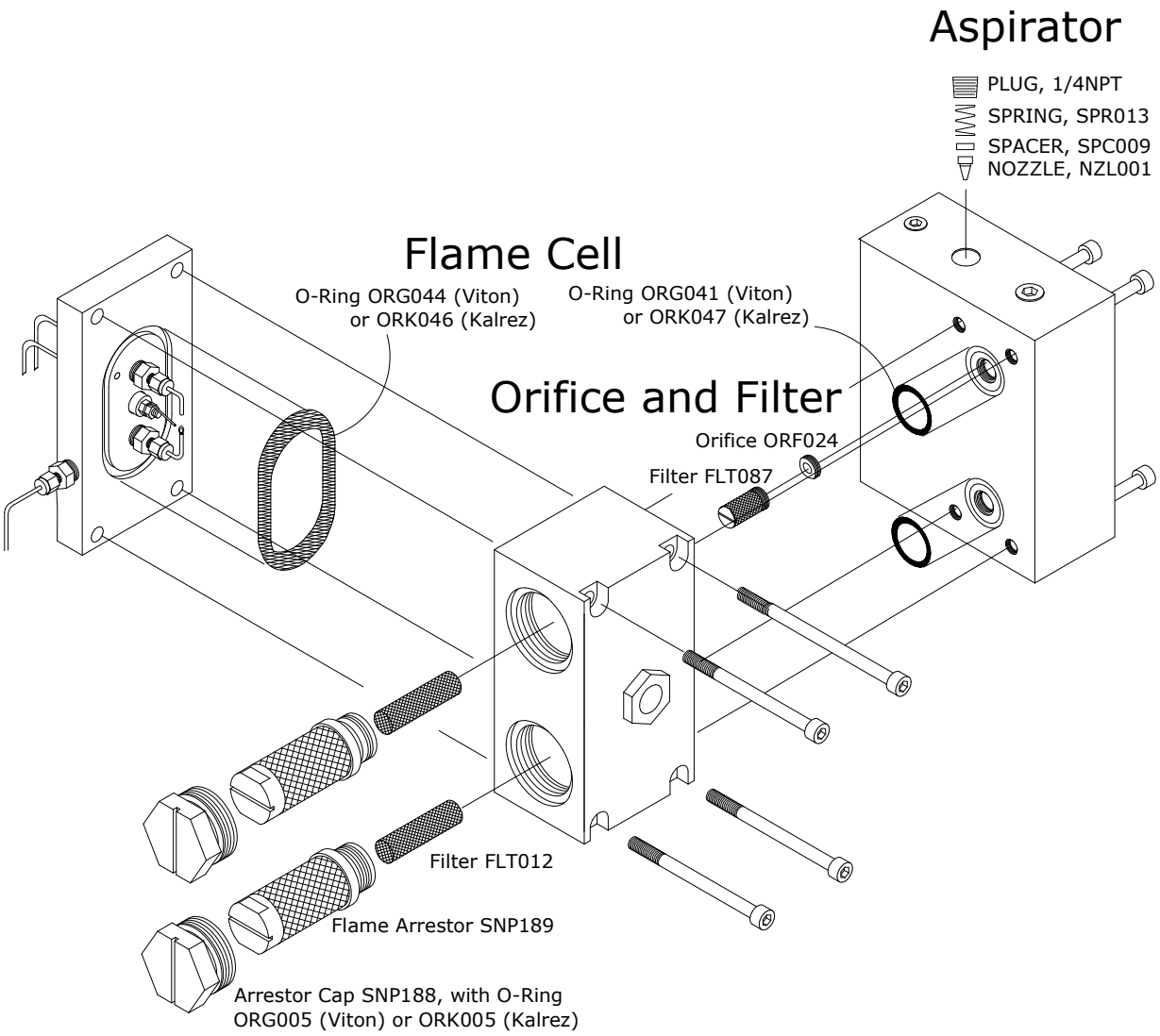
WARNING! The analyzer has a low-power high-voltage ignition transformer. When operating the controller with the ignition cable detached, attach a shorting clip from the ignition cable connection of the controller to the chassis ground.

WARNING! The sensor uses inlet and exhaust flame arrestors to protect the atmosphere being monitored. Observe precautions for hazardous locations, including:

11. Keep flame arrestors clean and installed at all times.
12. Flameproof joints and threads must be kept clean and free from any mechanical damage that can degrade the protection they provide.
13. Replace any components that show mechanical deformation, abrasion, or damage to the flameproof joints, threads or surfaces.
14. Test alarms regularly.

WARNING! Analyzers monitoring pressurized or hazardous atmospheres should not be disassembled until the atmosphere being monitored is safe and free of excess pressure, hazardous gases, or other substances or conditions that may be released during maintenance.

15. Make a record of analyzer maintenance.



Filters and Flame Arrestors

9.3. "Service Needed" Messages

When a condition exists that might lead to a fault, the SERVICE NEEDED relay activates. The ACKNOWLEDGE command deactivates the Service Needed relay and message, but does not fix the problem.

9.3.1. SERVICE NEEDED - CHECK FLOW – Check sample flow rate

The sample flow rate is continuously compared to the flow rate from the last successful calibration. If it deviates by more than the setting in the CHECK FLOW SET register, "Service Needed – Check Flow" activates.

1. At the earliest convenient time, troubleshoot the flow system. Isolate the problem to the air pressure, sample tubing, filters, O-rings, orifice, or aspirator. Clean and replace as needed.
2. Verify that the SAMPLE FLOW matches CAL SAMPLE FLOW within +/-CHECK FLOW SET.
3. After correcting the problem, use the ACKNOWLEDGE command to clear the message.

9.3.2. SERVICE NEEDED - CHECK RDNG – Check downscale reading

Whenever the reading drifts to within 3%LFL of the RDNG FAULT LEVEL, "Service Needed – Check Reading" activates. If the drift is not corrected, a DOWNSCALE FAULT occur at the RDNG FAULT LEVEL (typically -10%LFL).

1. Immediately verify that the fuel supply pressure is above the minimum and is not running out.
2. Check for leaks in the fuel system. If needed, tighten the fuel fitting at the side of the flame cell.
3. Record the FLAME TEMP from the GAUGES menu.
4. At the earliest convenient time, recalibrate the analyzer.
5. If the calibration fails, or if a CHECK ZERO message occurs after calibration, readjust the fuel regulator or flame thermocouple gain to get a 500°C FLAME TEMP.
6. If the fuel regulator has been adjusted, recalibrate the analyzer.
7. After correcting the problem, use the ACKNOWLEDGE command to clear the message.
8. Review the specifications for the influence of carbon dioxide and water vapor.

9.3.3. SERVICE NEEDED - CHK ZERO - Check zero calibration

If the flame temperature during zero calibration is less than 465 °C, or greater than 520 °C, the CHECK ZERO message is given and the SERVICE NEEDED relay is activated. Although the calibration is acceptable, the zero has drifted from the ideal value, and eventually might need readjustment of the fuel regulator pressure. If the drift continues, the flame temperature may go below 425°C and trip a flameout fault. Attempting to zero calibrate with the flame temperature below 460°C or above 525°C will give a ZERO FAIL fault.

1. Immediately verify the fuel supply pressure is above the minimum and the fuel supply is not running out.
2. Immediately verify that the air pressure to the zero calibration inlet is correct.
3. Record the ZERO FLAME TEMP from the CALIBRATION MENU.
4. At the earliest convenient time, ACKNOWLEDGE the message, and adjust the fuel regulator or flame thermocouple gain for a FLAME temperature of 500°C.
    a. Best practice is to adjust the fuel regulator first and then make adjustments to the thermocouple gain should fuel regulator adjusts fail to fix the issue
5. Recalibrate the sensor and verify that the ZERO FLAME TEMP is near 500°C.

The expected rate of drift in the ZERO FLAME TEMP is less than 25°C per year. Higher rates may indicate a problem with the fuel regulator, or the fuel quality. Propane fuel exposed to large temperature changes can cause reading drift.

9.3.4. SERVICE NEEDED - CHK SPAN Check span calibration

During span calibration, the rise in flame temperature above zero is tested against the expected temperature rise in the register SPAN TEMP SET. A temperature rise within ±20°C of SPAN TEMP SET is a successful calibration. If not within ±10°C, the calibration is successful, but a check span activates. At more than +/-20°C a span fail fault trips and check span activates.

The expected temperature rise for the standard test gas, 1.15% by volume Ethylene in Air, is approximately 125°C (60°C in air dilution types). A lower block temperature gives a higher span. An aging thermocouple results in a lower span. Leaks in the sampling system can produce a low span.

1. Record the SPAN FLAME TEMP temperature.
2. Immediately verify the concentration and type of the calibration span gas.
3. Verify that the span gas pressure at the analyzer inlet is 20 PSIG (1.4bar).
4. Check that the flame thermocouple and burner are clean.
5. Service the flow system to check for leaks.
6. ACKNOWLEDGE to clear the Service Needed message.
7. Recalibrate.
8. The "Service Needed – Check Span" should not occur.

9.3.5. SERVICE NEEDED - CHK DATA Check for data loss

Settings are protected from attempts to change them outside of their acceptable limits by rejecting the change and activating "Service Needed – Check Data."

Settings are continually tested to make sure that they are not corrupted. If the controller's self-diagnostics detects an unexpected change to a setting, the setting is automatically restored, and the Service Needed – Check Data activates.

Review that all settings are correct, then use the ACKNOWLEDGE command to clear the Service Needed message and relay, then recalibrate the analyzer

If this message is given repeatedly, investigate the analyzer power supply for power disturbances. Verify that the ignition spark discharges to the tip of the burner, and that the gap between the ignition electrode and burner tip is correct. If the problem remains, it is likely the controller has a fault that requires replacement.

If the settings cannot be reliably restored, a Data Lost fault trips and the controller enters into the Cold Start state.

9.3.6. SERVICE NEEDED - CHK TC3 Check for Cutoff Thermocouple TC3

700 series analyzers feature a type-k thermocouple known as TC3 or the "cutoff" thermocouple.[27]

The function for TC3 is as a secondary monitor of the block temperature. TC3s circuit is programmed to shut off all A/C power to the heater circuit should it reach 15 degrees celsius above the HSG TEMP set point. TC3 is a redundant check of TC2 making sure that there is no runaway heat going to the block which could damage the analyzer.

A fault in either the TC3 thermocouple or in the circuit for TC3 will result in a *CHK TC3* service needed message. This will NOT result in a fault meaning that the analyzer is able to perform all of its functions while in a CHK TC3 state.

[27] TC3 replaces the overtemperature thermostat seen in 670 series PrevEx

When a CHK TC3 occurs, first ensure that all connections involving TC3 are secure and that there are no open or loose connections. If everything has been checked and secured, a power cycle may get rid of the CHK alarm.

If power cycling does not solve the CHK TC3 contact Control Instruments.

9.3.7. SERVICE NEEDED - CHK RELAYS Check for monitoring of safety relays.

The 700 series PrevEx feature four safety relays, three of which (Warning, Danger, and Fault) are made using dual form A contact relays, while Cal is a single form A contact relay. A self-monitoring diagnostic check for the individual safety relays was added to ensure that the unit is never in a state where it's supposed to inform customers of upset conditions but doesn't. Each of the four safety relays (Warning, Danger, Fault, and Cal) are normally energized and de-energize when tripped.

Should the analyzer ever be in a state where the Warning, Danger, or Cal relay should be tripped (de-energized) but isn't, the Fault relay will trip letting the end user know that the analyzer is currently in a non-safe state. If the Fault relay should be tripped (de-energized) but is not, then the Cal relay will trip signaling an error with the fault relay causing the unit to be in a non-safe state. Should any of the Warning, Danger, Fault, or Cal Relays de-energize when they should be energized, then the "CHECK RELAYS" service needed message will appear and the Check relay will trip.

The cause for "CHECK RELAYS" can vary, but once you do receive this message it is recommended that you replace the controller at your earliest convenience.

9.3.8. SERVICE NEEDED - CHK 4-20mA Check for monitoring of both 4-20mA output chips.

The 700 series PrevEx features a self-diagnostic check for the dual 4-20mA circuit. A "Check 4-20mA" service needed message indicates an issue within either one or both circuits controlling the 4-20mA outputs. Should the "CHECK" message not clear with the addition of a 330-ohm terminating resistor in place on the unused terminal positions for the 4-20mA output, consult Control Instruments.

9.3.9. SERVICE NEEDED - CHK Dilution, Fuel, Enclosure Check for monitoring of fuel pressure, enclosure pressure, and dilution pressure.

These check messages relate to additional pressure transducers that may be introduced in the future as add-ons for the 700 series (as of now they are unavailable options). Should a check ever appear for any of these go to settings and make sure that all of "DILUTION MONITOR", "FUEL MONITOR", and "PURGE MONITOR" are all set to "Disabled". Should any of these settings be "enabled" a "COLD START" will be required to change the setting to "Disabled".

Scheduled Maintenance

There is a wide variety of applications, industry codes, and operating conditions. Base the maintenance schedule on both design and experience. The initial maintenance schedule shown below should be adapted as needed.

Procedure	Section	Frequency
Readings	9.5	Daily
Utilities	9.6	Weekly or monthy
Gauges	9.7	Monthly or quarterly

Recalibration	9.8	Monthly or more frequently if necessary.
Leak test	9.9	After any disassembly. Otherwise quarterly or yearly.
Alarm test	9.10	Yearly or more often if necessary.
Display Test	5.2	Annually, and before any change to settings.
System review	9.11	Semi-annual or when process or monitoring conditions change.
O-Rings	8.12	As needed (see section 8.12)
Purge restrictor	8.13	Annual
Flow system	8.14	Quarterly or yearly For CHECK FLOW, or LOW FLOW status
Cleaning	9.15	As needed for CHECK FLOW, or LOW FLOW status following flow system maintenance
Fuel regulator	9.16	Yearly, or for CHECK ZERO or ZERO FAIL status *More frequent adjustment should not be required.*

9.5. Readings

Readings should be checked frequently. Operators should have access to a display of the reading near the control panel or operator station used to control or monitor the process or atmosphere being monitored. Abnormally low readings are as important as abnormally high readings, and should be investigated. They might indicate an analyzer leak, or another fault that the analyzer itself is unable to detect.

The PEAK HI and PEAK LOW %LFL readings help with understanding the normal range of readings for a particular process or set of conditions. This can help identify a problem well before it might become a hazard, and can help optimize alarm levels. The lowest warning and danger alarm levels that do not produce false alarms are the best. Information gathered over time can be used during the annual review to help choose the best alarm levels.

9.6. Utilities

Keep all utilities, power, air and fuel, on continuously, so that the analyzer gives the best performance and requires the least maintenance. On regular intervals, verify the air pressure. The air supply likely has a filter that might require cleaning or replacement. If it is necessary to adjust the air pressure, check the air supply filter. Check the fuel pressure, and the quantity of fuel remaining in the supply, often enough that the fuel cannot lose pressure or unexpectedly run out. Before and after calibration, check the supply of calibration gas and reorder if necessary.

9.7. Gauges

Make an analyzer maintenance record at regular intervals, before and after calibration or maintenance. Investigate significant changes, especially in FLOW, ZERO FLAME TEMP and SPAN FLAME TEMP.

The SAMPLE FLOW rate on the GAUGES menu should not change by more than a few tenths of a liter per minute per week. The sample flow rate observed during startup, when the analyzer was first put into service in clean condition, is a good reference point. Reduced flow rates indicate need for filter changes. Inability to restore flow near its startup value indicates the need for cleaning.

The BLOCK TEMP should be stable to within +/- 3°C of the SET TEMP value.

The flame temperature from calibration, measured as ZERO FLAME TEMP and SPAN FLAME TEMP, should also be stable to within several °C before and after calibration. The CHECK ZERO and CHECK SPAN messages help identify significant changes.

9.8. Recalibration

Calibrations and tests using known gas concentrations should be performed regularly. This is the best way to verify proper operation. Operating history will determine frequency of calibrations. Zero and Span tests can be performed to verify correct response and help determine the most appropriate calibration schedule. Some codes, for example NFPA-86 Ovens and Dryers, require a monthly calibration to verify proper sensor response.

9.9. Leak test

The analyzer can not detect some types of leaks in the sampling system that might dilute the sample and cause false low readings. A leak test should be performed regularly. It should also be performed after maintenance involving disassembly of any part of the sampling system.

1. Use the GAUGES menu to display FLOW.
2. Remove the sample probe fitting at the analyzer inlet.
3. Cap off or otherwise block the analyzer sample inlet.
4. For air dilution types, turn off pressure and block flow to the dilution air inlet B.
5. Verify that the FLOW drops below 0.01 LPM.

9.10. Alarm test

The alarm relays are used for corrective action. Immediate and effective corrective action is needed to prevent the possibility of a fire or explosion. At regular intervals, perform a complete test of the relays and the corrective action.

1. Verify that WARNING and DANGER levels in ALARMS menu are correct.
2. Use RLY TEST in OUTPUTS menu to activate relays.
3. Verify that actions of each relay produce the desired overall result.
4. Verify that the **D**ANGER relay activates the safety shutdown.
5. Verify that the **F**AULT relay activates the safety shutdown.

9.11. System review

Semi-annually, or when process or monitoring conditions are changed, review the initial calibration information to ensure that the calibration remains valid, check maintenance records for potential problems and adjust the maintenance schedule as needed, and refresh training and education of operators and maintenance personnel. Some codes require an annual review by a factory-trained technician. Leak test failures and in some cases calibration failures or calibration check alarms may be an indication of an O-Ring failure.

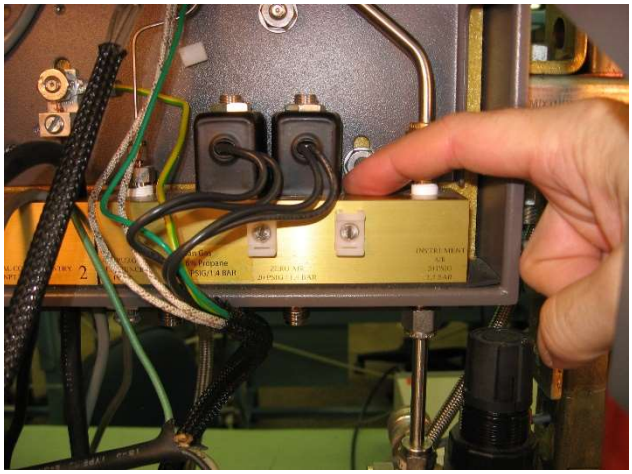
9.12. O-Rings

The table below indicates the expected life of the O-Rings by operating temperature.

BLOCK TEMP	Flame arrestor end caps O-rings	Interblock O-rings
60°C	ORG005 – 5 years	ORG041, ORG044 – 5 years
120°C	ORG005 – 5 years	ORG041, ORG044 – 5 years
200°C	ORG005 – 1 year	ORG041, ORG044 – 1 year
250°C	ORK005 – 6 month	ORK046, ORK047 – 1 year

9.13. Purge Restrictor

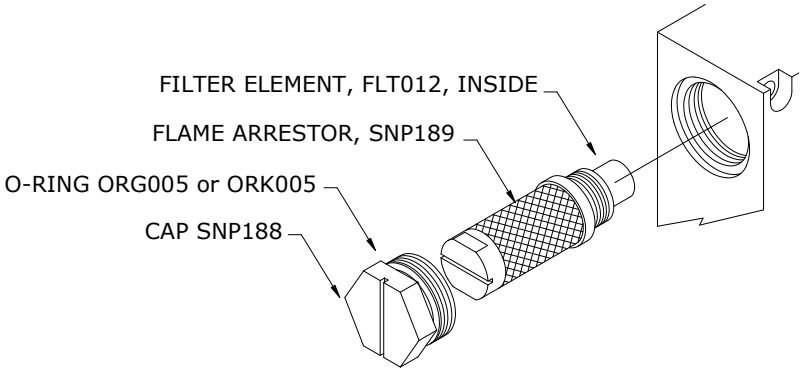
Over time, the air purge restrictor on the manifold can be clogged with oil, water or debris, reducing the purge flow that removes any leaking fuel from the enclosure. If purge flow cannot be detected, the restrictor should be replaced. Service kit SRV153 has parts and instructions.



9.14. Flow System Preventative Maintenance

Routine replacement of the disposable filter elements, and where needed, cleaning of flame arrestors, prevents loss of sample flow. Replacement of O-rings helps prevent leaks.

SNR701, SNR702 and SNR704 use Viton O-rings. These have good temperature and solvent resistance, and are inexpensive. They can be used up to 200°C. They should be replaced whenever they are disassembled during a service procedure. O-rings in SNR705 are Kalrez®. These are required for flame cell temperatures above 200°C. They are expensive, but have superior heat and solvent resistance, and so can be re-used if they are not damaged, deformed or shrunken. If the O-ring is elastic, has a uniform cross-section, has no abrasion, breaks or cuts, and no signs of shrinking or hardening, it can be re-used. When in doubt, replace the O-Ring.



1. Remove outer cover.
2. Remove inner cover (except SNR701 types). Cover is retained by four mounting pins that fit into keyhole-shaped slots on the cover. Lift cover straight up ¼ inch, until all four mounting pins line up with slot openings, then slowly pull the cover straight out from flame cell. Do not force the cover, or the mounting pins may be damaged.
3. Unscrew the flame arrestor caps located on the left side of the flame cell block.
4. Unscrew the flame arrestors from the flame cell.
5. Remove filter elements from inside the flame arrestor.
6. Clean the flame arrestor with compressed air. If necessary, soak the flame arrestor in solvent and dry completely before reinstalling. Do not clean mechanically, such as by sanding or scraping the surface. Abrasive cleaning will cause damage.
7. Insert replacement filter element (Part Number FLT012) into flame arrestors.
8. Screw flame arrestor clockwise into flame cell until hand-tight, then back out counterclockwise 1/8 of a turn. ***Do not over tighten.***
9. Replace the O-Rings on the flame arrestor caps. (Part Number ORG005, except for SNR705, which uses ORK005 Kalrez O-rings to operate above 200°C).
10. Screw flame arrestor caps into flame cell until O-Rings seal and cap is hand-tight.
    Do not over tighten: The flame arrestor cap is sealed by the O-Ring. When properly installed, there is a 1/32" (0.8 mm) gap between the arrestor cap and the flame cell surface. About half the O-ring width remains visible. Do not attempt to screw the arrestor cap completely into the flame cell until it bottoms out, because this can damage the threads.
11. Perform a leak test. Plug the sample inlet. Verify FLOW on GAUGES menu below 0.01 LPM.
12. After temperature stabilizes, perform a full calibration.

9.15. Cleaning

Purpose — To restore sample flow by removing contamination on the aspirator or sample flow orifice. This increases aspirator suction and decreases flow resistance.

Frequency — Cleaning is required if the Flow System Preventative Maintenance does not restore the sample flow and the sample and exhaust tubing are not clogged.

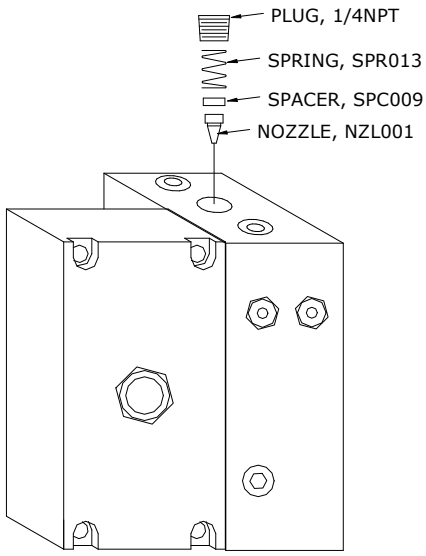
Clean the flame cell if an accumulation forms on the burner, the thermocouple, or the flame arrestors, causing a CHECK ZERO or CHECK SPAN message during calibration.

If a flow problem is not caused by a lack of air pressure, or a clog in the sample or exhaust tubing, the filters, or the flame arrestors, then the problem might be due to a lack of suction from the aspirator or a clog in the orifice. Over time, particles or condensate can accumulate. Some substances in the sample, such as silicones, may form dust when burned in the sensing flame. This dust can accumulate on the burner, thermocouple, or upper flame arrestor. The dust is removed to restore the sample flow, or to restore the ZERO FLAME TEMP and SPAN FLAME TEMP to normal values.

9.15.1. Aspirator cleaning or replacement

Parts Needed — Aspirator nozzle - if needed. Part number NZL001
Tool needed — 3/16 inch Allen key. Needle nose pliers.

1. Turn all utilities off.
2. Using a hex key, remove the ¼ NPT plug from the top of the heated block assembly. This is the middle plug on the top right hand side of the flame cell.
3. Remove the spring and spacer.
4. Use needle-nose pliers to remove the aspirator nozzle. Be careful not to damage the inside of the nozzle. If the nozzle is difficult to remove, you can also remove the ¼ NPT plug from the bottom of the block (directly below the aspirator) and use a soft wooden stick to push the nozzle from its seat.
5. Clean the bore and nozzle with a solvent such as acetone or alcohol, and let dry.
6. Replace nozzle with new if contamination cannot be removed.
7. Re-install the nozzle, spacer, and spring. Replace the ¼ NPT plug(s).
8. Restore utilities.
9. Re-ignite the analyzer
10. Allow analyzer time to stabilize.
11. Perform a full calibration.



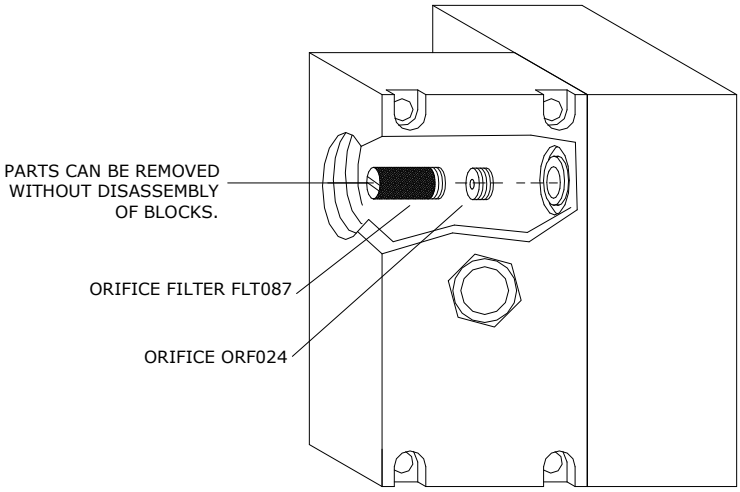
Oil or residue on the inside of the nozzle may indicate oil in the compressed air supply. Check the compressed air supply filters and replace if necessary.

9.15.2. Orifice cleaning or replacement

Parts Needed Orifice filter - if needed. Part number FLT087.
Orifice - if needed. Part number ORF024.
O-rings - if needed, quantity 1, part number ORG005 (Viton) or ORK005 (Kalrez)

Tools needed Screwdriver
Locking screw driver- 10” quick wedge screwdriver, part number TOL044
1/8 inch Allen key (Part number TOL046 or equal)

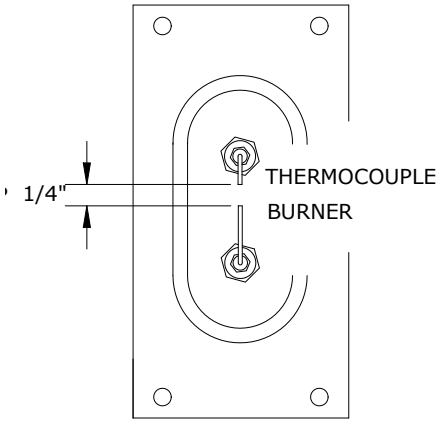
1. Verify the compressed air pressure. Insufficient air pressure causes reduced flow that will not be corrected by cleaning.
2. Turn all utilities off.
3. Remove upper flame arrestor cap.
4. Remove upper flame arrestor.
5. Use locking screwdriver to remove the orifice filter through the upper flame arrestor block opening.
6. Use Allen key to remove orifice.
7. Clean the orifice and orifice filter with a solvent such as acetone or alcohol, and let dry. Do not drill out.
8. Replace orifice or orifice filter with new if contamination cannot be removed.
9. Re-install the orifice.
10. Re-install the orifice filter.



CUTAWAY VIEW OF ORIFICE AND FILTER INSIDE FLAME CELL.
ACCESS IS THROUGH UPPER FLAME ARRESTOR CAP.

13. Re-install filter and flame arrestor. Screw flame arrestor clockwise into flame cell until hand-tight, then back out counterclockwise 1/8 of a turn. ***Do not over tighten.***
14. Check flame arrestor cap O-Ring and replace if needed.
11. Re-install flame arrestor cap and O-Ring. ***Do not over tighten:*** The flame arrestor cap is sealed by the O-Ring. When properly installed, there is a 1/32” (0.8 mm) gap between the arrestor cap and the flame cell surface. About half the O-ring width remains visible. Do not screw the arrestor cap completely into the flame cell until it bottoms out, it will damage the threads.
12. Restore utilities.
13. Re-ignite the analyzer.
14. Allow analyzer time to stabilize.
15. Perform leak check of sampling system.
16. Perform a full calibration.

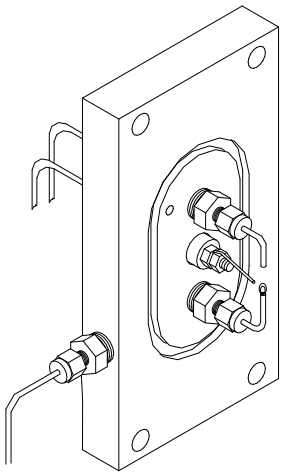
9.15.3. Flame cell cleaning



Parts Needed
- Orifice filter - if needed. Part number FLT087.
- Orifice - if needed. Part number ORF024.
- O-rings - if needed:
  - Qty 1 of ORG044 (Viton) or ORK046 (Kalrez)
  - Qty 2 of ORG041 (Viton) or ORK047 (Kalrez)
  - Qty 2 of ORG005 (Viton) or ORK005 (Kalrez)

Tools needed
- Screwdriver
- Locking screw driver- 10” quick wedge screwdriver, part number TOL044
- 1/8 inch Allen key, part number TOL046 or equal.
- 5mm Allen key for flame cell bolts.
- Alignment tool, part number TOL003 or TOL047.

1. Turn all utilities off.
2. Remove sample probe.
3. Remove flame arrestor caps, and remove flame arrestors.
4. Use locking screwdriver to remove the orifice filter.
5. Remove eight flame cell bolts. For best results, loosen bolts while blocks are warm.
6. Lift flame cell block straight out and away from burner plate, taking care not to bend the thermocouple or burner.
7. Clean flame cell parts as needed.
8. ***Align the thermocouple, burner and spark electrode.*** The thermocouple should be directly above the burner, with ¼ inch gap (1/2 inch for air dilution units). The spark electrode is located 1/10 inch (2.5mm) behind, and just above, the rear of the burner. Use the alignment tool, part number TOL003 or TOL047. Avoid excessive bending of the thermocouple, which can cause damage.
9. Replace flame cell O-Rings with new, if needed.
10. Re-install flame cell block, but **do not** bend the thermocouple or burner.
11. Re-install and tighten flame cell bolts.
12. Re-install the orifice filter.
13. Re-install filters and flame arrestors. Screw flame arrestor clockwise into flame cell until hand-tight, then back out counterclockwise 1/8 of a turn. ***Do not over tighten.***
14. Check flame arrestor cap O-Ring and replace if needed.
15. Re-install flame arrestor cap and O-Ring. ***Do not over tighten:*** The flame arrestor cap is sealed by the O-Ring. Properly installed, there is a 1/32” (0.8 mm) gap between the arrestor cap and the flame cell surface. Half the O-ring width remains visible. Do not attempt to screw the arrestor cap completely into the flame cell until it bottoms out, because this can damage the threads.
16. Restore utilities.
17. Re-ignite the analyzer
18. Allow analyzer time to stabilize.
19. Perform leak check of sampling system.
20. Perform a full calibration.





9.16. Fuel Regulator Adjustment

Purpose — Adjust sensing flame until FLAME TEMP on the GAUGES menu is 500°C when no flammable gases are present in the sample.

Frequency — Every twelve months
Following a CHECK ZERO Service Needed Message, when ZERO FLAME TEMP is below 485°C or above 512°C
Following a zero calibration fault, when the sensing flame is lit, but the flame temperature is below 460°C or above 525°C during a zero test.

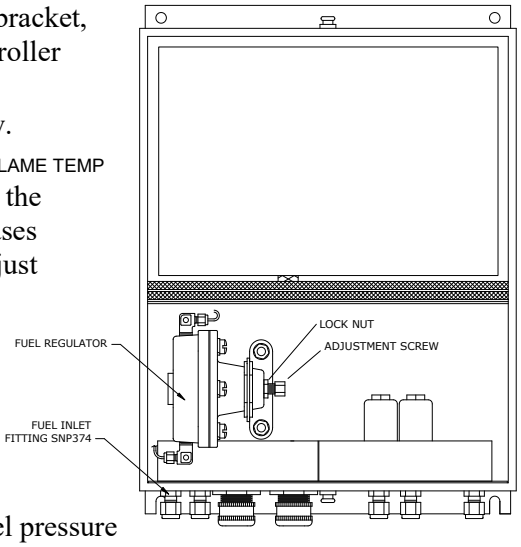
Parts Needed — Fuel inlet fitting, part number SNP374 - if needed.

The fuel regulator may exhibit a slow drift from time or temperature. Routine calibration will remove the error from this drift, but eventually it can accumulate, and a "Service Needed – Check Zero," or even a ZERO FAIL fault, might occur. Readjustment of the fuel regulator pressure can restore the sensing flame to its optimal size.

The fuel pressure should be above the minimum required pressure at the inlet to the analyzer.

There is a small sintered metal filter installed inside the fuel inlet fitting. This filter protects the fuel system from contamination. It should not normally require replacement if the fuel delivery system and fuel tubing is clean and free of contamination. In some cases, a contaminated fuel restrictor must be replaced in order to pass enough fuel through to the analyzer's sensing flame.

1. Verify that the flame cell temperature is stable. Variations in flame cell temperature can cause drift in the flame temperature that will not be corrected by fuel regulator adjustment.
2. Verify that the fuel pressure at the inlet is correct. Incorrect fuel pressure can cause drift in the flame temperature that will not be corrected by fuel regulator adjustment.
3. Check fuel purity. Impure fuels, or impurities in fuels that respond to temperature changes, will not be corrected by fuel regulator adjustment.
4. Remove the outer enclosure cover. (Leave the insulated flame cell cover in place).
5. Lift the controller, slide the upper retaining pins from the bracket, leave the lower retaining pins in place, and swing the controller down to give access to the fuel regulator.
6. Loosen the lock nut on the fuel regulator adjustment screw.
7. Slowly turn the fuel regulator adjustment screw until the FLAME TEMP is 500°C ± 5°C. Turn the adjustment screw clockwise into the regulator to increase, counterclockwise to decrease. Increases occur quickly, but decreases occur slowly. Do not over adjust when decreasing.
8. Tighten the lock nut on the fuel regulator adjustment screw.
9. Re-install outer cover.
10. Wait for the reading to stabilize.
11. Perform a full calibration.
12. Test the regulation of the fuel system by increasing the fuel pressure at the analyzer inlet by a few PSIG. The reading should not change. When finished, set fuel pressure to correct setting.



10. Troubleshooting

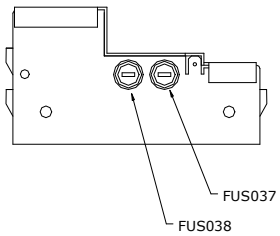
These requirements for proper operation are shown in their order of importance.

10.1. AC Power

All power and voltages for the analyzer's operation is derived from the AC power mains. Power must be supplied continuously within +10%/-15% of the rated voltage.

10.1.1. Fuse replacement

Parts Needed 5 Amp, Part number FUS037, for power to controller and heater.
½ Amp, Part number FUS038, for controller power.



1. Disconnect electrical power at the power switch or circuit breaker.
2. Remove the outer cover.
3. The controller is mounted by four pins that fit into slots on a bracket. Slide the controller up ¼ inch. Swing the top outwards so only the top pins come out of their slots. The lower pins must remain in their slots.
4. The fuses are on the right side of the controller. Use a screwdriver to turn the fuse holder cover counterclockwise 1/8th of a turn.
5. Remove and replace fuse with same rating and type as shown on label or fuse.
6. Re-mount the controller and replace the outer cover.

10.2. Flame Cell Temperature

The flame cell is heated by an AC-powered electric heater. Thermocouple TC2 in the flame cell block measures the BLOCK TEMP. Heating is controlled by BLOCK TEMP SET. A deviation more than a few degrees results in a fault. At startup, or after a change to the setting, a BLOCK TEMP LOW or HIGH fault exists until the flame cell reaches the new temperature. The time can vary from 15 minutes at 60°C, to as much as 90 minutes at 250°C.

The sensing flame requires a stable flame cell temperature. Temperature changes cause drift. Low temperatures can also cause condensation, leading to false low readings, or clogging. Always keep the flame cell heated and insulated with its covers in place. If power is to be turned off for some time, the fuel and air pressure should be turned off. In some cases the sample and exhaust tubing must be capped as well.

The TC2 and TC3 thermocouples are 0 millivolts at 0°C, and increases about +1 millivolt for every 25°C.

10.3. Air Inlet Pressure

Compressed air is used for suction to draw the sample into the flame cell. It also purges the enclosure to prevent the accumulation of leaking fuel or flammable gases. The compressed air pressure must be regulated to 20 PSIG (1.4 bar). Low or high air pressure causes flow faults.

10.4. Fuel Inlet Pressure

The sensing flame uses hydrogen or propane fuel. Low fuel pressure causes downscale reading drift.

External fuel line leaks not only waste fuel they can cause low fuel line pressure and analyzer reading drift sufficient to cause a downscale fault to occur. Test for leaks by shutting off the fuel at the tanks and watch the line pressure. Quick drops in pressure indicate fuel line leaks. A leak tight system will hold pressure for minutes without a significant drop in pressure. Use a soap solution at all fittings to locate possible leaks.

Propane fuel should not be stored outdoors, exposed to large temperature changes. Low temperatures cause low propane pressure. Large temperature changes increase the effect of impurities.

The fuel inlet fitting has a sintered metal filter inside. It prevents contamination from reaching the analyzer. It also limits the fuel flow that can leak into the enclosure. A contaminated restrictor drops the fuel pressure and causes low readings and reading drift. Tubing or fitting leaks in the analyzer enclosure can cause the pressure to drop, resulting in low readings. Check internal fitting for leaks especially the fuel fitting at the side of the flame cell.

Test the fuel restrictor and regulator. Attach a pressure gauge to the fuel inlet. Increase pressure by +5 PSIG and verify that the reading and flame temperature remain constant. Reading changes more than 2% means the fuel system is not regulating properly, and the analyzer readings will drift. Check for fuel leaks in the analyzer enclosure. Check or replace the fuel inlet fitting.

10.5. Spark

The sensing flame is ignited by a two-second spark discharge from a small electrode mounted to the spark plug in the flame cell to the grounded burner tube. The spark is generated by a high voltage (30 kV) transformer in the controller.

The spark electrode must be aligned about $1/10^{th}$ inch (2.5mm) behind, and slightly above, the rear of the burner tip. If the spark discharge strikes too low on the burner, the fuel will not ignite. If the spark discharge is too close or too far from the burner it can cause the controller to reset itself.

10.6. Flame

Flammable gases in the sample are burned in the sensing flame. This results in a temperature rise in the TC1 thermocouple located above the sensing flame. The sensing flame must be stable. The fuel supply pressure must be constant. The fuel, especially when propane is used, must be pure.

The TC1 thermocouple makes the flame temperature measurement. It must be properly aligned above the sensing flame, at the correct distance. Use care when aligning the thermocouple. Excessive bending causes damage.

A propane sensing flame appears as a small round blue flame about the size of a match head. It may have a small yellow tip at the top. Hydrogen flames are not visible to the human eye. Solvents and other flammables burning in a hydrogen flame cause a faint blue glow. Explosive concentrations in the flame cell can emit an orange or yellow burst of flame. Sometimes the TC1 thermocouple has a faint red glow when the flammability reading is high. A bright red thermocouple indicates very high flammability.

10.7. Sample Flow

The sample flow rate must be maintained to ensure that measurement is accurate and timely. If the flow rate decreases the measurement will become slower, especially if the sample tubing is long.

The sample orifice is used to measure the SAMPLE FLOW. It must be kept clean so that the measurement is accurate. In some cases, the orifice will require periodic cleaning. A sintered metal filter must be removed using a flat head holding screwdriver to get access to the sample orifice. The sample orifice can then be removed and cleaned.

A partially or completely clogged orifice may trigger a SAMPLE FLOW HIGH fault message. A partially clogged orifice may result in slow response to sample and calibration gas, flameout faults, inability to ignite or stay lit after an ignition attempt, or span calibration failure.

10.8. Calibration Flow

Zero gas must be clean, dry, instrument-grade air from a cylinder or properly filtered compressor. Oil in unfiltered compressed air can be vaporized in a high temperature analyzer, burn in the sensing flame, and cause a false zero reading.

Zero and span calibration gases are injected under 20 PSIG (1.4 bar) pressure through sintered metal restrictors located inside the inlet fittings. Each restrictor must be kept clean and in place. If the calibration flow is too low or too high, errors will occur. An excess of calibration flow must be free to flow out of the sample tubing.

Fuel pressure must be steady, within specification, and not running low for a successful calibration. Fuel line leaks that cause reading drift will be captured during calibrations resulting in reading error until a proper calibration is done.

Positive reading shifts after a zero calibration (when no combustibles are present) may be an indication of a partially clogged sample line. Compare SAMPLE FLOW in the GAUGES menu to CAL SAMPLE FLOW. Flow increases during calibration that exceed 0.1 LPM indicate sample line clogging that restricts the free flow of excess calibration out through the sample line. Clean or replace the sample line to restore proper calibration accuracy.

10.9. Troubleshooting by status message

STATUS	Observations	Causes and/or Corrective Action
BLOCK TEMP LOW	HEAT “on” continuously	• Proper voltage levels not present at terminal. • Insufficient warm-up time. Wait up to 45 to 90 minutes after power is turned on. • Heater failure • Heater relay failure in controller • Loose controller connections on Terminals 31-37. • Insulation and/or enclosure covers not in place. • Overtemperature cutoff tripped – requires manual reset. • TC2 thermocouple measuring circuit failure.
TC2 FAIL	HEAT is “off” And HSG TEMP in GAUGES menu has negative temperature reading	TC2 thermocouple open circuit or loose connection Tighten all connections
BLOCK TEMP HIGH	HEAT LED is “OFF” And HSG TEMP on GAUGES menu reading above SET TEMP by more than +5°C.	• Mis-wiring of AC power to heater terminals instead of AC input terminals. • Failure of controller circuit. • Failure of heater relay, or short circuit to AC heater. • TC2 thermocouple measuring circuit failure.

STATUS	Observations	Causes and/or Corrective Action
CHK FLOW or LOW FLOW	CHK FLOW message Service Needed Relay Activated Or Fault LED indicator “on” and FLOW measurement in GAUGES menu less than LOW FLOW setting (1.4 LPM typical)	• Sample and/or Exhaust filters clogged. Change filter(s). • Flame arrestors starting to clog. Remove, clean and reinstall. • Low compressed air pressure. Verify/set to 20psi. • Sample line starting to clog. Disconnect sample line at sensor. Clean line. • Exhaust line starting to clog. Disconnect exhaust line at sensor. Clean line. • Pressure transducer tubes loose or slightly leaking on back of electronics. Tighten and/or replace. • Pressure transducer tubes reversed. Verify correct connections. • Failure of measuring circuitry or pressure sensor. Call for service. • Aspirator nozzle plugging, clean. • Sample flow orifice loose, inspect and tighten.
DOWNSCLE	READING < -10% LFL	• Drift from low fuel supply pressure. • Drift from impurities in propane fuel supply. • External or internal fuel line leak, check and tighten. • Improper zero calibration without zero gas. Zero on slight upscale reading. • Contaminated zero gas, such as oil in compressed air line. • Insufficient zero gas pressure. • Contaminated thermocouple needs cleaning.
FLOW HIGH	Fault LED indicator “on” and FLOW measurement in GAUGES menu at or above the limit	• Air pressure too high, causing excessive suction in aspirator and too high a flow value: SNR701 < 3.4 LPM SNR702 < 3.1 LPM SNR704 < 2.8 LPM SNR705 < 2.7 LPM • Process pressure too high, causing excessive sample flow. • Differential pressure between sample inlet and exhaust too high, causing excessive sample flow. • Clogged sample orifice, causing excessive pressure drop at normal flow rate. • Flow Transducer failure, causing high electronic signal, call for service.
FLAMEOUT	GAUGES/FLAME ≈ GAUGES/HSG TEMP Indication of a sensing flame failure – without which the sensor will not read sample.	• Low fuel, fuel leak, or insufficient fuel inlet pressure. Replace/set to specified value. • Ignition circuit not lighting the flame. Repeat RESET. • No visible spark during ignition. Call for service. • Lack of oxygen in sample stream. Call for service. • Air pressure low or not connected. • Fuel regulator failure. Call for service.

STATUS	Observations	Causes and/or Corrective Action
CHK RDNG	-10% < READING < -7%	• Same as DOWNSCLE.
CHK ZERO Or ZEROFAIL	CHK ZERO message Service Needed Relay Activated or Fault indicator “on”	• Zero calibration gas not connected. • Incorrect zero gas or incorrect pressure. • Zero gas oxygen concentration not 20.9%volume. Must use clean, dry compressed air. • Insufficient fuel pressure at fuel inlet. • Drift from fuel regulator requires readjustment until FLAME measurement is 500°C under zero conditions.
CHK SPAN Or SPANFAIL	CHK SPAN message Service Needed Relay Activated or Fault LED indicator “on”.	• Span gas not connected to inlet. • Incorrect span gas type or concentration. • Insufficient span gas pressure. • Incorrect CAL RDNG setting. • CAL TIME too short to allow time to respond. • Incorrect alignment between thermocouple and burner. • Thermocouple requires cleaning to remove accumulated debris. • Sample line leaks in analyzer. Tighten fittings and/or replace O-Rings.
CHK TC3	CHK TC3 Message Service Needed Relay Activated	• Cutoff Thermocouple TC3 open or loose connection • Ensure all TC3 connections are secured
CHK RELAYS	CHK RELAYS Message Service Needed Relay Activated	• Perform Relay Tests for all Safety Relays and check for Continuity across each contact and the common • If wiring the relays externally you can switch the wires from the bad contact to the still working contact to be able to continue production while replacing the controller
CHK 4-20mA	CHK 4-20mA Message Service Needed Relay Activated	• Ensure there is a 330 ohm terminating resistor across the terminal positions for any 4-20mA circuit not being used
CHK DATA	CHK DATA message Service Needed Relay Activated	• Self-diagnostics has detected an unintentional change in a setting. See section 9.3.5. • Check all register settings. Perform a calibration.
DATALOST	DEVICE TYPE “Unknown” Other settings may be wrong Fault LED and relay active.	• Self-diagnostics detected an unintentional change in a setting. See section 9.10. • Attempt to change a setting to an invalid value. • Check all settings and perform a calibration.
CIRCUIT FAILURE	Memory failure, power failure, board failure	• Cycle power “off” and then “on.” Check all settings. Perform a calibration. Controller may require factory repair or replacement

10.10. Data Lost and Cold Start

When power is turned on, the settings are restored from nonvolatile memory. During operation, the settings are tested continuously to make sure they are not corrupted. If the settings are corrupted and cannot be restored, a DATA LOST fault trips, and the controller will be in a "Cold Start" state.

To recover from DATA LOST :

1. Enter the correct DEVICE TYPE in the SETTINGS menu.
2. If necessary, correct the SERIAL NUMBER number in the SETTINGS menu.
3. Check all settings
   Factory settings can only be changed during a cold start – see the Registers Table.
4. Once settings are correct, use the ALARM RESET command to clear the COLD START.
   Following an alarm reset, user settings can be changed, factory settings cannot.
5. Use the ACKNOWLEDGE command to clear the CHECK DATA service-needed message.
6. Verify all settings.
7. Perform a FULL CALIBRATION.

10.11. Drift

Most causes of drift are relatively minor, and do not cause error outside of the specified accuracy. The specifications for pressure, flow, temperature and oxygen effects should be reviewed to determine if the drift is due to one of these specified effects. Some causes of drift are more significant, and can activate faults or service-needed messages.

10.11.1. Initial drift

Drift is sometimes observed during the first 24 to 48 hours after initial startup. It may be due to air trapped in the fuel line, or other causes. The analyzer then stabilizes and drift stops.

10.11.2. Fuel regulation

Loss of fuel regulation can cause a significant amount of drift, even flameout. The fuel pressure must be above the minimum at the fuel inlet. The fuel inlet fittings, which contains a sintered metal filter, must be clean. This problem is diagnosed when an increase in the fuel supply pressure of +10% or more causes more than +1% to +2% LFL increase in the reading.

10.11.3. Fuel purity

Impure fuels cause drift. The contamination may be in the fuel itself. Propane fuels are more likely to have purity problems than hydrogen, especially if the propane supply is subjected to temperature changes. Or the contamination may be residue of sealant, flux, oil, etc., in the fuel line. This contamination may be slowly released into the fuel, and can vary in magnitude over time from changes in temperature and pressure. Clean fuel tubing and pure fuels are essential.

10.11.4. Sample line contamination

An obstruction or clog in the sample tubing can cause drift.

Heated processes sometimes produce vapors from high boiling point substances that can condense in the sampling system, especially if the sample tubing is unheated or insufficiently heated. This contamination, when subjected to temperature changes, can release back into the sample stream and interfere with the readings. If this occurs, the sample tubing temperature must be increased.

During calibration, test gases exit through the sample tubing. A flow change during calibration can indicate a clog in the sample tubing. Compare the SAMPLE FLOW to the CAL SAMPLE FLOW in the FLOW SYSTEM menu.

During calibration, a clog in the exhaust flame arrestor or the orifice filter can cause cause the flow to decrease. Compare the SAMPLE FLOW to the CAL SAMPLE FLOW in the FLOW SYSTEM menu. Check that the RESPONSE TIME remains fast.

10.11.5. Thermocouple or burner contamination

An accumulation of material on the thermocouple or burner, in particular silicones, can slightly insulate the thermocouple from the sensing flame and cause abnormally low zero readings. Such contamination usually presents itself as zero and span drift, and lower-then-expected readings.

10.12. "Service Needed – Check" Relay

The Check is by factory default a normally de-energized relay that energizes to trip. This relay is also by factory default an open relay contact under normal operations that closes when tripped. This is different than the four safety relays (Fault, Warning, Danger, and Cal) which are energized under normal operation and de-energize to trip, with a normally closed contact that opens when tripped.

Under the CONTROLS menu there is a register called CHECK RELAY NE, this register can change the state of the Check relay during normal operation. Out of the factory; this register is disabled. When enabled, the Check relay becomes energized under normal operation that de-energizes to trip, with a normally closed contact that opens to trip.

Should wiring cause an inverted signal compared to what is expected (a check message appearing on your PLC when there is none on the controller side), the CHECK RELAY NE register may need to be switched from the current to the opposite setting (from DISABLED to ENABLED, or vice versa). This action is locked behind a COLD START.

10.13. Cold Start

NOTE: Cold Start should only be done after consulting with Control Instruments as performing a Cold Start allows the user to change factory default settings that should remain untouched. This should only be done when absolutely necessary as a troubleshooting tool.

To perform a Cold Start, the user needs to assign COLD START to one of the 4 available Control inputs. This is done by going into the CONTROLS menu and under either register CONTROL 1, 2, 3, or 4 setting the register value to be COLD START.[28] To get out of a COLD START perform an ALARM RESET.

[28] It is recommended to set an unused CONTROL input to COLD START when possible.

11. Spare Parts

11.1. Parts for Preventative Maintenance

For maintaining the flow system of analyzers operating up to 200°C

FLT012	4 per year	Filter element, flame cell sample and exhaust
ORG005	4 per year	O-Ring, 1 3/16" ID, Viton, Flame arrestor cap

For maintaining the flow system of analyzers operating above 200°C (SNR705)

FLT012	4 per year	Filter element, flame cell sample and exhaust
ORK005 (ORK005M)	2 every 6 months	O-Ring, 1 3/16" ID, Flame arrestor cap, Kalrez® or equal

11.2. Parts for Routine Maintenance

Replacement O-rings for re-assembly of the flame cell, for analyzers operating up to 200°C

ORG044	1	O-Ring, Viton, 2.484" ID, Flame cell
ORG041	2	O-Ring, Viton, 0.359" ID, Flame cell to aspirator block

Replacement O-rings for the flame cell, for analyzers above 200°C (SNR705)

ORK046	1	O-Ring, 2.484" ID, Flame cell, Kalrez® or equal (ORK046M)
ORK047	2	O-Ring, 0.359" ID, Aspirator block, Kalrez® or equal (ORK047M)

11.3. Parts for Maintenance and Repair

These parts are needed occasionally during maintenance and repair and should be kept on hand.

ORF024	Orifice, Sample flow measurement
FLT087	Sintered metal filter to protect orifice
NZL001	Nozzle, aspirator
SNP189	Flame arrestor
FUS037	Fuse, 5 amp, 5mm x 20mm, quick acting, IEC 127 code F
FUS038	Fuse, ½ amp, 5mm x 20mm, time lag, IEC 127 code T

11.4. Burners and restrictors

BRN071	Hydrogen burner assembly for 700 Series analyzers (SNR70x) (FOR REFERENCE ONLY, NOT TO BE USED DURING ROUTINE MAINTENANCE.CALL FOR INQUIRIES ABOUT REPLACEMENT PARTS IF NEEDED)
BRN050	Hydrogen burner assembly for air dilution 700 Series (SNR70xD)
BRN006	Propane burner for SNR701
RSP018	Propane Fuel Restrictor for SNR701
)	
RSH002	Hydrogen Fuel Restrictor for air dilution 700 Series (SNR70xD)

11.5. Parts for Repair

These parts may be needed to replace a failed component.

THC002R	Thermocouple, flame and block
THC008R	
VLV079	Solenoid valve, zero or span, 120 VAC
VLV083	Solenoid valve, zero or span, 230 VAC
PRV058R	Fuel Regulator

HTR077R	Heater, 385 watt, 120 VAC HTR078R Heater, 385 watt, 230 VAC
TOL003	Tool, flame thermocouple bending and alignment
TOL047	Tool, thermocouple alignment
TOL044	Quick-wedge locking screwdriver for orifice filter service
TOL046	1/8” Hex socket driver
CBL229	Ignition cable, replacement, SNR705 only
SPK031	Ignition cable, replacement, SNR701 – SNR704
SPK021	Spark plug, flame ignition
SNP366	Calibration gas restrictor/filter
SNP374	Fuel inlet restrictor/filter
HSG274	Inner Cover with Insulation Material, SNR702 – SNR705

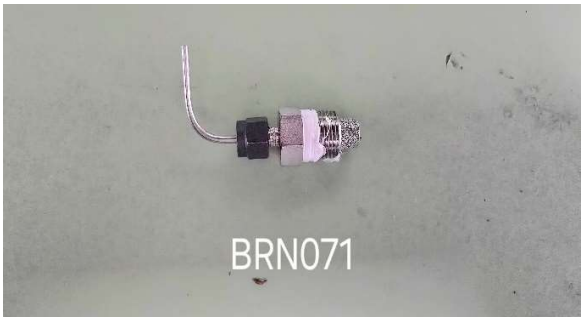
11.6. Controllers

When ordering a replacement controller, include the analyzer’s serial number

HSG237L	Controller for 670 Series, replacement (32 character display)
HSG341R	Controller for all 700 Series analyzers (SNR70x, SNR70xD)

Spare Part Photos
The coin in the photographs is approximately 2 centimeters, or 3/4 inch.

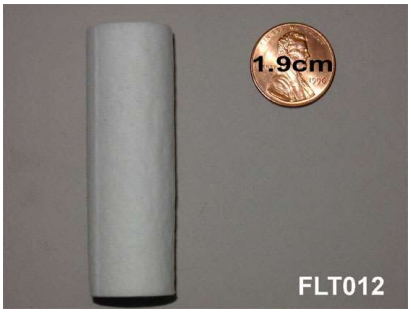
Burner assembly[29]: part # BRN071



Ignitor cable: part # CBL229 for SNR705



Filter element: part # FLT012



Controller: part# HSG341R/HSG237L



Fuses: part#s FUS037 and FUS038





Heater elements - part#s HTR077R (120VAC) and HTR078R (220 VAC)



Use HTR077R to replace HTR057 heater



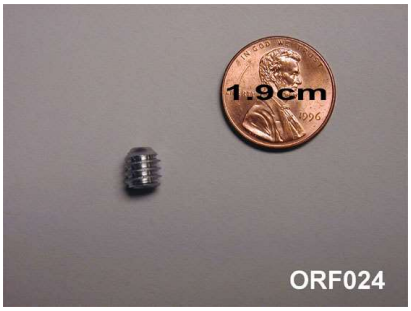
Use HTR078R to replace HTR058 heater

[29] Air dilution types (D) use a shorter burner, part# BRN050, having a 0.5 inch (1.3 cm) gap.

Aspirator nozzle: part# NZL001



Sample flow orifice: part# ORF024
Orifice inner diameter is 0.025 inches (#72 drill).



O-Ring, Flame Arrestor Cap, viton: part# ORG005 and Kalrez: part#ORK005
Viton O-Rings (ORG) are for lower temperatures. Kalrez (ORK) is required for high temperatures.





O-Ring, block seals, viton: part# ORG041 and Kalrez: part# ORK047





O-Ring, flame cell seal, viton: part# ORG044 and Kalrez: part# ORK046





Fuel Regulator: part# PRV058R



Flame Arrestor: part# SNP189



Fuel Restrictor, hydrogen[30]: part# RSH001 (or RSH002) and propane: part# RSP018

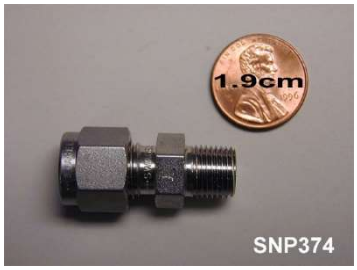




Calibration Gas Inlet Restrictor: part# SNP366



Fuel Inlet Restrictor / Filter: part# SNP374

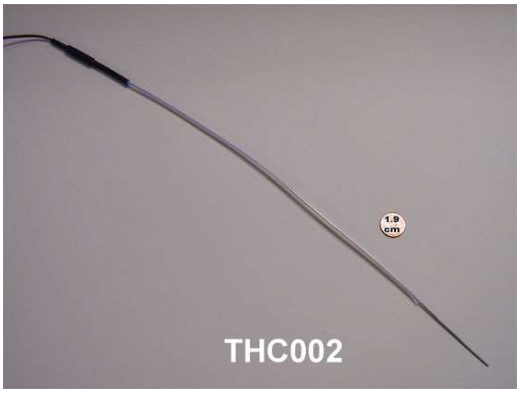


[30] Air dilution type analyzers use hydrogen fuel restrictor part number RSH002.

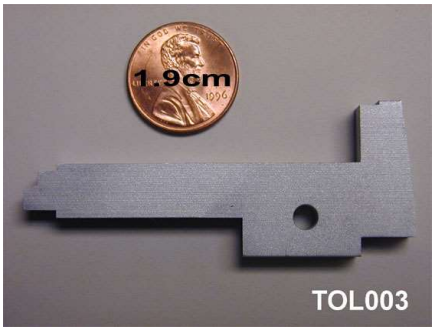
Spark Plug Assembly: part# SPK021



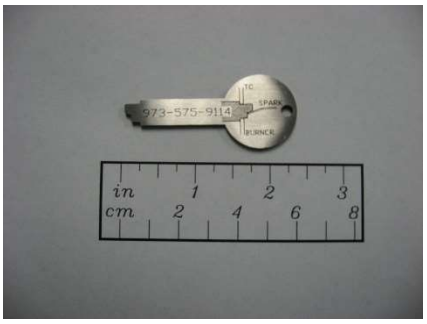
Thermocouple assembly for TC1 and TC2: part# THC002R



Tool: part# TOL003
Used to set the standard burner-thermocouple gap,[31] and spark gap



Tool: part# TOL047



Solenoid valve, calibration: part#s VLV079 (120 VAC) and VLV083 (230 VAC)





[31] Air dilution types (D) use a shorter burner, part# BRN050, having a 0.5 inch (1.3 cm) gap.

Air Regulator, Part# PRV069,

for Air-dilution Types only



Ignitor cable: part # SPK031

for SNR701, SNR702 & SNR704 only



Cutoff Thermocouple, Part# THC008,

for TC3 Replacement



Inner Cover w/ Insulating Material, Part# HSG274



Serial Communications

Data can be transferred between one or more analyzers, and a single host device, using the Modbus Remote Terminal Unit (RTU) protocol. The host device sends polls, to which the analyzer replies.

12.1. Serial Communications Specifications

Specification		Description
Supported commands	Command 3	"Read Output (Holding) Registers"
	Command 6	"Preset Single Register"
	Command 16	"Preset Multiple Registers."
Command 16 limit	Limited to one register in length (special case).	
Command 3 limit	Limited to 127 registers in length.	
Broadcast mode	Supported for commands 6 and 16.	
Serial port hardware	Half-duplex RS485	
Data format	8 bits, no parity, 1 stop bit	
Baud rates	Baud rates from 9600 to 38400 are supported	
Standard baud rate	19200	
Receive-transmit delay	Minimum: 5 characters between receipt of poll and reply.	
	Maximum: 250 milliseconds, if maximum poll rate is reached	
Error handling	Invalid start register address	Command ignored, no reply
	Read more than 127 registers	Command ignored, no reply
	CRC error, poll corrupted	Command ignored, no reply
	Read over page boundary	Command ignored, no reply
	Read invalid page	Command ignored, no reply
	Data written outside valid limits	Reply is made but data is ignored, "Check Data"
	Read of unsupported register	0x2020 Data returned in reply (ASCII blank)

12.2. Serial Communications Notes

The modbus master should be secured from unauthorized or unintentional changes. Networked access is not recommended.

WARNING! Serial communications must not be used for critical safety functions such as process shutdown.

Writing illegal data outside the register's acceptable limits is acknowledged with a reply, but the data is ignored and a "Service Needed – Check Data" occurs. The user of the Modbus master should verify completion of write commands by a subsequent read of the register.

Data written to the COMMANDS register is cleared as soon as the command is executed. Reading the COMMANDS register returns 00H data, even if the register has just been written. The host should verify COMMANDS by reading the STATUS register.

12.3. Register Conversion Factors

The register table specifies either a lookup table or engineering units for each register. To convert, either match the data with a "lookup table" of values found in the register description, or multiply by the conversion factor for the register's engineering units in the table below.

For conversion of register data	
Engineering Units	**Conversion Factor**
%LFL Reading	Multiply by 0.00390625 %LFL
Temperature ºC	Multiply by 0.2442 °C
Flow Rate LPM	Multiply by 0.00172936 Liters per minute
Milliamps mA	Multiply by 0.1 Milliamps
LookupTenthsMA	Multiply by 0.1 Milliamps
PressureHPA	Multiply by 0.01 hPa (data stored as pascals)
PressureKPA	Multiply by 0.01 kPa (data stored as dekapascals)
PressureBar	Multiply by 0.0001 Bar (data stored as dekapascals)
VersionNbr	Multiply by 0.01, e.g 701 = Version 7.01
TenthsSecs	Multiply by 0.1 Seconds
Minutes	Minutes, no conversion
Percent	Multiply by 0.01 Percent
None	Decimal number. No conversion.
Register data in Engineering Units are signed sixteen bit numbers (two's complement"). Positive numbers are 0 and up to 32767. Above 32767, subtract 65536.	

12.4. Registers

<table>
<tr><td colspan="4">THESE REGISTERS ARE DISPLAYED AUTOMATICALLY ON THE FRONT PANEL AND DO NOT APPEAR ON MENUS</td></tr>
<tr><th>REGISTER NUMBER</th><th>SIXTEEN CHARACTER NAME</th><th>DESCRIPTION</th><th>EIGHT CHARACTER NAME</th></tr>
<tr><td>1</td><td>FLAMMABILITY</td><td>The flammability reading in percent Lower Flammable Limit (%LFL)
Lower Limit: -128% LFL, Upper Limit: 127% LFL, Default Value: 0% LFL</td><td>READING</td></tr>
<tr><td>2</td><td>STATUS</td><td>Engineering Unit: Lookup, Lower Limit: 0, Upper Limit: 23, Default Value: 11
Device status messages, codes, and their meanings:
<table>
<tr><th>CODE</th><th>MESSAGE</th><th>DESCRIPTION</th></tr>
<tr><td>0</td><td>Normal Operation</td><td>No faults, no alarms</td></tr>
<tr><td>1</td><td>Warning Alarm</td><td>Reading above warning level</td></tr>
<tr><td>2</td><td>Danger Alarm</td><td>Reading above danger level</td></tr>
<tr><td>2</td><td>Danger Flooded</td><td>Flameout after reading exceeded danger level</td></tr>
<tr><td>3</td><td>DownscaleReading</td><td>Negative reading too low below zero (-10%LFL)</td></tr>
<tr><td>4</td><td>Zero Gas On</td><td>Zero calibration or zero test in progress</td></tr>
<tr><td>5</td><td>Span Gas On</td><td>Span calibration or span test in progress</td></tr>
<tr><td>6</td><td>Purging</td><td>Purging of calibration gases in progress</td></tr>
<tr><td>7</td><td>Calibration Ends</td><td>Recovery period following calibration</td></tr>
<tr><td>8</td><td>Zero Fail</td><td>Zero calibration failed, result unacceptable</td></tr>
<tr><td>9</td><td>Span Fail Low</td><td>Span calibration failed, result too low</td></tr>
<tr><td>10</td><td>Span Fail High</td><td>Span calibration failed, result too high</td></tr>
<tr><td>11</td><td>Warmup</td><td>Power-on time delay for self diagnostics, purge,</td></tr>
<tr><td>12</td><td>Self Test</td><td>Self diagnostic at power on</td></tr>
<tr><td>13</td><td>Self Test Fail</td><td>Failed self diagnostic test</td></tr>
<tr><td>14</td><td>Data Lost</td><td>Nonvolatile settings have been lost (cold start)</td></tr>
<tr><td>15</td><td>12VDC Fail</td><td>Power supply is too high or too low</td></tr>
<tr><td>16</td><td>Dilution Low</td><td>Dilution failed low</td></tr>
<tr><td>17</td><td>Dilution High</td><td>Dilution failed high</td></tr>
<tr><td>18</td><td>Block Temp Low</td><td>Temperature of analyzer is too low</td></tr>
<tr><td>19</td><td>Block Temp High</td><td>Temperature of analyzer is too high</td></tr>
<tr><td>20</td><td>Sample Flow Low</td><td>Sample flow rate too low</td></tr>
<tr><td>21</td><td>Sample Flow High</td><td>Sample flow rate too high</td></tr>
<tr><td>22</td><td>Fuel Pressure Lo</td><td>Fuel pressure too low</td></tr>
<tr><td>23</td><td>Fuel Pressure Hi</td><td>Fuel pressure too high</td></tr>
<tr><td>24</td><td>TC1 Fail</td><td>Thermocouple for sensing flame failed</td></tr>
<tr><td>25</td><td>TC2 Fail</td><td>Thermocouple for temperature control failed</td></tr>
<tr><td>26</td><td>Flameout</td><td>Sensing flame is out</td></tr>
<tr><td>27</td><td>ROM Fail</td><td>Continuous self-diagnostic has failed</td></tr>
</table></td><td>STATUS</td></tr>
</table>

COMMANDS MENU

THESE REGISTERS ARE DISPLAYED AUTOMATICALLY ON THE FRONT PANEL AND DO NOT APPEAR ON MENUS * ADJUSTABLE SETTING USER CAN CHANGE			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
* 8	COMMANDS	Commands always accessible to the operator, execute a command to control the device. Engineering Unit: Lookup, Lower Limit: 0, Upper Limit: 0x1700, Default Value: 0 MESSAGE — DESCRIPTION Alarm Reset — Clear Alarms Acknowledge — Clear "Service Needed - Check" NOTE: Calibration commands accessible to the operator when enabled by security. These commands temporarily bypass new warning & danger alarms and trip the calibration in progress relay: Zero Adjust — Adjust flammability readings within +/-3%LFL to 0 %LFL Zero Calibration — Zero calibration with gas, make adjustment to 0%LFL Span Calibration — Span calibration with gas, make adjustment to calibration reading Full Calibration — Perform zero & span calibration with gases, make adjustments Zero Test — Test using zero gas without adjustment Span Test — Test using span gas, without adjustment Full Test — Launch zero and then span tests, no adjustments	COMMANDS

LIST OF ALL COMMANDS - ACCESSIBLE THROUGH MODBUS			
HEX	**DECIMAL**	**NAME**	**DESCRIPTION**
0x0000	0	No Command	
0x0100	256	Acknowledge	Deactivates the "Service Needed" relay and message
0x0200	512	Alarm Reset	Clears latched alarm. Re-ignites sensing flame. Restarts air/fuel cutoff timer
0x0300	768	Zero Calibration	Zero calibration with gas, make adjustment
0x0400	1024	Span Calibration	Span calibration with gas, make adjustment
0x0500	1280	Full Calibration	Zero and span calibrations with gases, make adjustments
0x0600	1536	Zero Test	Test using zero gas without adjustment
0x0700	1792	Span Test	Test using span gas, without adjustment
0x0800	2048	Full Test	Zero and then span tests, no adjustments
0x0900	2304	Save Settings	Allow "factory settings" to be changed
0x0A00	2560	Cold Start	Enter "cold start" state to allow changes to factory settings
0x0B00	2816	Menu	Remote activation of "menu" pushbutton
0x0C00	3072	Select	Remote activation of "select" pushbutton
0x0D00	3328	Do Nothing	Unused command code – "flashlight" command via Modbus no longer supported
0x0E00	3584	Zero Adjust	Adjust flammability readings to 0 %LFL by up to +/-3%LFL
0x0F00	3840	Lock	Lock commands written through MODbus no longer have any effect - see "Security"
0x1000	4096	Unlock	Unlock commands written through MODbus no longer have any effect - see "Security"
0x1100	4352	Check Utilities	Activate "Service Needed" "Check Utilities"
0x1200	4608	Test Warning Rly	Trip warning relay until alarm reset
0x1300	4864	Test Danger Rly	Trip danger relay until alarm reset
0x1400	5120	Test Fault Relay	Trip fault relay until alarm reset
0x1500	5376	Test Cal Relay	Trip cal relay until alarm reset
0x1600	5632	Test Check Relay	Trip "Service Needed - Check" relay until alarm reset
0x1700	5888	Gas Alarm Test	Turn on span valve with alarm enabled

GAUGES MENU

* ADJUSTABLE SETTING USER CAN CHANGE			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
38	PEAK HIGH %LFL	The highest flammability reading since the last "reset" command, in %LFL Lower Limit: -128%LFL, Upper Limit: 128%LFL, Default Value: -128%	PEAK HI
39	PEAK LOW %LFL	The lowest flammability reading since the last "reset" command, in %LFL Lower Limit: -128%LFL, Upper Limit: 128%LFL, Default Value: 128%	PEAK LOW
9	BLOCK TEMP	The temperature of the flame cell block, in Degrees C Limits depend on device type	HSG TEMP
* 24	BLOCK SETTING	The temperature of the flame cell block, in Degrees C Limits depend on device type TYPE / MINIMUM / MAXIMUM / DEFAULT SNR701 (T6) 50°C 65°C 60°C SNR702 (T4) 100°C 120°C 120°C SNR702 (T3) 100°C 153°C 120°C SNR704 (T3) 100°C 185°C 180°C SNR704 (T2) 100°C 210°C 200°C SNR705 (T2) 100°C 280°C 250°C	SET TEMP
15	SAMPLE FLOW	Sample flow rate, in Standard Liters per Minute (LPM) Default Value: 0	FLOW
11	FLAME TEMP	Current flame temperature (Thermocouple TC1 reading), in Degrees C Default Value: 0	FLAME
29	ZERO FLAME TEMP	Flame temperature measured during the last successful zero calibration, in Degrees C Lower Limit: 460°C, Upper Limit: 525°C, Default Value: 500°C	RAW ZERO
30	SPAN FLAME TEMP	Increase in flame temperature measured during the last successful span calibration, in Degrees C Lower Limit: 37°C, Upper Limit: 175°C, Default Value: 80°C	RAW SPAN

FLOW SYSTEM MENU

* ADJUSTABLE SETTING USER CAN CHANGE #FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
15	SAMPLE FLOW	Current sample flow rate, in liters per minute (LPM) Lower Limit: 0.00 LPM	FLOW
70	CAL SAMPLE FLOW	Sample flow rate measured during the last successful calibration, in liters per minute Lower Limit: 0.00 LPM	NEW
* 23	LOW FLOW LEVEL	Flow rate at which a low sample flow fault trips, in liters per minute Lower Limit: 1.4 LPM, Upper Limit: 2.4 LPM, Default Value: 1.8 LPM	LOW FLOW
* 42	CHECK FLOW SET	Change in flow rate since the last calibration that activates "Service Needed - Check Flow", in liters per minute Lower Limit: 0.2 LPM, Upper Limit: 0.6 LPM, Default Value: 0.4 LPM	CHK FLOW
93	ORF PRESSURE	Current pressure measured across the sample flow orifice, in Kilopascal (kPa) Default Value: 0 kPa	NEW
# 94	ORF PRESS ADJ	An adjustment to the sample flow orifice pressure sensor, in Kilopascal Lower Limit: -4 kPa, Upper Limit: 4 kPa, Default Value: 0 kPa	NEW

DILUTION MONITOR MENU
WHEN ENABLED BY DEVICE SETTTINGS

* ADJUSTABLE SETTING USER CAN CHANGE
FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING

REGISTER NUMBER	SIXTEEN CHARACTER NAME	DESCRIPTION	EIGHT CHARACTER NAME
13	DILUTION PRESS	Current dilution pressure, in pressure bar Default Value: 0 kPa	NEW
* 66	DILUTION SETTING	Dilution pressure setting, in pressure bar Lower Limit: 0.00, Upper Limit: 1.72, Default Value: 0.00	NEW
* 67	DILUTION LIMIT	Setting for allowable deviation between dilution pressure & setting, in pressure bar Lower Limit: 0.03, Upper Limit: 0.34, Default Value: 0.17	NEW
# 96	DILN PRESS ADJ	Adjustment to the dilution pressure sensor, in pressure bar Lower Limit: -0.10, Upper Limit: 0.10, Default Value: 0.00	NEW

FUEL MONITOR MENU
WHEN ENABLED BY DEVICE SETTTINGS

* ADJUSTABLE SETTING USER CAN CHANGE
FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING

REGISTER NUMBER	SIXTEEN CHARACTER NAME	DESCRIPTION	EIGHT CHARACTER NAME
12	FUEL PRESSURE	Current fuel pressure, in pressure bar Lower Limit: 0.00, Upper Limit: 0.00, Default Value: 0.00	RAW RDNG
* 68	FUEL SETTING	Fuel pressure setting, in pressure bar Lower Limit: 0.00, Upper Limit: 1.72, Default Value: 0.00	NEW
* 69	FUEL LIMIT	Setting for the allowable deviation between fuel pressure & setting, in pressure bar Lower Limit: 0.03, Upper Limit: 0.34, Default Value: 0.17	NEW
# 95	FUEL PRESS ADJ	Adjustment to the fuel pressure sensor, in pressure bar Lower Limit: -0.10, Upper Limit: 0.10, Default Value: 0.00	NEW

PURGE MONITOR MENU
WHEN ENABLED BY DEVICE SETTTINGS

* ADJUSTABLE SETTING USER CAN CHANGE
FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING

REGISTER NUMBER	SIXTEEN CHARACTER NAME	DESCRIPTION	EIGHT CHARACTER NAME
14	ENCLOSURE PRESSURE	Current enclosure pressure, in Pressure hPa Default Value: 0 hPa	NEW
# 97	ENCL PRESS ADJ	An adjustment to the enclosure pressure sensor, in Pressure hPa Lower Limit: -2.5 hPa, Upper Limit: 2.5 hPa, Default Value: 0 hPa	NEW

CALIBRATION MENU

* ADJUSTABLE SETTING USER CAN CHANGE
FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING

REGISTER NUMBER	SIXTEEN CHARACTER NAME	DESCRIPTION	EIGHT CHARACTER NAME
* 7	CAL READING	Setting for the span calibration, in %LFL Lower Limit: 40%LFL, Upper Limit: 80%LFL, Default Value: 60%LFL	CAL READING
91	ZERO AS FOUND	Flammability reading recorded during zero calibration prior to adjustment, in %LFL	NEW
92	SPAN AS FOUND	Flammability reading recorded during span calibration prior to adjustment, in %LFL	NEW
101	RESPONSE TIME	Response time recorded during last full calibration, in Tenths of Secs	NEW
* 20	SPEED SETTING	Speed of response & noise filter, 0 = Off, 1 = Low, 2 = Medium, 3 = High Lower Limit: 1 or 0, Upper Limit: 3, Default Value: 2	SPEED
* 10	ZERO ADJUSTMENT	Amount of adjustment that's been made to zero flammability reading through "Zero Adjust" command, in %LFL Lower Limit: -3 %LFL, Upper Limit: 3 %LFL, Default Value: 0 %LFL	NEW
* 28	CALIBRATION TIME	Duration of each phase of calibration, in minutes Lower Limit: 1, Upper Limit: 4, Default Value: 1	CAL TIME
* 56	SPAN TEMP SET	Setting for target flame temperature rise for span calibration, in Degrees C Lower Limit: 50°C, Upper Limit:150°C, Default Value: 120°C	SPAN °C
11	FLAME TEMP	Current flame temperature (Thermocouple TC1 reading), in Degrees C	FLAME
29	ZERO FLAME TEMP	Flame temperature recorded during last successful zero calibration, in Degrees C Lower Limit: 460°C, Upper Limit: 525°C, Default Value: 500°C	RAW ZERO
30	SPAN FLAME TEMP	Rise in flame temperature recorded during last successful span calibration, in Degrees C Lower Limit: 37°C, Upper Limit:175°C, Default Value: 80°C	RAW SPAN
43	RDNG LAST CAL	Flammability reading recorded during last successful span calibration, in %LFL Lower Limit: 40%LFL, Upper Limit:80%LFL, Default Value: 60%LFL	LAST CAL

ALARMS MENU

* ADJUSTABLE SETTING USER CAN CHANGE # FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
* 5	WARNING LEVEL	Setting for "high flammability" warning alarm, in %LFL Lower Limit: 5% LFL, Upper Limit: 60% LFL, Default Value: 40% LFL	WARNING
* 6	DANGER LEVEL	Setting for "high high flammability" danger alarm, in %LFL Lower Limit: 10% LFL, Upper Limit: 60% LFL, Default Value: 50% LFL	DANGER
* 61	RATE ALARM	Setting to enable or disable the rate-of-rise alarm, 0 = disabled,1 = enabled Engineering Unit: On/Off Lower Limit: 0, Upper Limit: 1, Default Value: 1	RATE
* 72	WARNING LATCH	Setting to latch warning alarm and relay until manual reset 0 = autoreset (disabled), 1 = latched (enabled) Engineering Unit: On/Off Lower Limit: 0, Upper Limit: 1, Default Value: 1	NEW
* 73	DANGER LATCH	Setting to latch danger alarm and relay until manual reset 0 = autoreset (disabled), 1 = latched (enabled) Engineering Unit: On/Off Lower Limit: 0, Upper Limit: 1, Default Value: 1	NEW
* 74	FAULT LATCH	Setting to latch fault and fault relay until manual reset 0 = autoreset (disabled), 1 = latched (enabled) Engineering Unit: On/Off Lower Limit: 0, Upper Limit: 1, Default Value: 1	NEW
# 99	RDNG FAULT LEVEL	Setting to trip a fault on low "downscale" flammability readings between -10%LFL and -20%LFL, in %LFL Lower Limit: -20% LFL, Upper Limit: -10% LFL, Default Value: -10% LFL	NEW
* 35	AUTO IGNITION	Setting for number of automatic re-ignition attempts from 0 to 120, at fifteen second intervals Lower Limit: 0, Upper Limit: 120, Default Value: 0	AUTO IGN

ALARM TESTS MENU

WHEN ENABLED BY SECURITY, THESE TESTS TRIP RELAYS OR ACTIVATE THE SPAN VALVE THE TEST LAST FOR UP TO ONE HOUR, IT IS ENDED BY AN ALARM RESET OR CALIBRATION COMMAND			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
8	COMMANDS	Execute a command to control the device (see commands menu for full list): Lower Limit: 0, Upper Limit: 23, Default Value: 0	COMMANDS
	Test Warning Rly	Turn on warning relay for up to 15 minutes, or until an alarm reset command	
	Test Danger Rly	Turn on danger relay for up to 15 minutes, or until an alarm reset command	
	Test Fault Relay	Turn on fault relay for up to 15 minutes, or until an alarm reset command	
	Test Cal Relay	Turn on calibrating relay for up to 15 minutes, or until an alarm reset command	
	Test Check Relay	Turn on "Service Needed - Check" relay for up to 15 minutes, or until an alarm reset command	
	Gas Alarm Test	Open span valve for up to 15 minutes, or until an alarm reset command	
	Alarm Reset	Reset command halts testing	

4-20MA OUTPUTS MENU

* ADJUSTABLE SETTING USER CAN CHANGE # FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
50	MILLIAMP OUTPUT	Flammability reading currently being output, including the effect of the calibration & fault overrides and milliamp test, in %LFL Lower Limit: -10% LFL, Upper Limit: 125% LFL, Default Value: 0% LFL	REM MA
* 25	4MILLIAMP ADJUST	Adjustment to the 4-milliamp output for the 0% LFL flammability reading Default = 0, Adjustable from -2.0 Lower Limit: -20% LFL, Upper Limit: 20% LFL, Default Value: 0% LFL	4 MA ADJ
* 26	MILLIAMP CAL SET	Setting to signal that a calibration is in progress, it sends this %LFL flammability reading to the milliamp output Adjustable from -10% LFL to 117% LFL, Defaults to 113% LFL When set to 0% LFL the override is disabled and has no effect NOTE: In earlier versions below 7.01 this register was displayed as milliamps	MA CAL
* 32	MILLIAMP FAULT	Setting to signal that a fault exists, it sends this %LFL flammability reading to the milliamp output Adjustable from -10% LFL to 117% LFL, Defaults to 106% LFL. When set to 0% LFL the override is disabled and has no effect NOTE: In earlier versions below 7.01 this register was displayed as milliamps	MA FAULT
* 44	MIN MILLIAMP	Setting for the lowest milliamp output in %LFL Is used to test the milliamp output or to keep the milliamp output above a minimum value Lower Limit: -10% LFL, Upper Limit: 117% LFL, Default Value: -10% LFL	MA TEST

CONTROLS MENU

* ADJUSTABLE SETTING USER CAN CHANGE # FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
* 36	CONTROL 1	Command that executes when the Control 1 input is activated NOTE: See Commands list for codes and their meanings	CONTROL1
* 37	CONTROL 2	Command that executes when the Control 2 input is activated	CONTROL2
* 33	CONTROL 3	Command that executes when the Control 3 input is activated NOTE: Controllers without Control 3 & Control 4 hardware still display these registers on the menu	NEW
* 34	CONTROL 4	Command that executes when the Control 4 input is activated	NEW
* 58	RELAY SETTING	Setting to control the "Air/Fuel Cutoff" relay NOTE: "Horn" function has been removed and is replaced with the "Off" setting 0 = Off, 1 = Air Fuel, 2 = On	RLY CNFG
# 105	CHECK RELAY NE	Setting to "Air/Fuel Cutoff" relay state "normally energized" Setting 1 = enabled makes the relay normally energized Setting 0 = disabled makes the relay normally de-energized	NEW

COM PORT MENU

* ADJUSTABLE SETTING USER CAN CHANGE			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
47	CURRENT BAUD	Code for the current baud rate Can differ from the default setting if autobaud is enabled 14 = 38400, 15 = 38400, 16 = 19200, 17 = 9600	BAUDRATE
* 21	MODBUS ADDRESS	Setting for MODbus address from 1 to 32 Default =1	ADDRESS
* 22	DEFAULT BAUD	Setting for baud rate used when automatic baud rate adjustment is disabled	SET BAUD
* 48	BAUD AUTO ADJUST	Setting to automatically search & set baud rate, overriding Default Baud setting 0 = Disabled, 1 = Enabled, Default = Enabled	AUTOBAUD
* 19	ID NUMBER	Setting for user-defined identification, can be any number from 0 to 99 Default = 0	ID NBR

SETTINGS MENU

THESE REGISTERS ARE DISPLAYED AUTOMATICALLY ON THE FRONT PANEL AND DO NOT APPEAR ON MENUS * ADJUSTABLE SETTING USER CAN CHANGE # FACTORY SETTING ONLY ADJSUTABLE DURING COLD START SETTING			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
# 62	DEVICE TYPE	Analyzer type - this register is written once at factory, from 1 to 23 Defaults to 0 to indicate "Data Lost" The device type setting should match the analyzer nameplate 1 = SNR671 2 = SNR671D 3 = SNR672P 4 = SNR672 5 = SNR672D 6 = SNR672K 7 = SNR674P 8 = SNR674PL 9 = SNR674 10 = SNR674D 11 = SNR675 12 = SNR675D 13 = SNR675K 14 = Unused, causes DATA LOST fault 15 = Unused, causes DATA LOST fault 16 = SNR701 17 = SNR702 18 = SNR704 19 = SNR705 20 = SNR701D 21 = SNR702D 22 = SNR704D 23 = SNR705D	TYPE
40	SOFTWARE VERSION	Version number, for example 701, which is displayed as "7.01"	VERSION
# 63	SERIAL NUMBER	Last three digits of the serial number Example: Serial number 101 for year built 2022 displays as "22-101" Serial number adjusts from 100 to 9999, the year built adjusts from 2001 to 2100	SERIAL
# 65	YEAR BUILT		NEW
* 64	LANGUAGE	Display language: 0 = English, 1 = German Lower Limit: 0, Upper Limit: 1, Default Value: 0	LANGUAGE
* 98	FLAME TC GAIN	Percent gain of the TC1 flame thermocouple from 95 to 115 Lower Limit: 95% Upper Limit: 115% Default Value: 100%	NEW
# 106	TC TIME CONST	Time constant of the flame thermocouple from 9 to 12 seconds in tenths of seconds (90 to 120), Defaults to 10.7 seconds	NEW
# 100	MINIMUM SPEED	The minimum speed setting, when enabled, 0 = OFF, 1 = Low	NEW
# 103	DILUTION MONITOR	Setting for the dilution monitor, 1 = enabled, 0 = disabled	NEW
# 102	FUEL MONITOR	Setting for the fuel monitor, 1= enabled, 0 = disabled	NEW
# 104	PURGE MONITOR	Setting for the purge monitor, 1 = enabled, 0 = disabled	NEW

SECURITY MENU

* ADJUSTABLE SETTING USER CAN CHANGE			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
* 60	LOCK SETTING	Setting to lock out calibrations and tests and/or to lock parameters NOTE: See "Security" section of manual, the function has changed Previously was a read-only indication of calibration lock only, 0 = unlocked, 1 = Hardware cal lock, 2 = Software cal lock, 3 = Both cal locks Lower Limit: 0, Upper Limit: 3, Default Value: 0	CAL LOCK
27	LOCK STATE	Current state of the lock(s), 0 = Unlocked, 1 = LockCal & Tests, 2 =LockSettings, 3 = LockAll Lower Limit: 0, Upper Limit: 3, Default Value: 0	NEW

REGISTERS MENU

THESE REGISTERS DO NOT APPEAR ON THE FRONT PANEL			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
45	RELAY TESTS	Indication for relays currently under test	RLY TEST
46	DUMMY REG	Register available for user, such as communications test Performs no function	DUMMY
59	_____	Current state of the "Service Needed - Check" messages 1 = active, 0 = inactive Flags = 1 when the check is active Bit 0 = Check Flow Bit 1 = Check Zero Bit 2 = Check Span Bit 3 = unused check, always 0 Bit 4 = Check Data Bit 5 = Check Reading Bit 6 = unused check, always 0 Bit 7 = unused check, always 0 Bit 8 = Check Dilution Bit 9 = Check Fuel Bit 10 = Check Enclosure Bit 11 = Check Utilities Bit 12 = Check Relays Bit 13 = Check 4-20mA Bit 14 = Check TC3 Bit 15 = unused check, always 0 Note: Bit 3 was “LED flash” but is no longer supported	SERVICE
71	BLOCK TEMP LIMIT	Cutoff temperature in ^{o}C, automatically set to +15°C above the block temperature setting	NEW
75 to 90	DISPLAY BUFFER	The front panel display buffer - two lines by sixteen text characters - can be read via serial communications	NEW

THESE "LEGACY" REGISTERS ARE READ-ONLY OUTPUTS PROVIDED FOR COMPATIBILITY WITH 670 SERIES HOST APPLICATIONS THEY DO NOT CONTROL ANY FUNCTIONS - THEY ARE NOT SHOWN ON THE FRONT PANEL			
REGISTER NUMBER	**SIXTEEN CHARACTER NAME**	**DESCRIPTION**	**EIGHT CHARACTER NAME**
3	FULL SCALE	Full scale value of the flammability reading - always 100	FULLSCLE
4		Indication of the current state of the relays - tripped/clear or active/inactive Read-only register. Not accessible through the front panel. The flag for the horn function always clear. Format is 00000AKC-TISZ0FDW. Three additional relays have been added: A = Air Fuel cutoff K = Service Needed Check C = Calibrating (CAL)	RELAYS
16		Reference, for monitoring the power supply, typically 506	A/D REF
17	ALARM LATCH	Indication of relays which latch until cleared by the alarm reset This function is now controlled by registers 72, 73 and 74	LATCHES
18	SAFETY RELAYS	Indication of relays which "de-energize to trip" - always 0x0007	FAILSAFE
31	20MA ADJUST	Adjustment to the milliamp output at the 20-milliamp point, no longer supported Always 0	20MA ADJ
41	LINEARITY SET	Constant for linearizing readings 80%LFL for dilution types, 40%LFL for standard types	LINEAR
49	RELAY STATES REM	Safety relays (18) and Relays states (4) combined Copy of relays in low byte position and failsafe register in high byte position Not accessible through the front panel. Read-only: format 0000**0**111_TISZ**0**FDW The flag representing the Horn is now always 0.	REM RLY
51 to 55	REM DISPLAY1-5	Eight-character display buffer no longer supported - reads DISABLED	
57	MILLIAMP SCALE	Full scale value of the milliamp output - always 100	MA SCALE
106		Highest register number - There are no valid registers above 105	
257+	IDENTIFICATION	Unsupported legacy information that has been replaced by new registers. Attempt to read these registers will fail. Data read will be 0x2020 (ASCII blanks)	

13. Revision History

Revision	Revision Date	Revised by	Reviewed by	Description
0	Jan-23	JI	CGS	Document Created
A	Mar-23	JI	CGS	Added Warning in French in section 1.4.2 about not disconnecting wiring to power while still on
B	Dec-23	JI	CGS	Updated to meet requirements of EN60079-29-1, 60079-2, and FM project PR465068 for ATEX approval, with UKEx and IECEx removed per FM request
C	Feb-24	JI	CGS	Reintroduced IECEx and UKEx information to manual with completion of QAN for ATEX, IECEx and UKEx requirements
D	Aug-25	JPE	RG	2.6.1 Control of Flame Cell Temperature updated to include TC3 2.6.5 Calibration and Reading Calculations updated with formulae 2.6.8 4 to 20 milliamp Output Signal: output conversion table added for clarity; use of terminating resistor added for non-use of 4-20mA outputs 2.6.12 Flameout temperature corrected to 425C (was 450C) 2.6.14 Table for Analyzer outputs and behaviors for all states added 2.7 Air Dilution Options: removal of K device type for 700 series analyzers due to inclusion of MINIMUM SPEED in SETTINGS 2.8.15 Humidity Limits added 3 Specifications: Response time table edited for clarity; Added Sample Pressure Range and Humidity Effect; Updated Storage Pressure Range to 500 to 1500 millibar (was -500 to +500 mbar gauge); Maximum Allowable Voltages and Clamping Voltage added 4.16 Security: Placing a cover on phototransistors added as a measure 4.2 Location and Mounting: Statement added: "Ensure that the analyzer

					exhaust does not dilute the sample" 4.4 Sample and Exhaust tubing updated to specify correct sample line lengths to prevent exhaust from influencing readings 5.2 Menu and Select Push Buttons updated to reflect behavior with software 7.01 and above 5.16 Settings Menu and 5.17 Security Menu added 5.18 Flashlight no longer accessible through Modbus 6.6, 7.2: Span Ends behavior clarified, and recovery time updated to 60 seconds (was 30 seconds) 6.11 Verification of LFL versus expected reading added 7.1.4 Calibration - Limits to Adjustment and 7.1.5 Calibration - Aborting Calibration Added 7.1.5 Substances added to Response Factor tables 9.3.3 Updated flame temperature ranges (Service Needed – CHK ZERO) 9.3.6 CHK TC3, 9.3.7 CHK RELAYS, 9.3.8 CHK Dilution, Fuel, Enclosure Service Needed Checks added 9.4 Display Test Added 9.15.3 Thermocouple gap from burner specified for air dilution units (1/2 inch) 10.9 Troubleshooting by Message updated with CHK TC3, CHK RELAYS, and CHK 4-20mA Messages 10.12 Service Needed - "Check" Relay and 10.13 Cold Start sections added Spare part pictures and labels updated Wiring diagram labels updated for CHW341 2.6.2 CAL RDNG and RDNG LAST CAL lower limit updated to 40%LFL (was 25%)

Drawing Index:

H7FTA718	Fax Troubleshooting Worksheet
C7FTA114	FTA Mounting
C7FTA132	Concentric Probe Installation
C7FTA169	Dual Tube Installation
D7FTA225	Assembly Diagram
C5FTA203	Wiring Diagram
B7FLT067	End of Line Filter
B7CAL130	Calibration Kit
C7SRV135	“Y” Purge Installation
C7HTR070	Heat Trace Mounting
H7FTA708	Maintenance Instructions

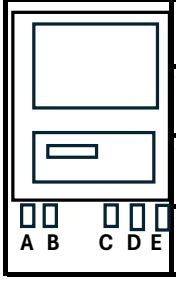
Control Instruments Troubleshooting Form
service_request@controlinstruments.com

Customer Name **Location** **Date**			**Person to contact** **Phone Number** **Email**		
Gauges	*Block temp		**4-20mA Outputs**	Milliamp output	
	*Block temp setting			4 Milliamp adjust	
	*Flame temp			Milliamp cal set	
Flow System	*Sample flow			Milliamp fault	
	*Cal sample flow			Min Milliamp	
	*Low Flow level		**Controls**	Control #1	
	*Check flow set			Control #2	
	*Orifice pressure			Control #3	
	*ORF press adj			Control #4	
Calibration	*Cal Reading			Relay Setting	
	*Zero as found			Check Relay NE	
	*Span as found		**Com Port**	Current baud	
	*Speed setting			Modbus address	
	Response time			Default Baud	
	*Zero adjustment			Baud auto adjust	
	*Calibration time			ID number	
	*Span temp set		**Settings**	*Device type	
	*Flame temp			*Software version	
	*Zero flame temp			*Serial number	
	*Span flame temp			*Year built	
	*RDNG last cal			Language	
Alarms	Warning level			*Flame gain adj	
	Danger level			TC time const	
	Rate alarm			Minimum speed	
	Warning latch			Dilution monitor	
	Danger latch			Fuel monitor	
	Fault latch			Purge monitor	
	Rdng fault level		**Security**	Lock setting	
	Auto ignition			Lock state	

Please include a description of the issue you are experiencing:

☐ Low Flow ☐ Flameout ☐ Cal Issue ☐ Check Alarms ☐ Erratic Readings ☐ LFL Drift ☐ Communication issue ☐ 4-20 mA Issue ☐ Other

Utilities	**A**	*Fuel inlet Pressure		PSIG
	B	*Dilution air inlet Pressure		PSIG
	C	*Span Inlet Pressure		PSIG
	D	*Zero Inlet Pressure		PSIG
	E	*Compressed air inlet Pressure		PSIG
	☐	120 VOLTS AC AT 50/60 Hz		
	☐	OPT230VAC, 220 VOLTS AC AT 50/60Hz		
	☐	Calibration Gas: 1.15% Ethylene in Air =		%LFL



Is the process under pressure or vacuum?

What is the value and the unit of measure?

Where is the sample pickup tube located?

Where does the exhaust tube go to?

***Current LED Display**

Check all that apply otherwise leave blank

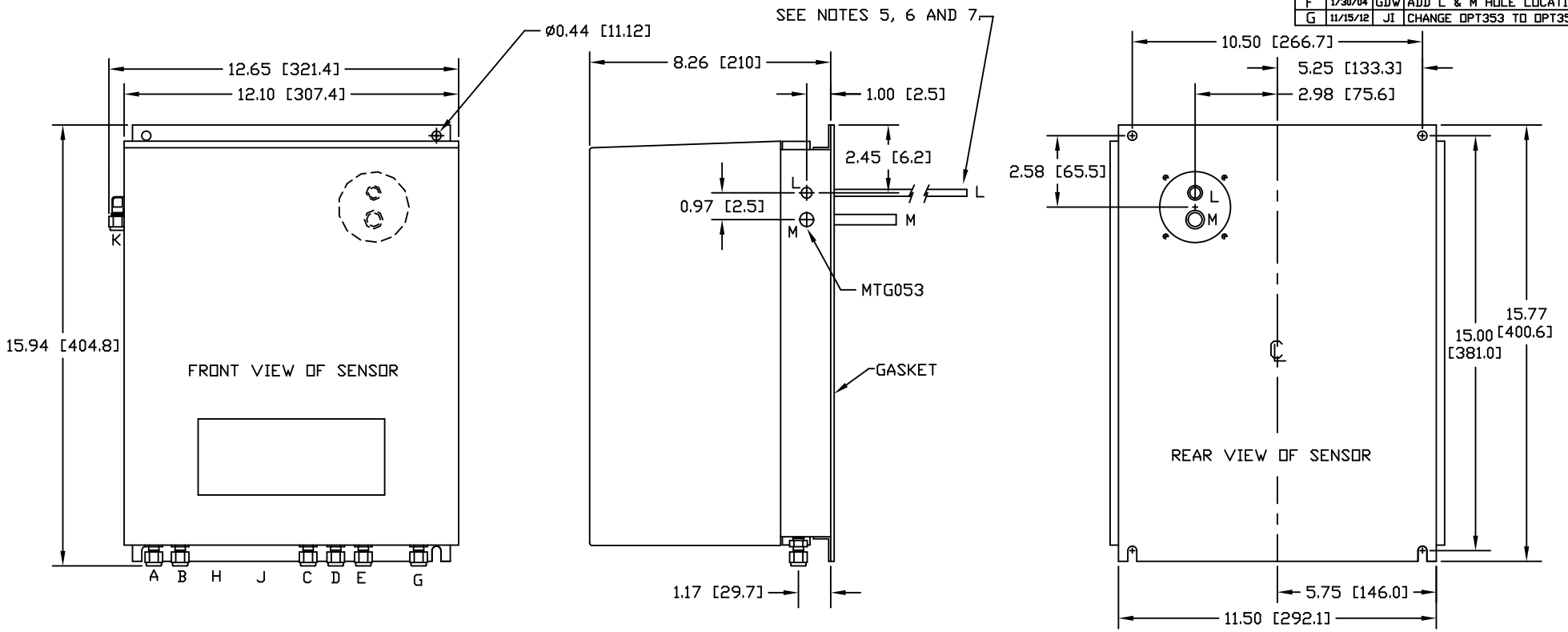
	Active	Flashing
SCAN	☐	☐
FAULT	☐	☐
DANGER	☐	☐
WARNING	☐	☐
CAL	☐	☐
CHECK	☐	☐
IGNITION	☐	☐
HEATER	☐	☐

Questions? Call 973-575-9114
Email: service_request@controlinstruments.com

*** Required Information** (more information may be requested depending on the issue)

C7FTA114

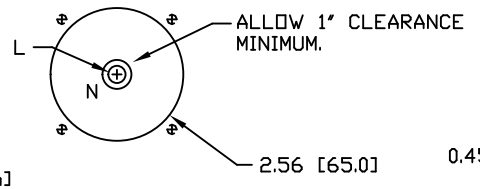
SYM	DATE	BY	REVISIONS
A	8/24/99	GDW	CORRECTED DRAWING
B	3/1/00	GDW	FMR APPROVAL
C	3/1/00	GDW	FMR APPROVAL
D	6/5/00	GDW	REMOVE DUCT VIEW
E	4/10/01	GDW	REMOVE "F" FITTING
F	1/30/04	GDW	ADD L & M HOLE LOCATION
G	11/15/12	JI	CHANGE OPT353 TO OPT353HP



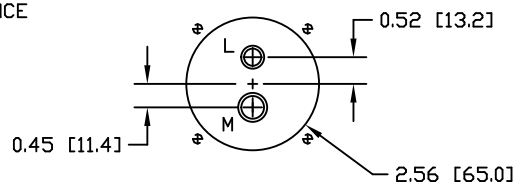
CONNECTIONS:

A. FUEL	1/4" TUBE
B. DILUTION AIR INLET (OPT353HP)	1/4" TUBE
C. PURGE PRESSURE REFERENCE (OPTIONAL)	1/4" TUBE
D. SPAN GAS	1/4" TUBE
E. ZERO GAS	1/4" TUBE
F.	
G. COMPRESSED AIR.	1/4" TUBE
H. POWER INPUT.	3/4 NPT
J. COMMUNICATION WIRING.	3/4 NPT
K. PURGE EXHAUST (NO CONNECTION)	1/4" TUBE
L. SAMPLE INLET (4 FT SUPPLIED)	1/4" TUBE
M. EXHAUST (2 FT SUPPLIED)	3/8" TUBE
N. EXHAUST, HTR063 (3.5 FT SUPPLIED)	3/4" TUBE

HTR063
CONCENTRIC PROBE



SAMPLE/EXHAUST LOCATION
MTG054 WITHOUT HTR063



NOTES:

1. ALL DIMENSIONS = INCH [mm]
2. SAMPLE AND EXHAUST CAN BE TWO SMALL OR ONE LARGE CLEARANCE HOLE. A 2" [51mm] DIAMETER HOLE WILL CLEAR BOTH HOLES.
3. INSULATE OR SUPPORT TUBES SO THEY DO NOT CONTACT COOL SURFACES.
4. MTG053 HAS NO SAMPLE/EXHAUST TUBES SUPPLIED.
5. THE SAMPLE PROBE SHOULD BE LOCATED WHERE IT WILL GIVE A REPRESENTATIVE READING OF THE PROCESS.
6. THE EXHAUST TUBE SHOULD BE LOCATED WHERE IT WILL NOT DILUTE THE SAMPLE.
7. CUT SUPPLIED SAMPLE AND EXHAUST TUBES TO LENGTH.
8. SENSOR WEIGHT 35-40 LBS.

REQ	ITEM	PART NO.	DESCRIPTION

LIST OF MATERIALS

DO NOT CHANGE WITHOUT AGENCY APPROVAL

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE

TOLERANCES
DEC. ± 0.01" (0.25) FRACT. ± ANG. ±

PREVEX SERIES FTA MOUNTING

CLIENT

SPEC. NO.

CONTROL INSTRUMENTS Corporation
25 Law Drive
Fairfield, N.J. 07004 U.S.A.

DWN.	ENGR.	SCALE
S.A.A.	G.D.W.	1=3
DATE 8/9/99	DATE 8/9/99	REVISION G

DRAWING NO. C7FTA114

C7FTA132

SYM	DATE	BY	REVISIONS
A	4/5/00	JI	UPDATE SDC SHOWN
B	9/1/00	GW	ADD PROBE INSULATION
C	2/24/04	GW	CORRECT OF TYPO'S
D	8/26/25	JPE	UPDATE NOTE 2

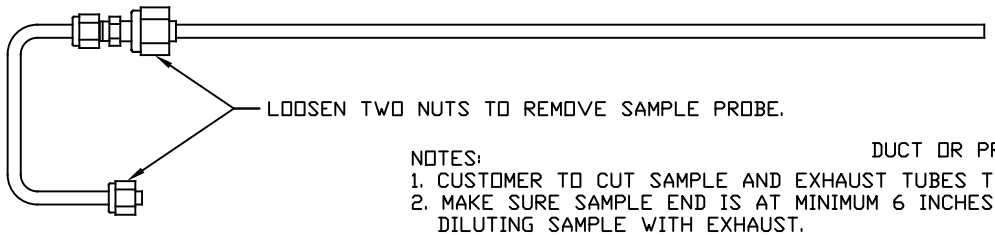
FLUSH MOUNTING SENSOR TO PROCESS IS RECOMMENDED.
IF FLUSH MOUNTING IS NOT POSSIBLE, MINIMIZE THE DISTANCE BETWEEN THE SENSOR AND PROCESS. THIS WILL MINIMIZE ANY COLD SPOTS IN THE SAMPLE TUBE. COLD SPOTS CAN CAUSE SAMPLE CONDENSATION RESULTING IN AN ARTIFICIALLY LOW SAMPLE READING.

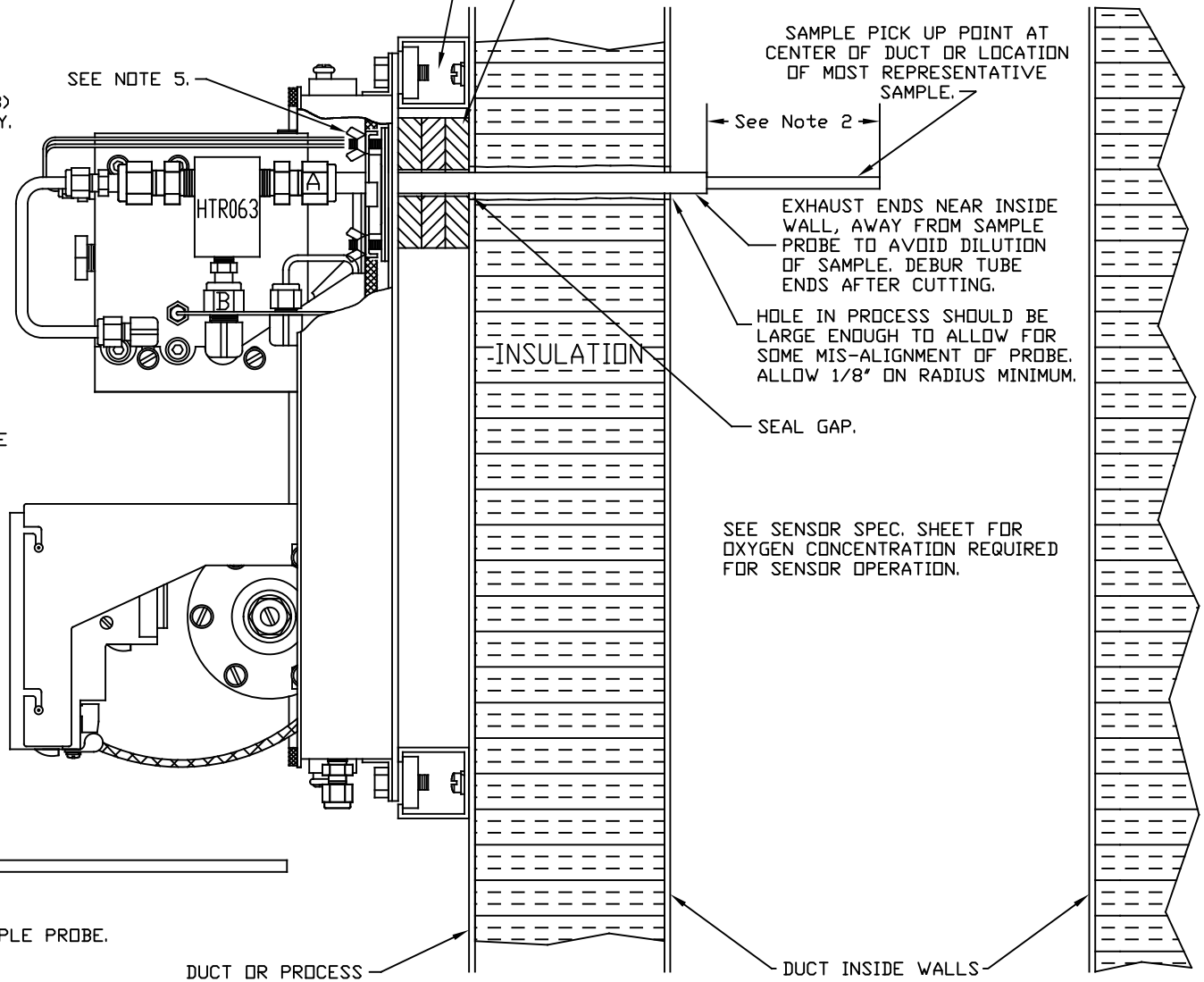
INSTALLING SAMPLE AND EXHAUST

1. REMOVE BODY OF HEAT EXCHANGER PROBE(LABELED HTR063) BY LOOSENING NUT MARKED "B" AND LIFTING OUT ASSEMBLY.
2. CUT 3/4" EXHAUST TUBE TO LENGTH. DEBUR INSIDE AND OUTSIDE OF BOTH ENDS.
3. SLIDE EXHAUST TUBE THROUGH HOLE IN BACK OF SENSOR, MAKING SURE NOT TO DAMAGE O-RING SEAL. LUBRICATE O-RING IF NEEDED.
4. PLACE BODY OF PROBE(REMOVED IN STEP 1) IN PLACE, TIGHTEN NUT "B" FINGER TIGHT.
5. SLIDE EXHAUST TUBE INTO FITTING ON PROBE BODY UNTIL IT SEATS INSIDE FITTING. TIGHTEN FITTING ONTO EXHAUST TUBE. LOOSEN WING NUTS IF NEEDED TO ALIGN TUBE WITH FITTING.
6. TIGHTEN NUT "B".
7. CUT 1/4" SAMPLE TUBE TO LENGTH. DEBUR INSIDE AND OUTSIDE OF CUT END.
8. SLIDE END OF SAMPLE TUBE INTO PROBE ASSEMBLY, IN SIDE OPPOSITE EXHAUST TUBE. SLIDE HOOKED END OF TUBE INTO FITTING ON HEATED BLOCK. TIGHTEN THIS NUT AND THE NUT ON THE PROBE BODY.
9. SEAL EXHAUST TUBE TO DUCT.

TO REMOVE SAMPLE PROBE OR EXHAUST TUBE.

1. UNSCREW FITTING MARKED "A".
2. REMOVE SAMPLE TUBE BY LOOSENING TWO NUTS.
3. LOOSEN NUT MARKED "B".
4. PUSH EXHAUST TUBE APPROXIMATELY 3/8" TOWARDS BACK OF SENSOR.
5. LIFT PROBE ASSEMBLY UP AND OUT OF SENSOR.
6. PULL EXHAUST TUBE OUT OF SENSOR IF NEEDED. IF EXHAUST TUBE IS STUCK, WING NUTS CAN BE REMOVED AND THE SEAL ASSEMBLY AND BRACKET REMOVED WITH THE EXHAUST TUBE.





NOTES:
1. CUSTOMER TO CUT SAMPLE AND EXHAUST TUBES TO LENGTH.
2. MAKE SURE SAMPLE END IS AT MINIMUM 6 INCHES [15.24cm] AWAY FROM EXHAUST TO AVOID DILUTING SAMPLE WITH EXHAUST.
3. SOLVENTS WITH HIGH FLASH POINTS OR HIGH DEW POINTS WILL CONDENSE WHEN EXPOSED TO COLD SPOTS. KEEPING THE SAMPLE LINE HOT IS IMPORTANT TO ENSURE RELIABLE READINGS AND PREVENTING CONDENSATE FROM PLUGGING THE SAMPLE LINE.
4. USE AN END OF LINE FILTER FOR SAMPLES WITH PARTICULATE.
5. LOOSEN WING NUTS TO MAKE ADJUSTMENTS IN POSITION OF EXHAUST PROBE.
6. USE SUPPLIED GASKET TO CREATE A SEAL BETWEEN THE BACK OF THE SENSOR AND THE PROCESS IF FLUSH MOUNTING.

REQ	ITEM	PART NO.	DESCRIPTION

LIST OF MATERIALS

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE
TOLERANCES DEC. ± FRACT. ± ANG. ±

HTR063 PROBE INSTALLATION

CLIENT | SPEC. NO.

CONTROL INSTRUMENTS Corporation
25 Law Drive
Fairfield, N.J. 07004 U.S.A.

DWN.	ENGR.	SCALE
GDW	GDW	0.5=1
DATE 12/2/99	DATE 12/2/99	REVISION D

DRAWING NO. C7FTA132

C7FTA169

SYM	DATE	BY	REVISIONS
A.	8/26/2025	JPE	Updated note regarding distance sample end should be away from exhaust

FLUSH MOUNTING SENSOR TO PROCESS IS RECOMMENDED. IF FLUSH MOUNTING IS NOT POSSIBLE, MINIMIZE THE DISTANCE BETWEEN THE SENSOR AND PROCESS. THIS WILL MINIMIZE ANY COLD SPOTS IN THE SAMPLE TUBE. COLD SPOTS CAN CAUSE SAMPLE CONDENSATION RESULTING IN AN ARTIFICIALLY LOW SAMPLE READING.

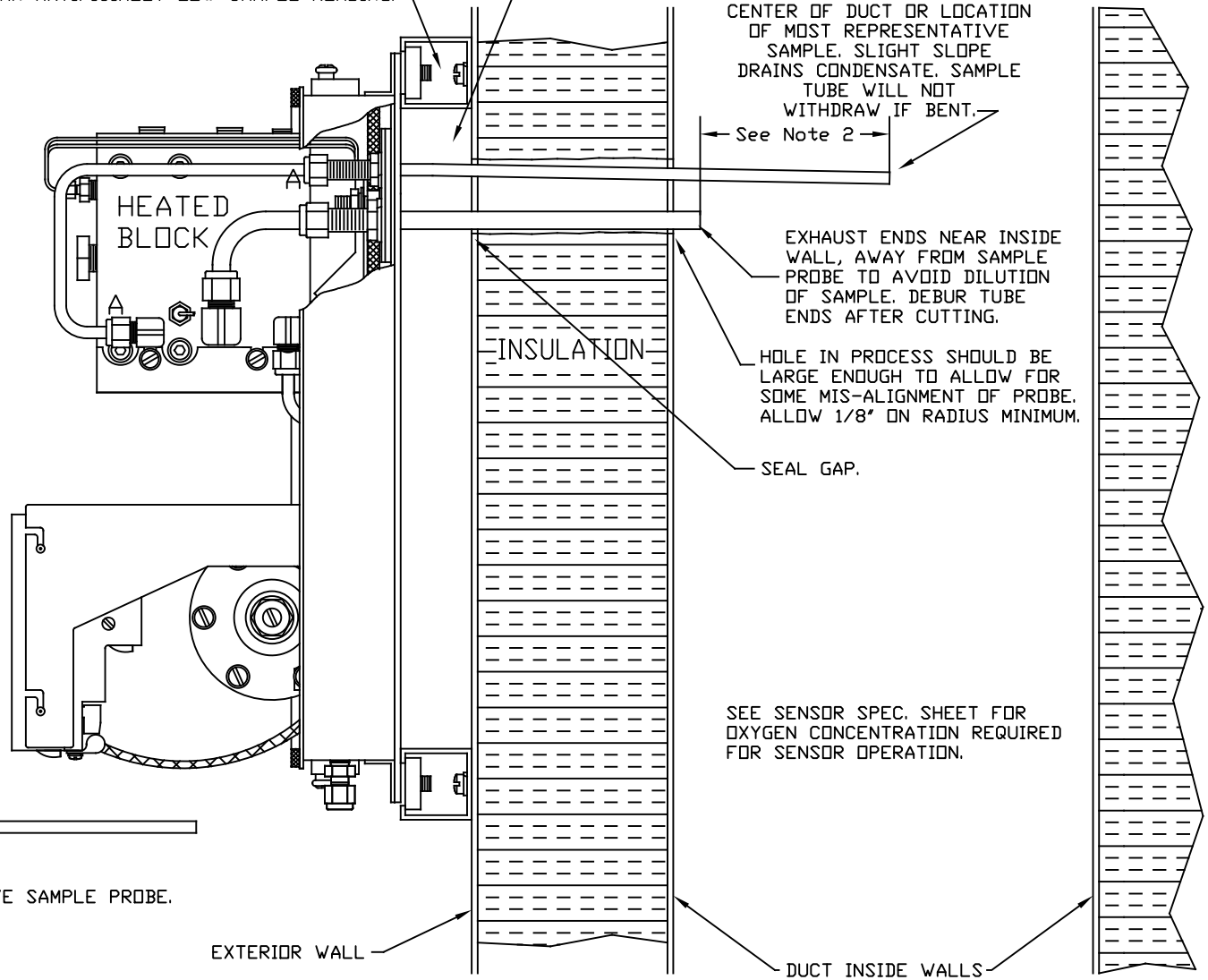
INSULATION OVER ANY EXPOSED TUBING.

SAMPLE PICK UP POINT AT CENTER OF DUCT OR LOCATION OF MOST REPRESENTATIVE SAMPLE. SLIGHT SLOPE DRAINS CONDENSATE. SAMPLE TUBE WILL NOT WITHDRAW IF BENT.

See Note 2

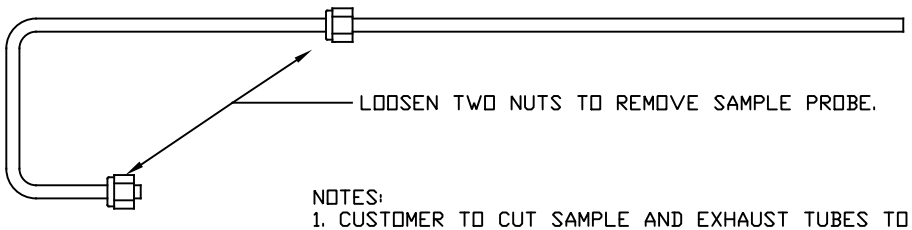
INSTALLING SAMPLE AND EXHAUST

1. CUT 3/8" EXHAUST TUBE TO LENGTH. DEBUR INSIDE AND OUTSIDE OF BOTH ENDS.
2. SLIDE EXHAUST TUBE THROUGH FITTING ON BACK OF SENSOR. ALIGN SHORT END WITH FITTING ON BLOCK. LOOSEN 4 SCREWS ON BRACKET IF NEEDED. TIGHTEN FITTING NUTS ON TUBING.
3. CUT 1/4" SAMPLE TUBE TO LENGTH. DEBUR INSIDE AND OUTSIDE OF CUT END.
4. SLIDE END OF SAMPLE TUBE THROUGH FITTING ON BACK OF SENSOR. PLACE OTHER END OF TUBE IN FITTING ON BLOCK. TIGHTEN BOTH FITTING NUTS ONTO THE TUBE.
5. SEAL TUBES TO DUCT.



TO REMOVE SAMPLE PROBE.

1. REMOVE SAMPLE TUBE BY LOOSENING TWO NUTS, MARKED "A".
2. PULL SAMPLE TUBE OUT OF SENSOR.



NOTES:
1. CUSTOMER TO CUT SAMPLE AND EXHAUST TUBES TO LENGTH.
2. MAKE SURE SAMPLE END IS AT MINIMUM 6 INCHES [15.24cm] AWAY FROM EXHAUST TO AVOID DILUTING SAMPLE WITH EXHAUST.
3. SOLVENTS WITH HIGH FLASH POINTS OR HIGH DEW POINTS WILL CONDENSE WHEN EXPOSED TO COLD SPOTS. KEEPING THE SAMPLE LINE HOT IS IMPORTANT TO ENSURE RELIABLE READINGS AND PREVENTING CONDENSATE FROM PLUGGING THE SAMPLE LINE.
4. USE AN END OF LINE FILTER FOR SAMPLES WITH PARTICULATE.
5. LOOSEN WING NUTS TO MAKE ADJUSTMENTS IN POSITION OF EXHAUST PROBE.
6. USE SUPPLIED GASKET TO CREATE A SEAL BETWEEN THE BACK OF THE SENSOR AND THE PROCESS IF FLUSH MOUNTING. IF NOT FLUSH MOUNTING, SEAL PROCESS BY OTHER MEANS.

REQ	ITEM	PART NO.	DESCRIPTION

LIST OF MATERIALS

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE
TOLERANCES DEC. ± FRACT. ± ANG. ±

PrevEx DUCT INSTALATION

CLIENT

SPEC. NO.

CONTROL INSTRUMENTS Corporation
25 Law Drive
Fairfield, N.J. 07004 U.S.A.

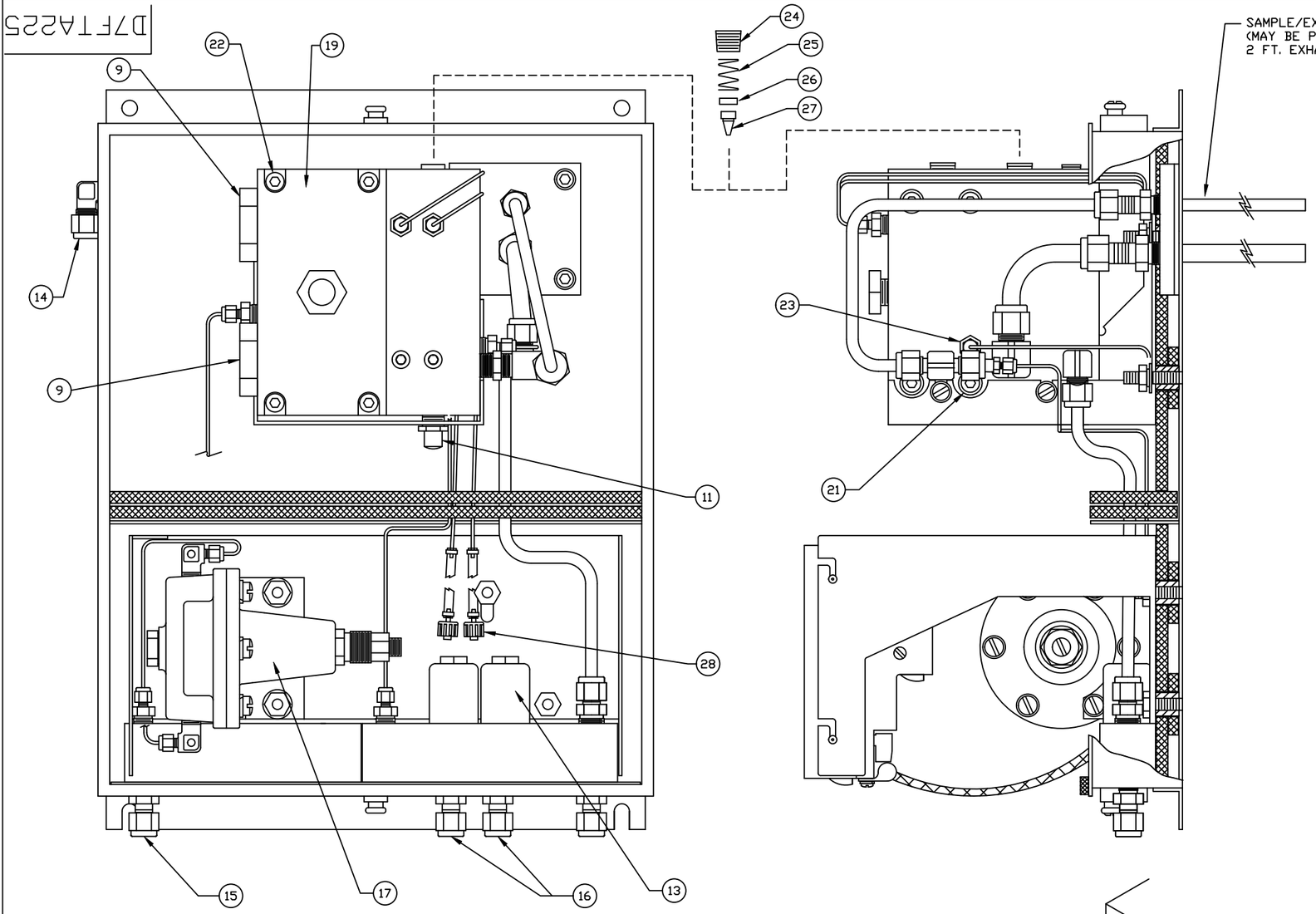
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DATE 2/27/03	DATE 2/27/03	REVISION A

DRAWING NO. C7FTA169

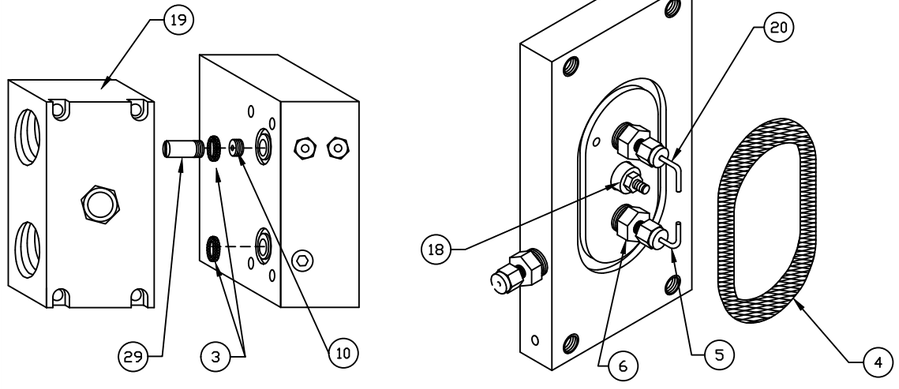
D7FTA225

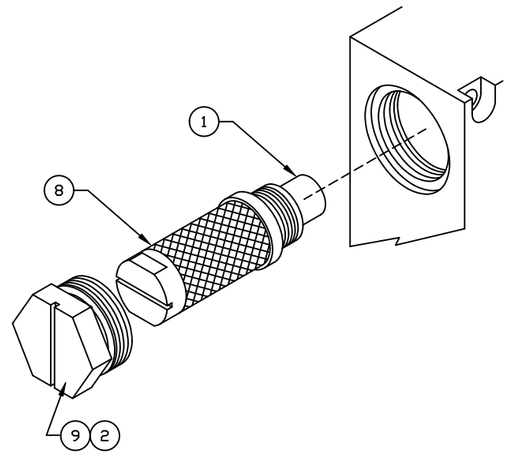
SYM	DATE	BY	REVISIONS

SAMPLE/EXHAUST TUBES SUPPLIED WITH ANALYZER
(MAY BE PACKED SEPARATELY)
2 FT. EXHAUST, 4 FT. SAMPLE, PREBENT.



NOTE: NOT ALL PARTS WILL APPEAR ON ALL SENSORS.





ITEM	PART #	DESCRIPTION
30	THC008	THERMOCOUPLE, OVER TEMPERATURE
29	FLT087	ORIFICE FILTER
28	FNMLL1/16TR	FITTING, OPD PNEUMATIC (RED)
28	FNMLL1/16TBL	FITTING, OPD PNEUMATIC (BLACK)
27	NZL001	NOZZLE, ASPIRATOR
26	SPC009	SPACER, ASPIRATOR
25	SPR013	SPRING, ASPIRATOR
24	PLG002	PLUG, 1/4 NPT
23	THC002R	THERMOCOUPLE, BLOCK TEMPERATURE
23	FS1BZSS000	NUT, THERMOCOUPLE FITTING
22	MSC023	MACHENE SCREW, FLAME CELL TO BURNER PLATE
21	MSC007	MACHINE SCREW, BYPASS TO FLAME CELL
20	THC002R	THERMOCOUPLE, FLAME TEMPERATURE
19	SNP219H	FLAME CELL BLOCK
18	SPK021	SPARK PLUG
17	PRV058R	FUEL REGULATOR
16	SNP366	RESTRICTOR, CALIBRATION GAS
15	SNP374	RESTRICTOR, FUEL INLET
14	FS40022000	AIR PURGE OUTLET
13	VLV083	SOLENOID VALVE OPERATOR, 230VAC
13	VLV079	SOLENOID VALVE OPERATOR, 120VAC
11	HTR078R	HEATER, 230VAC
11	HTR077R	HEATER, 120VAC
10	ORF024	ORIFICE, SAMPLE FLOW
9	SNP188	FLAME ARRESTOR CAP (2X)
8	SNP189	FLAME ARRESTOR (2X)
6	RSP018	FUEL RESTRICTOR, PROPANE
6	BRN071	FUEL RESTRICTOR, HYDROGEN(EXC.W/OPT353HP)
5	BRN006	BURNER, PROPANE FUEL
4	ORK046	O-RING, FLAME CELL, KALREZ, (SNR675 ONLY)
3	ORK047	O-RING, FLAME CELL TO BYPASS BLOCK (2X) KALREZ,(SNR675 ONLY)
2	ORK005	O-RING, FLAME ARRESTOR COVER, KALREZ, (SNR675 ONLY)
4	ORG044	O-RING, FLAME CELL
3	ORG041	O-RING, FLAME CELL TO BYPASS BLOCK (2X)
2	ORG005	O-RING, FLAME ARRESTOR COVER(2X)
1	FLT012	FILTER ELEMENT (2X)

REQ	ITEM	PART NO.	DESCRIPTION
LIST OF MATERIALS			

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE
TOLERANCES DEC. ± FRACT. ± ANG. ±

PREVEX SERIES ASSEMBLY

CLIENT

SPEC. NO.

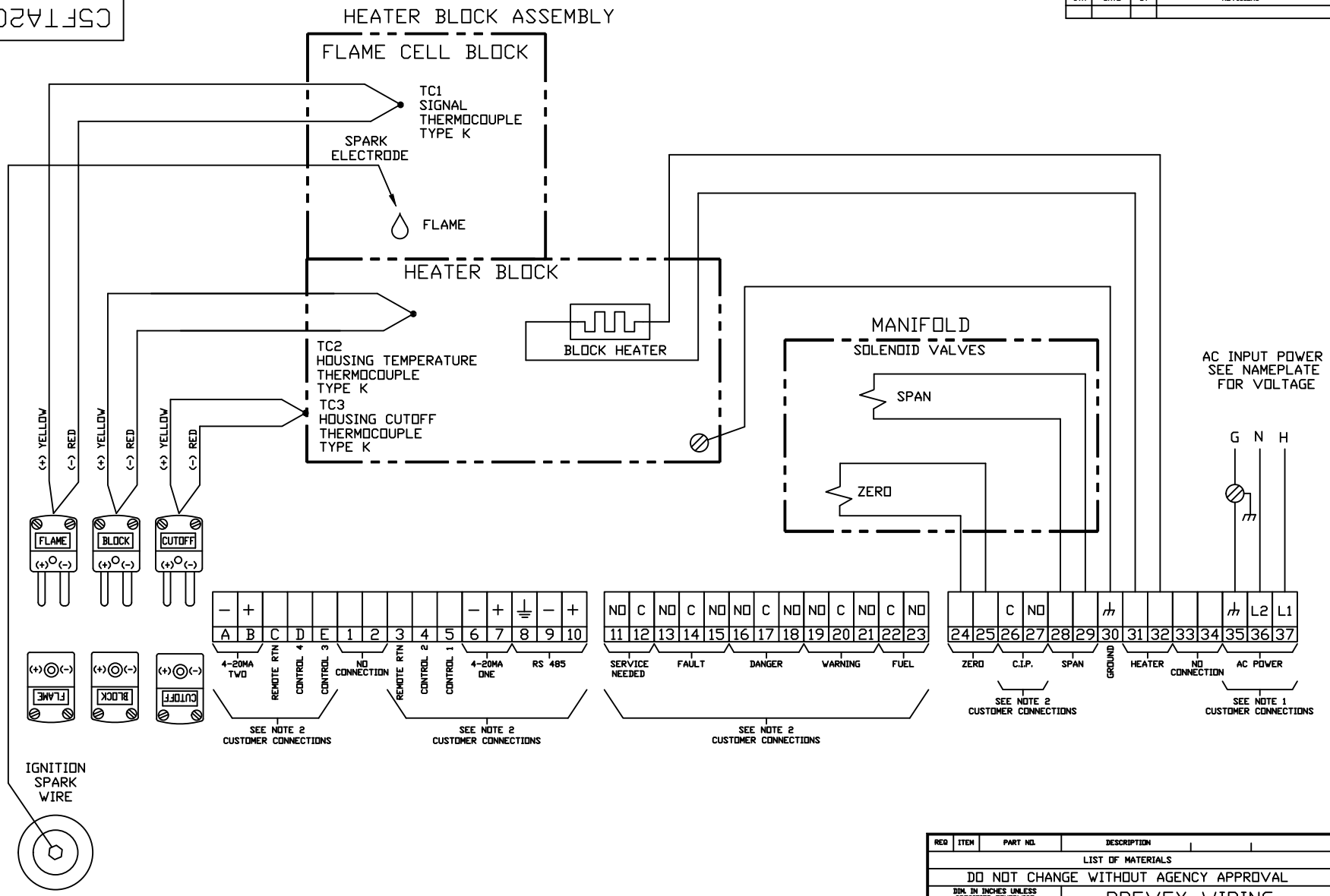
CONTROL INSTRUMENTS Corporation
25 Law Drive
Fairfield, N.J. 07004 U.S.A.

DWN.	ENGR.	SCALE
GW	GW	0.5=1
DATE 8/12/24	DATE 8/12/24	REVISION 0

DRAWING NO. D7FTA225

C5FTA203

SYM	DATE	BY	REVISIONS



NOTE 1. POWER IS FACTORY CONFIGURED EITHER FOR 120 VAC OR 230 VAC BUT NOT BOTH.
USE OF INCORRECT VOLTAGE WILL CAUSE DAMAGE. SEE NAMEPLATE FOR CORRECT VOLTAGE.
NOTE 2. TERMINALS A - E, 3 - 10, 11 - 23, 26 - 27 AND 35 - 37 ARE CUSTOMER CONNECTIONS.
ALL OTHER TERMINALS ARE FACTORY WIRED AND COMPLETE.

REQ	ITEM	PART NO.	DESCRIPTION
			LIST OF MATERIALS

DO NOT CHANGE WITHOUT AGENCY APPROVAL

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE
TOLERANCES DEC. ± FRACT. ± ANG. ±

PREVEX WIRING
700 SERIES

CLIENT | SPEC. NO.

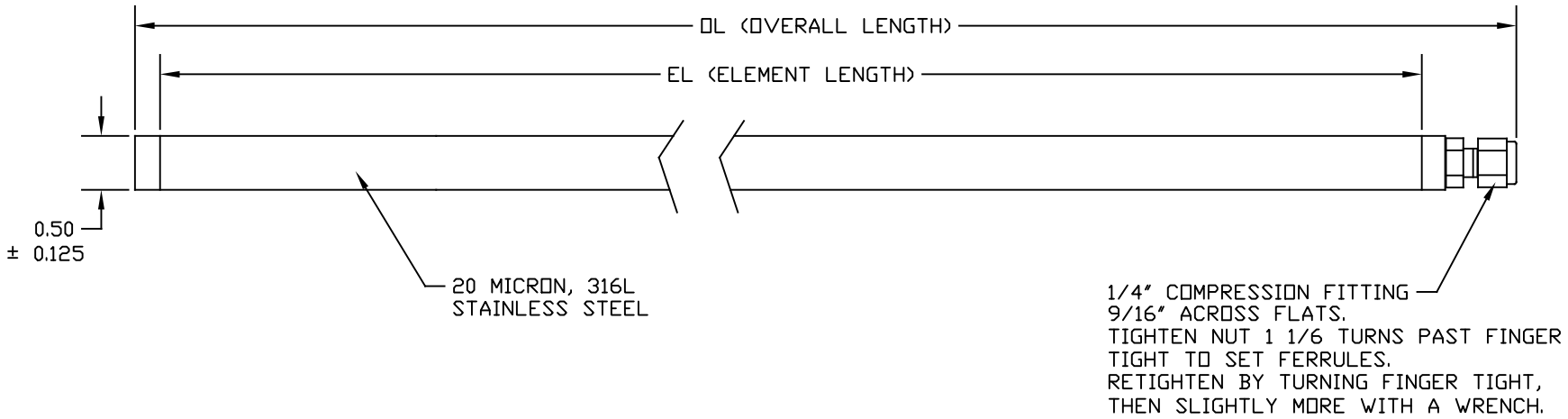
CONTROL INSTRUMENTS Corporation
25 Law Drive
Fairfield, N.J. 07004 U.S.A.

DWN.	ENGR.	SCALE
GW	GW	
DATE 4/11/22	DATE 4/11/22	REVISION 0

DRAWING NO. C5FTA203

B7FLT067

SYM	DATE	BY	REVISIONS
A	9/4/2025	JPE	Added Note 1



FILTER NUMBER	DIMENSION EL	OL
FLT067	12"	13"
FLT083	6"	7"

Notes
1. MAKE SURE CONNECTION TO SAMPLE END IS AT MINIMUM 6 INCHES [15.24CM] AWAY FROM EXHAUST TO AVOID DILUTING SAMPLE WITH EXHAUST

REQ	ITEM	PART NO.	DESCRIPTION
LIST OF MATERIALS			

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE
TOLERANCES DEC. ± FRACT. ± ANG. ±

END OF LINE FILTER

CLIENT

SPEC. NO.

CONTROL INSTRUMENTS Corporation
25 Law Drive
Fairfield, N.J. 07004 U.S.A.

DWN.	ENGR.	SCALE
GW	GW	
DATE 4/30/03	DATE 4/30/03	REVISION A

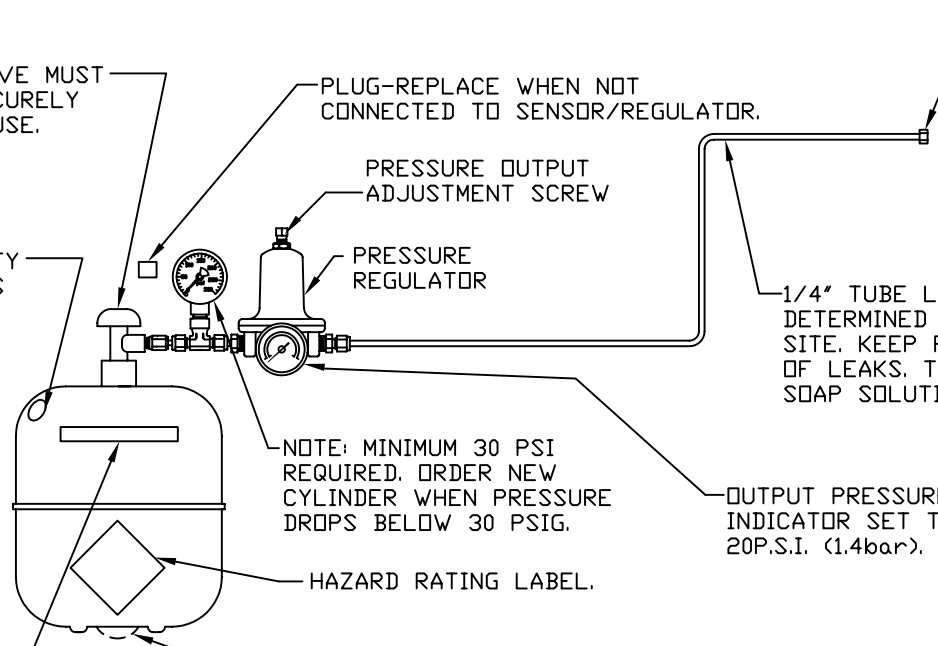
DRAWING NO. B7FLT067

B7CAL130

SYM	DATE	BY	REVISIONS
A	12/6/04	GW	ADD CAL. MINUTES

INSTRUCTIONS:

1. ATTACH REGULATOR INLET TO CALIBRATION CYLINDER.
2. ATTACH REGULATOR OUTLET TO SENSOR CALIBRATION INLET USING 1/4"O.D. TUBING. (NOT PROVIDED)
3. OPEN CYLINDER "SHUT OFF" VALVE.
4. SET REGULATOR OUTPUT PRESSURE TO 20p.s.i. (1.4bar).
5. LEAK TEST ALL LINES AND FITTINGS.



NOTES:

1. CYLINDERS ARE DISPOSABLE NOT TO BE REFILLED. EMPTY IN SAFE AREA BEFORE DISCARDING.
2. HARDWARE KIT DOES NOT INCLUDE CYLINDER OR TUBING.

DIMENSIONS 9"X16 1/2" (23cmX42cm) CONTAINS 6.4 FT^3 @ 250psi (182 LITERS @ 17bar) OF USABLE GAS.
CYLINDER: 7LB (3.2 KG)
REGULATOR: 1LB (0.45 KG)

CONTAINES ABOUT 45 MINUTES OF CALIBRATION GAS WHEN USED WITH A PREVEX ANALYZER.

CONTENTS OF CYLINDER	REPL. CYL#	HARDWARE KIT#
1.15% ETHYLENE IN AIR	CYL018	PRV022
100% COMPRESSED AIR	CYL016	PRV022

LIST OF CALIBRATION KITS

REQ	ITEM	PART NO.	DESCRIPTION

LIST OF MATERIALS

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE
TOLERANCES
DEC. ± FRACT. ± ANG. ±

CALIBRATION KIT

CLIENT

SPEC. NO.

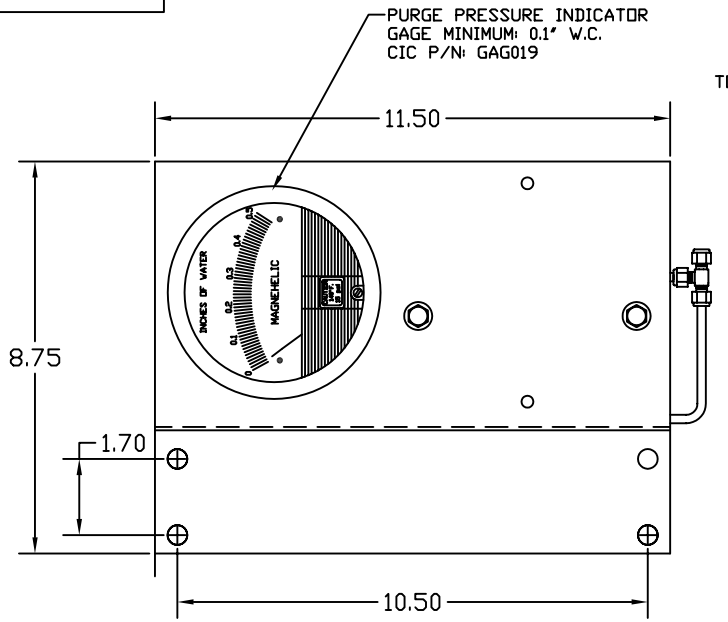
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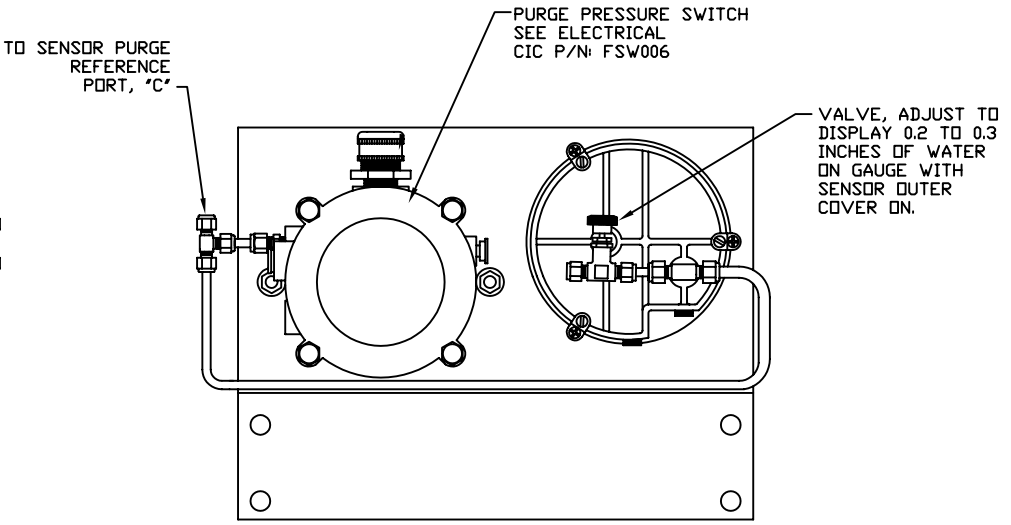
DWN. GW	ENGR. GW	SCALE A
DATE 2/15/02	DATE 2/15/02	REVISION

DRAWING NO. B7CAL130

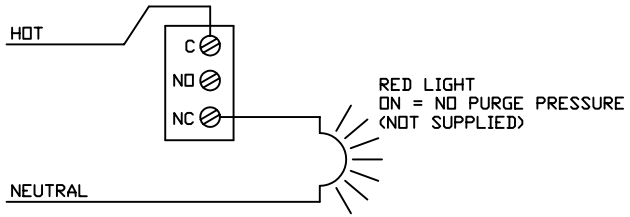
C7SRV135

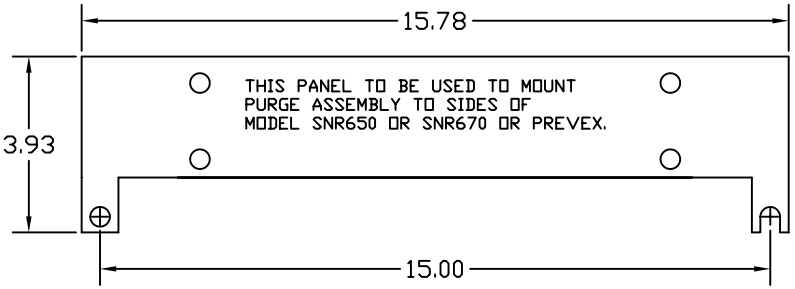
SYM	DATE	BY	REVISIONS
A	3/31/03	JI	ADDED PREVEX
B	4/4/05	GW	ADD VLV027





CONTACT RATING:
15 AMP @ 125 VAC MAX
1/8 HP @ 125 VAC
CONTACT OPEN ON DESCENDING
PRESSURE @ 0.1" W.C.





REQ	ITEM	PART NO.	DESCRIPTION
LIST OF MATERIALS			

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE
TOLERANCES: DEC. ±, FRACT. ±, ANG. ±

INSTALLATION ELECTRICAL "Y" PURGE

CLIENT

SPEC. NO.

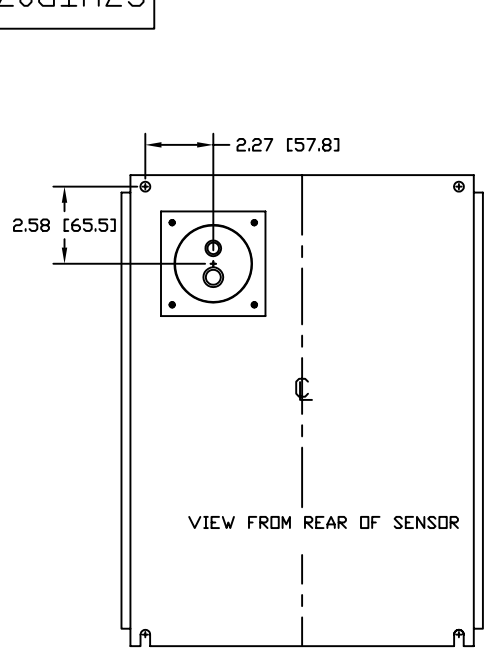
CONTROL INSTRUMENTS
Corporation
25 Law Drive
Fairfield, N.J. 07004 U.S.A.

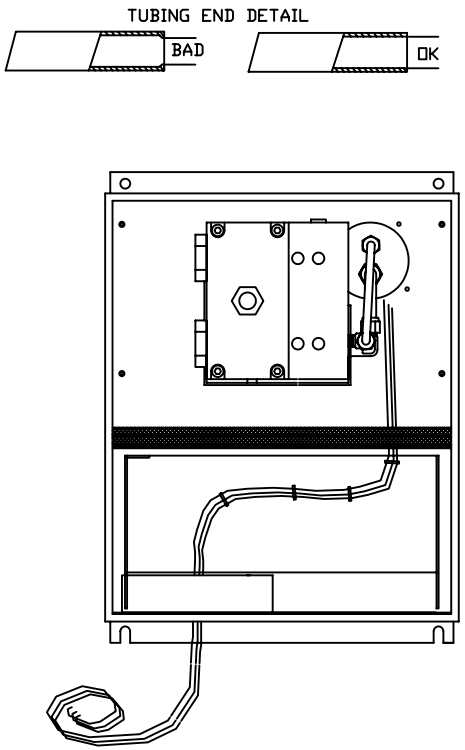
DWN.	ENGR.	SCALE
P.K.P	P.K.P.	.5=1
DATE 06/02/00	DATE 06/02/00	REVISION B

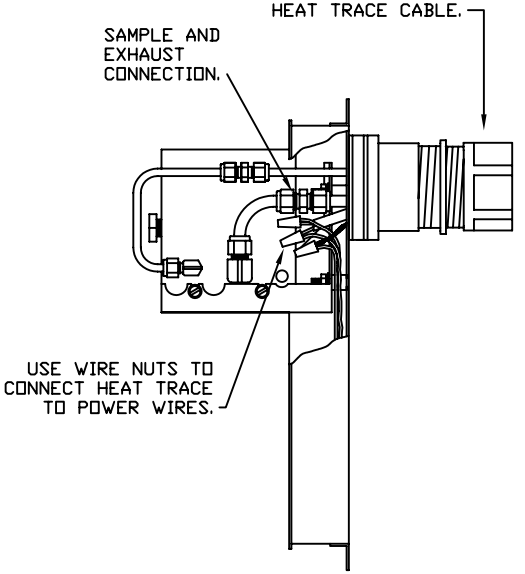
DRAWING NO. C7SRV135

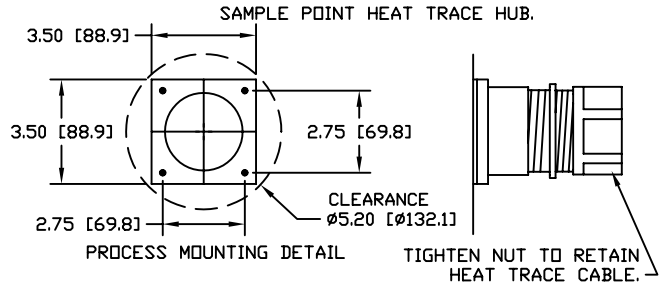
C7HTR070

SYM	DATE	
A	6/27/00	GW CREATE DRAWING
B	2/24/05	GW ADD NOTE 7.
C	3/15/07	GW ADD NOTE 8, 9.
D	12/3/09	GW REMOVE HTR021.









GENERAL NOTES:
1. ALL DIMENSIONS = INCH [mm]

GUIDELINES FOR INSTALLING HEAT TRACE

1. A HEAT TRACED TUBE BUNDLE IS USED WHEN THE SENSOR CANNOT BE MOUNTED AT THE SITE TO BE SAMPLED.
2. THE PURPOSE OF THE HEAT TRACED TUBE BUNDLE IS TO TRANSPORT THE SAMPLE AND EXHAUST VAPORS TO AND FROM THE SOURCE WITHOUT LOSING ANY OF THE SAMPLE TO CONDENSATION.
3. CARE MUST BE TAKEN TO GUARD THE SAMPLE AND EXHAUST TUBES AGAINST COLD SPOTS.
4. CARE MUST BE TAKEN TO GUARD THE ELECTRICAL ENDS OF THE HEATER WIRES FROM WEATHER AND SHORTS.
5. USE HIGH TEMPERATURE ELECTRICAL TAPE WHEN PREPARING HEATER WIRE ENDS.
6. PREVENT EXHAUST GASES FROM MIXING WITH SAMPLE INTAKE.
7. SEE H7HTR066 TO SEAL HEAT TRACE ENDS.
8. INCLUDES SAMPLE POINT HUB, GASKET AND HEAT TRACE CABLE GRIP.
9. HEAT TRACE NOMINAL DIAMETER: 1.75"
   MINIMUM BEND RADIUS: 15"
   POWER CONSUMPTION: 20 WATTS/FT.

REQ	ITEM	PART NO.	DESCRIPTION

LIST OF MATERIALS

DIM. IN INCHES UNLESS SPECIFIED OTHERWISE
TOLERANCES
DEC. ± 0.01" (0.25) FRACT. ± ANG. ±

SNR670 SERIES HEAT TRACE MOUNTING

CLIENT | SPEC. NO.

CONTROL INSTRUMENTS Corporation
25 Law Drive
Fairfield, N.J. 07004 U.S.A.

DWN.	ENGR.	SCALE
G.D.W.	G.D.W.	1=3
DATE 8/3/00	DATE 8/3/00	REVISION D

DRAWING NO. C7HTR070

1 Maintenance Instructions

Menu Settings

Press Menu to move between menus → Press Select to enter menu

Press Menu again to display registers → Press Select again to display register's value

Press Select again to change a setting → ! Wait until flashing stops to keep the new setting; ! Press Menu to escape and keep the old setting

Press Menu to go to next register

Press Select on EXIT register to leave menus

*After 3 minutes of inactivity the display automatically exits the menu

Menus, Registers & Explanations

COMMANDS *Reset latched alarms* *Acknowledge service needed messages* *Initiate and control calibrations*	Alarm Reset	Clear latched alarms and peak readings, retry ignition
	Acknowledge	Clear service needed messages
	Zero Adjust	Adjust flammability readings within +/-3% LFL to 0% LFL
	Zero Calibration	Start calibration of zero using air = 0% LFL
	Span Calibration	Start span calibration using test gas, typically 1.15% Ethylene = 60%
	Full Calibration	Automatically zero and then span using test gases
	Zero/Span/Full Test	Test similar to calibration, except no adjustment is made
GAUGES *Display the reading and status, flow and temperatures, and other operating parameters*	PEAK HIGH %LFL	Highest flammability reading since the last "reset" command, in % LFL
	PEAK LOW %LFL	Lowest flammability reading since the last "reset" command, in % LFL
	BLOCK TEMP	Temperature of the flame cell block, °C
	BLOCK SETTING	Temperature setting of the flame cell block, °C
	SAMPLE FLOW	Sample flow rate, in Standard Liters per Minute (LPM)
	FLAME TEMP	Current flame temperature, thermocouple TC1 reading, in °C
	ZERO FLAME TEMP	Flame temperature measured during the last successful zero calibration, in °C
	SPAN FLAME TEMP	Increase in flame temperature measured during last successful span calibration, in °C
FLOW SYSTEM *Analyzers operating flow readings and setpoints*	SAMPLE FLOW	Current sample flow rate, in liters per minute (LPM)
	CAL SAMPLE FLOW	Sample flow measured during calibration, in LPM
	LOW FLOW LEVEL	Flow rate at which a low flow alarm trips, in LPM
	CHECK FLOW SET	Change in flow rate that activates "Service Needed, Check Flow", in LPM
	ORIFICE PRESSURE	Orifice pressure drop measured, in Kilopascal (KPA)
	ORF PRESS ADJ	An adjustment to the sample flow pressure sensor, in KPA
CALIBRATION *Calibrate the analyzer reading* *Display the calibration settings*	CAL READING	Flammability of the span gas, for example, 1.15% Ethylene = 60% LFL
	SPEED SETTING	Speed of response & noise filter, 0 = Off, 1 = low, 2 = Medium, 3 = High
	CALIBRATION TIME	Duration of the calibration, minutes
	SPAN TEMP SET	Setting for target flame temperature rise for span calibration, in °C
	ZERO FLAME TEMP	Flame temperature recorded during last successful zero calibration, in °C
	SPAN FLAME TEMP	Rise in flame temperature recorded during last successful span calibration, in °C
ALARMS *Configure alarms*	WARNING LEVEL	High flammability warning alarm, in % LFL
	DANGER LEVEL	High, high flammability danger alarm, in % LFL
	RATE ALARM	Enable or disable the rate-of-rise alarm, 0 = disabled,1= enabled
	WARNING LATCH	Latch warning alarm & relay until manual reset, 0 = autoreset (disabled), 1= latched (enabled)
	DANGER LATCH	Latch danger alarm & relay until manual reset, 0 = autoreset (disabled), 1= latched (enabled)
	FAULT LATCH	Latch fault & fault relay until manual reset, 0 = autoreset (disabled), 1= latched (enabled)
	RDNG FAULT LEVEL	Trip a fault on low downscale flammability readings
ALARM TESTS *Used for commissioning and testing of outputs*	ALARM RESET	Reset command halts testing
	TEST WARNING/DANGER/ FAULT/CAL/CHECK RELAY	Turn on warning/danger/fault/Cal/Check relay for up to one hour, or until an alarm reset command
	GAS ALARM TEST	Open span valve for up to 60 minutes, or until an alarm reset command
4-20MA OUTPUTS *Test the 4-20 milliamp output & configure milliamp controls*	4MILLIAMP ADJUST	Adjustment to the 4-milliamp output for the 0% LFL
	MILLIAMP CAL SET	Signal that a calibration is in progress
	MILLIAMP FAULT	Signal that a fault exists
	MIN MILLIAMP	Lowest milliamp output, in % LFL
CONTROLS *Input/output setup*	CONTROL 1/2/3/4	Command that executes when the Control 1/2/3/4 input is activated
	RELAY SETTING	Control the Air/Fuel Cutoff relay
	CHECK RELAY NE	Check relay state normally energized
SETTINGS *Analyzer ID*	DEVICE TYPE	Analyzer type
	SOFTWARE VERSION	Firmware version
	SERIAL NUMBER	Last 3 digits of the serial number
DILUTION/FUEL/ PURGE MONITOR	When enabled by device settings	

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2 Maintenance Instructions

Flame Cell Maintenance

! Before disassembly make sure you have replacements for ALL O-Rings! Always replace O-Rings after removal, or if they are brittle or damaged.

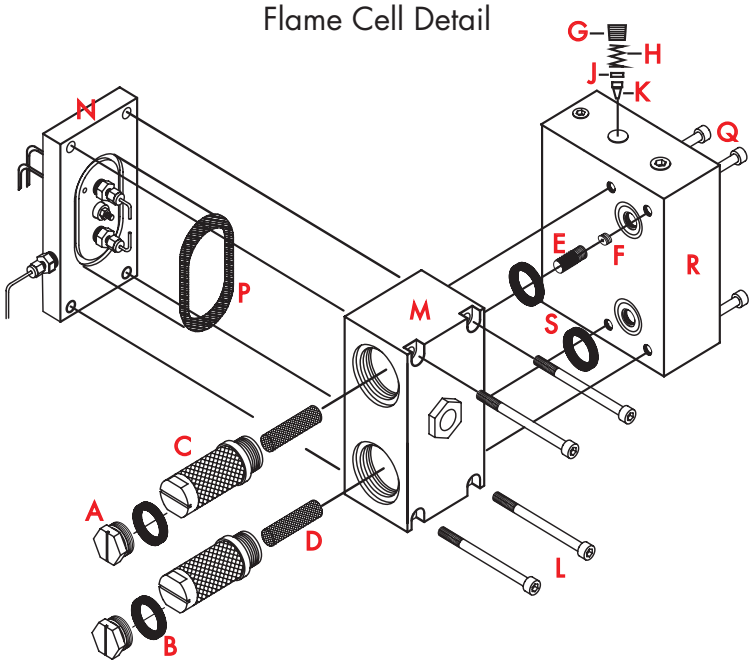
! Check for leaks and recalibrate after any maintenance procedure to verify proper operation.

! Spark ignition of the sensing flame may be inhibited up to 15 minutes after power-on or restoration of flow.

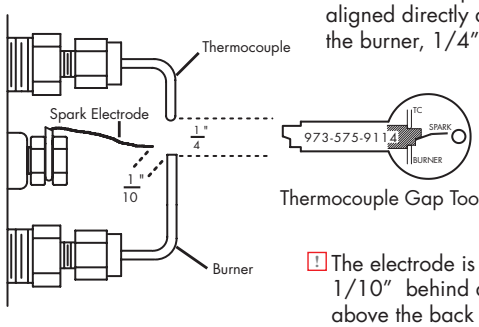
Disassembly Sequence	Preventative Maintenance*	Low Flow Troubleshooting*	Reassembly Sequence	Part Name, Number (Quantity)
Step 1	X	Step 1	Step 9	Sample/Exhaust Tube and Leak Check
Step 2	Step 1	Step 2	Step 8	A Flame Arrestor Cap, SNP188 (2)
Step 3	Step 2	Step 3	Step 7	B Flame Arrestor O-Ring, ORG005 or ORK005M ** (2)
Step 4	Step 3	Step 4	Step 6	C Flame Arrestor, SNP189 (2): Hand tight -1/8th of a turn
Step 5	Step 4	Step 5	Step 5	D Filter Element, FLT012 (2)
Step 6		Step 6	Step 4	E Orifice Filter, FLT087: use 10" quick wedge screwdriver, TOL044, to remove
X		Step 7	X	F Sample Flow Orifice, ORF024: use 1/8" Allen Key, TOL046, to remove
X		Step 8	X	Aspirator Nozzle G Plug, PLG002: use 1/4" Allen Key to remove H Spring, SPR013 J Spacer, SPC009 K Nozzle, NZL001
Step 7			Step 3	Flame Cell L Bolts, MSC023 (4): use 5mm Allen Key to remove M Flame Cell Block, SNP219 N Burner Plate, SNP215 P O-Ring, ORG044 or ORK046 **
Step 8			Step 2	Aspirator Q Bolts, MSC007 (4): use 5mm Allen Key to remove R Aspirator Block, SNP359 S O-Ring, ORG041 or ORK047M ** (2)
			Step 1	Thermocouple Alignment Detail: use Gap tool TOL047

* For each step in these procedures, remove, inspect, & clean part; if necessary replace ** ORK - Kalrez (SNR705) ORG - Viton (all other models)

Flame Cell Detail



Thermocouple Detail



Thermocouple Gap Tool

! The thermocouple is aligned directly above the burner, 1/4" away.

! The electrode is located 1/10" behind and above the back edge of the burner.

PrevEx 700 SERIES

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3 Maintenance Instructions

Monthly Maintenance

Calibration

- Calibrate safely - never calibrate if combustible gases or vapors might be present.
- Turn calibration cylinders on and set pressure at analyzer inlet to 20 PSIG. Do not over or under pressurize.
- Use Full Calibration on COMMANDS menu to initiate the automatic calibration sequence.
- For each analyzer, observe the reading during the calibration process:
  1. Cal LED is on, zero solenoid energizes.
  2. Reading is near 0% LFL.
  3. After about one minute (CALIBRATION TIME on CALIBRATION menu) reading adjusts to 0% LFL.
  4. Cal LED remains on, zero solenoid de-energizes, span solenoid energizes.
  5. Reading climbs up to near CAL READING, typically about 60% LFL.
  6. After about one minute reading adjusts to CAL READING.
  7. Cal LED remains on, span solenoid de-energizes. Zero solenoid energizes for about 30 seconds to purge, reading falls back to 0% LFL.

Gauges

- Regularly check the settings and readings on the GAUGES and FLOW SYSTEM menus.

Alarms

- Use ALARM TESTS menu to activate relays. Verify that the FAULT and DANGER relays activate the E-Stop (shut-down or machine stop).

Notes

- 20 PSIG compressed air allows the analyzer to use fuel safely. Keep the fuel and power off until the air is on and checked. Compressed air drives the sample flow. Keep air pressure steady and at 20 PSIG or else Sample Flow Low may occur.
- The sensing flame is fed by a small flow of fuel. The fuel pressure at the inlet must be high enough to make a stable sensing flame or else the reading may drift.
- The analyzer uses AC power for its heater and electronics. When first turned on, the status will be Block Temp Low and the heater LED will be on until the analyzer temperature, BLOCK TEMP, reaches its operating temperature setting, BLOCK SETTING. When analyzer reaches operating temperature, the heater LED cycles on/off. Analyzer may still be in fault condition Flameout.
- When in Flameout, flame temperature less than 450°C, use Alarm Reset command to re-ignite sensing flame. Spark ignition of the sensing flame may be inhibited up to 15 minutes after power-on or restoration of flow.
- Check the analyzer for leaks. Plugging the sample inlet tube should reduce the SAMPLE FLOW to near 0 LPM, as indicated in the GAUGES menu.

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